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Tim Netzer
Daniel Plaumann

Geometry of Linear Matrix Inequalities

A Course in Convexity and Real Algebraic Geometry with a View Towards Optimization

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A Course in Convexity and Real Algebraic Geometry with a View Towards Optimization

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For Miranda, Rainer, Frederik and Alwin

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Introduction and Preliminaries

1

In this first chapter we give an introduction and outline of the topics from the book. We also introduce basic notions and results from linear algebra, convexity theory and real algebraic geometry.

1.1 Introduction

Optimization, an important branch of mathematics, is becoming ever more influential with steadily increasing computational power. Convex optimization is considered to be its most accessible form, allowing for efficient numerical and geometrical approaches, such as interior point and subgradient methods, as well as duality theory. Many of the solution methods are of polynomial complexity, at least up to arbitrary precision, see [5, 13] for a comprehensive general overview.

For a long time, *linear programming* has been, and still is, the most important subclass of convex programming, and a thorough theory of linear programming has been established, combined with highly effective solution procedures (see for example [19, 20]). The feasible sets of linear programming are *polytopes/polyhedra*, convex sets defined by finitely many affine linear inequalities. The study of polyhedra from a geometric point of view reaches far back in history, but it also remains a challenging branch of modern research, see for example [106]. Polyhedra are of an intrinsically combinatorial nature, which allows to apply special geometric and algebraic methods to study them. On the other hand, this clearly hinders applicability of linear programming to problems which are not of such combinatorial flavor.

Semidefinite programming, a much more recent development, can come to the rescue here (see Sect. B.3 from the Appendix for an exact definition). As a still restrictive class of convex optimization problems, it allows for efficient numerical solution schemes, mostly based on interior point methods with barrier function (see for example [66, 101, 102, 105]). Furthermore, several restrictions imposed by linear

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programming disappear. One of the first significant applications stems from [36], where the stable set problem for graphs was translated into a semidefinite program, solvable in polynomial time for perfect graphs. In recent times, semidefinite programming has found numerous other applications [3, 12, 21, 56, 76].

The feasible sets of semidefinite programming are called *spectrahedra*. They are sets defined by linear matrix inequalities, i.e., by the condition that a symmetric matrix with affine linear entries is positive semidefinite. This involves the *spectr*um of matrices and generalizes poly*hedra*, which explains the name [86]. Spectrahedra are very special convex sets. They share some important properties with polyhedra, but also generalize these to a great extend. Furthermore, linear images of spectrahedra, so-called *spectrahedral shadows*, are equally useful for optimization, and they extend the class of convex sets even more.

The study of spectrahedra and their shadows as geometric objects is much younger, among the earliest works seem to be [59, 77, 86] from the 1990s. Again, the area has seen tremendous progress in recent years, combining results and methods from such diverse areas as (real) algebraic geometry, convexity theory, operator algebra, discrete geometry and optimization. Among the landmark results are surely Ramana and Goldman [86] with a first systematic study of the geometry of spectrahedra, the relaxation techniques by Lasserre [56,57] and Parrilo [76], Helton and Vinnikov's [43] classification of spectrahedra in dimension two (which also gives a solution to the Lax conjecture [60,61]), the positive results of Helton and Nie on spectrahedral shadows [41, 42], and Scheiderer's classification of spectrahedral shadows in dimension two [98], as well as his obstructions in higher dimensions [99]. All these insights are not only fascinating from a geometric point of view, but can also help to develop more efficient optimization algorithms and extend their field of application.

Of particular relevance in optimization are questions as the following: Given a certain set (often described by polynomial inequalities), what is a good description of its convex hull? Can semidefinite programming be performed on this convex hull, i.e., is it a spectrahedron or a spectrahedral shadow? If so, find explicit descriptions by linear matrix inequalities. If not, find good inner or outer approximations by sets on which semidefinite programming can be performed. Try to make the optimization as efficient as possible, by finding semidefinite presentations of smallest possible size and complexity.

The goal of this book is to give an up-to-date presentation of the most important results on the geometry of spectrahedra and their projections. Even though the motivation comes from computational problems and in particular from optimization, our focus will mostly be on geometry and theoretical foundations. We tried to keep the exposition as elementary as possible, to make the book useful for pure and applied mathematicians, and hopefully even for engineers with a solid background in mathematics. Many of the results should be accessible with good a knowledge of linear algebra, some require more background from algebra and (real) algebraic geometry. We provide some of this non-standard background material in the appendix. We point out that [9] partially covers similar subjects, combining chapters

by different authors. That book, however, is much more focused on applications and less on the geometric aspects that we explain here.

This book is organized as follows. Chapter 2 is devoted to a systematic study of spectrahedra. We give definitions and explain first important properties in Sects. 2.1 and 2.2. These sections also contain large classes of examples. In Sects. 2.3 and 2.4 we explain the connections between spectrahedra, hyperbolic polynomials and determinantal representations. This culminates in a proof of the Helton-Vinnikov Theorem on two-dimensional spectrahedra in Sect. 2.5. Section 2.6 explains connections between graphs and spectrahedra, proving Brändén's results on elementary symmetric polynomials and derivative cones. Section 2.8 contains an overview of the theory of non-commutative spectrahedra and their relations to operator systems.

Chapter 3 treats linear images of spectrahedra, so-called spectrahedral shadows. After introducing the concept in Sect. 3.1, we review in Sect. 3.2 the many constructions they permit. The important Lasserre-Parrilo relaxation method is explained in Sect. 3.3. Based on this, we prove Scheiderer's result on spectrahedral shadows in the plane in Sect. 3.4. Section 3.5 contains the results of Helton and Nie on exactness of relaxations, and they are applied to hyperbolicity cones in Sect. 3.6. After presenting some general obstructions to the relaxation method in Sect. 3.7, the main part of the book closes with Sect. 3.8 and Scheiderer's examples of convex semialgebraic sets that are not spectrahedral shadows.

The Appendix contains two parts. In Sect. A we collect necessary background material on real algebraic geometry, most notably Archimedean Positivstellensätze for polynomials and matrix polynomials. Section B explains some concepts and results from convexity theory and optimization.

This book presents many wonderful results of our colleagues. Among them are Petter Brändén, João Gouveia, Bill Helton, Igor Klep, Mario Kummer, Jean Lasserre, Pablo Parrilo, Mihai Putinar, Claus Scheiderer, Konrad Schmüdgen, Markus Schweighofer, Rainer Sinn, Andreas Thom, Victor Vinnikov and Cynthia Vinzant. We thank them all for numerous discussions on the topic of this book. A preliminary version was used as lecture notes for courses at the Universities of Innsbruck, Konstanz and Leipzig. We thank our students for the interest, and for many remarks and questions that helped improving the exposition a lot. All graphics in this book were created with Mathematica [104].

1.2 Preliminaries

Let us fix some preliminaries that we will use throughout this book.

A subset $S \subseteq \mathbb{R}^n$ is *convex* if for all $u, v \in S$ and $\lambda \in [0, 1]$ we have

$$\lambda u + (1 - \lambda)v \in S.$$

So S must contain all line segments between its points (Fig. 1.1). For any given set $S \subseteq \mathbb{R}^n$, there is a smallest convex superset, called *the convex hull* $\mathrm{conv}(S)$ of S. It



Fig. 1.1 A non-convex set and its convex hull



Fig. 1.2 A convex set and a (truncated) convex cone

consists of all *convex combinations*

$$\sum_{i=1}^{m} \lambda_i u_i$$

where $m \in \mathbb{N}$, $u_i \in S$, $\lambda_i \geqslant 0$ and $\sum_{i=1}^{m} \lambda_i = 1$. Sometimes it is helpful to employ Carathéodory's Theorem (see for example [3] Theorem 2.3), which says that convex combinations of length $m = n + 1$ suffice when taking convex hulls in $\mathbb{R}^n$.

One reason why convex sets are so important in optimization is that the supremum/infimum of an affine linear function on a set S coincides with its supremum/infimum on $\mathrm{conv}(S)$. Since convex sets have a much simpler geometry than general sets, this makes optimization over convex sets much easier to handle.

Just like projective geometry often behaves much more regular than affine geometry, *convex cones* often behave more regular than general convex sets (Fig. 1.2). A convex cone is a subset $C \subseteq \mathbb{R}^n$ such that for all $u, v \in C$ and $\lambda, \gamma \geqslant 0$ we have

$$\lambda v + \gamma u \in C.$$

So C must be closed under *positive combinations* instead of just convex combinations. The smallest convex cone containing a set S is called *the conic hull* $\mathrm{cone}(S)$ of S. It consists of all positive combinations of elements from S, and Carathéodory's Theorem says that positive combinations of length n always suffice in $\mathbb{R}^n$. A general procedure how to pass from convex sets to convex cones and vice versa (without losing too much information) is described in more detail in Remark 2.31 (1) below. Good general references for convexity theory are for example [3, 90] or [103] .

Let us continue with some linear algebra. If R is a ring, we write

$$\begin{aligned}
\mathrm{Mat}_{s\times t}(R) &= \text{space of all matrices of size } s \times t \text{ with entries in } R,\\
\mathrm{Mat}_s(R) &= \text{ring of all square matrices of size } s \text{ with entries in } R,\\
\mathrm{GL}_s(R) &= \text{group of invertible square matrices of size } s \text{ over } R,\\
\mathrm{Sym}_s(R) &= \text{space of all symmetric matrices of size } s \text{ with entries in } R.
\end{aligned}$$

Unless specified otherwise, matrices will always be assumed to have real entries. We write Mat_s, Sym_s, etc. to denote $\mathrm{Mat}_s(\mathbb{R})$, $\mathrm{Sym}_s(\mathbb{R})$, etc. Sometimes we will write Her_s for the set of all complex Hermitian matrices, i.e., matrices with $A^* = A$, where $A^* = (\overline{A})^T$ denotes the conjugate transpose of A. On Sym_s and Her_s we will often use the standard inner product, defined as

$$\langle P, Q\rangle := \mathrm{Tr}(PQ).$$

It is well-known that real symmetric matrices can be diagonalized: for $M \in \mathrm{Sym}_s$ there exists an *orthogonal* matrix $U \in \mathrm{Mat}_s$ (i.e., $U^{-1} = U^T$) with

$$U^{-1}MU = U^TMU = \mathrm{diag}(\lambda_1, \ldots, \lambda_s).$$

In this case, the diagonal entries λ_i are the eigenvalues of M, and hence uniquely determined (up to ordering). When relaxing orthogonality of U to invertibility, the λ_i in

$$U^TMU = \mathrm{diag}(\lambda_1, \ldots, \lambda_s)$$

are not uniquely determined anymore, they can in fact all be normalized to $-1, 0, 1$. However, the number of positive and negative λ_i is still uniquely determined, by *Sylvester's Law of Inertia*. If M has p positive and n negative signs in a diagonalization, the pair (p, n) is called the *signature* of M.

Recall that a real symmetric matrix $M \in \mathrm{Sym}_s$ is called *positive semidefinite (psd)* if

$$v^TMv \geqslant 0$$

holds for all $v \in \mathbb{R}^s$. It is called *positive definite* if this inequality is strict for all $v \neq 0$. We write

$$\begin{aligned}
\mathrm{Sym}_s^{+} &= \text{convex cone of positive semidefinite matrices of size } s\\
\mathrm{Sym}_s^{++} &= \text{convex cone of positive definite matrices of size } s.
\end{aligned}$$

Note that Sym_s^+ is the closure of Sym_s^{++} and Sym_s^{++} is the interior of Sym_s^+ in the finite-dimensional space Sym_s. For a symmetric matrix M, we will write $M \succcurlyeq 0$ if it is positive semidefinite, and $M \succ 0$ if it is positive definite. We extend this to a partial order on Sym_s by writing $M \succcurlyeq N$ if $M - N \succcurlyeq 0$.

Exercise 1.1
Let M be a real symmetric matrix of size s and rank r.

1. Show that the following are equivalent:
   (i) M is positive semidefinite.
   (ii) All eigenvalues of M are nonnegative.
   (iii) There exists a real $r \times s$-matrix Q with $M = Q^T Q$.
   (iv) There exists a symmetric and positive semidefinite $s \times s$-matrix P with $M = P^2$.
   (v) All principal minors of M are nonnegative.
   (vi) All principal minors of size at most r of M are nonnegative.
2. If $P^T P = Q^T Q$ with P and Q of the same size $r \times s$, there exists an orthogonal $r \times r$-matrix U such that $P = UQ$.
3. For $M \in \mathrm{Sym}_s^+$, the matrix P in 1(iv) is uniquely determined (and commonly denoted $\sqrt{M}$). $\sqrt{M}$ lies in the subalgebra of Mat_s generated by M.
4. For $M \in \mathrm{Sym}_s^+$ and $v \in \mathbb{R}^s$, show that $v^T M v = 0$ implies $Mv = 0$.
5. For $M \in \mathrm{Sym}_s^+$, show that $\mathrm{Tr}(M) = 0$ implies $M = 0$.

Finally, we will need some basic notions from real algebraic geometry. We always denote by $\mathbb{R}[x]$ the real polynomial ring in the variables $x = (x_1, \ldots, x_n)$. For $p_1, \ldots, p_r \in \mathbb{R}[x]$ we write $p = (p_1, \ldots, p_r)$ for the tuple and define

$$\mathcal{S}(p) = \mathcal{S}(p_1, \ldots, p_r) = \left\{u \in \mathbb{R}^n \mid p_1(u) \geqslant 0, \ldots, p_r(u) \geqslant 0\right\},$$

a *basic closed (semialgebraic)* set (Fig. 1.3). A general *semialgebraic set* is a finite Boolean combination of basic closed sets.

One of the most important results on semialgebraic sets is the projection theorem, also closely related to quantifier elimination and Tarski's transfer principle in real algebraic geometry (see [10, 63, 83] for details and proofs).

Theorem 1.2 (Projection Theorem)
If $S \subseteq \mathbb{R}^m \times \mathbb{R}^n$ is semialgebraic and $\pi : \mathbb{R}^m \times \mathbb{R}^n \to \mathbb{R}^n$ is the canonical projection, then $\pi(S) \subseteq \mathbb{R}^n$ is semialgebraic.

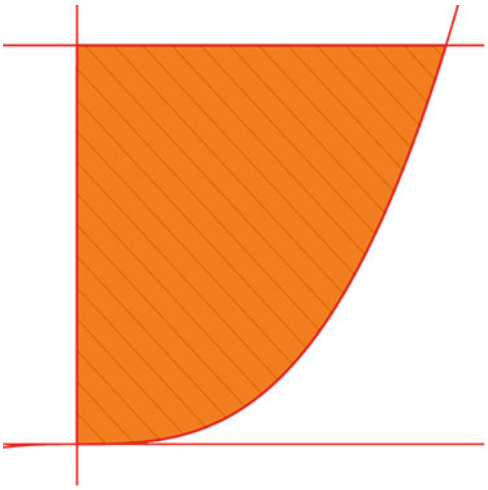


Fig. 1.3 Basic closed set in $\mathbb{R}^2$, defined by $x_1 \geqslant 0, 1 - x_2 \geqslant 0, x_2 - x_1^2 \geqslant 0$

The result immediately extends to arbitrary polynomial mappings instead of projections. Note however, that the projection of a (basic) closed set is not necessarily (basic) closed.

Exercise 1.3
Show that Sym_s^+ is a basic closed semialgebraic set, and that Sym_s^{++} is a semialgebraic set.

Exercise 1.4
Let $S \subseteq \mathbb{R}^n$ be a semialgebraic set. Using the projection theorem, show the following:

1. $\mathrm{conv}(S)$ and $\mathrm{cone}(S)$ are semialgebraic.
2. The closure $\mathrm{clos}(S)$ and the interior $\mathrm{int}(S)$ are semialgebraic.

Exercise 1.5
Find a basic closed semialgebraic set S and a projection π, such that $\pi(S)$ is closed but not basic closed.

Exercise 1.6
Show that $\mathbb{Z} \subseteq \mathbb{R}^1$ is not semialgebraic, and neither is $\{(r, e^r) \mid r \in \mathbb{R}\} \subseteq \mathbb{R}^2$.



Linear Matrix Inequalities and Spectrahedra

2

In this chapter we introduce the notion of a spectrahedron, and thoroughly study its properties. We will see many examples, and learn methods to determine whether a given set is a spectrahedron or not. In most cases we will also obtain procedures to explicitly construct defining linear matrix inequalities.

2.1 Spectrahedra

A polyhedron is a subset of $\mathbb{R}^n$ described by a finite number of affine linear inequalities. Compact polyhedra are exactly the polytopes, i.e., the convex hulls of finitely many points. These are the simplest (and the most extensively studied) convex sets. Spectrahedra, our basic objects, are generalizations of polyhedra that comprise many more sets, but still share some of the good properties of polyhedra. Just as polyhedra are the feasible sets of linear progamming, spectrahedra are the feasible sets of semidefinite programming.

Definition 2.1

A *spectrahedron* in $\mathbb{R}^n$ is the inverse image of Sym_s^+ under an affine linear map $\mathbb{R}^n \to \mathrm{Sym}_s$, for some $s \in \mathbb{N}$.

Many of the basic geometric properties we explain in this section can already be found in [86]. Spectrahedra are clearly convex, since the inverse image of any convex set under an affine linear map is again convex (Fig. 2.1). Occasionally, we may find it convenient not to fix coordinates and consider a spectrahedron in a finite-dimensional real vector space V given as the inverse image of $\mathrm{Sym}(W)^+$ under an affine linear map $\Phi\colon V \to \mathrm{Sym}(W)$, for some finite-dimensional real vector space W. Here, $\mathrm{Sym}(W)$ denotes the space of symmetric bilinear forms on W, and $\mathrm{Sym}(W)^+$ denotes the convex cone of positive semidefinite forms.

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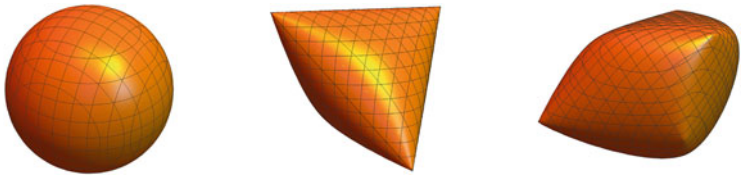


Fig. 2.1 Some spectrahedra

When working with matrices, we can write things out explicitly: An affine linear map $\Phi\colon \mathbb{R}^n \to \mathrm{Sym}_s$ is given by an $(n+1)$-tuple of symmetric matrices $M_0, M_1, \ldots, M_n \in \mathrm{Sym}_s$ via

$$\Phi(u) = M_0 + \sum_{i=1}^{n} u_i M_i,$$

and the corresponding spectrahedron is the set

$$\Phi^{-1}(\mathrm{Sym}_s^+) = \left\{u \in \mathbb{R}^n \mid M_0 + u_1 M_1 + \cdots + u_n M_n \succcurlyeq 0\right\} \subseteq \mathbb{R}^n.$$

We call the expression

$$M(x) = M_0 + x_1 M_1 + \cdots + x_n M_n$$

a *linear matrix polynomial*. We can view $M(x)$ as a polynomial of degree one with matrix coefficients, i.e., as an element of degree at most one in the space $\mathrm{Sym}_s[x]$. Alternatively, we can think of it as a matrix with polynomial entries

$$M(x) = \begin{bmatrix} \ell_{11}(x) & \cdots & \ell_{1s}(x) \\ \vdots & \ddots & \vdots \\ \ell_{s1}(x) & \cdots & \ell_{ss}(x) \end{bmatrix}$$

with $\ell_{11}, \ell_{12}, \ldots, \ell_{ss} \in \mathbb{R}[x]$ of degree at most one, an element of the space $\mathrm{Sym}_s(\mathbb{R}[x])$. If not specified otherwise, linear matrix polynomials will always be real and symmetric. If $M(x)$ is any linear matrix polynomial in n variables, we write

$$\mathcal{S}(M) = \left\{u \in \mathbb{R}^n \mid M(u) \succcurlyeq 0\right\}$$

for the spectrahedron defined by M, just as for ordinary polynomials. Spectrahedra are therefore the sets of solutions to *linear matrix inequalities*.

Examples 2.2

1. Any polyhedron is also a spectrahedron. For if $P = \mathcal{S}(\ell_1, \ldots, \ell_s)$ is a polyhedron defined by polynomials $\ell_1, \ldots, \ell_s \in \mathbb{R}[x]$ of degree one, then $P = \mathcal{S}(M)$ for the diagonal matrix polynomial $M = \mathrm{Diag}(\ell_1, \ldots, \ell_s)$. This also shows that semidefinite programming is a generalization of linear programming.
   While any real symmetric matrix can be diagonalized, the same is not true for matrix polynomials, because the finitely many coefficient-matrices need not be *simultaneously* diagonalizable (see also Exercise 2.4). Therefore, not every spectrahedron is a polyhedron.
2. The linear matrix polynomial

$$M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + x_1 \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} + x_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 - x_1 & x_2 \\ x_2 & 1 + x_1 \end{bmatrix}$$

   in two variables x_1 and x_2 defines the closed unit disc in $\mathbb{R}^2$ as a spectrahedron. This is clear by looking at the minors of M, of which only the determinant is relevant here. This is however not the only possibility to define the disc, as can be seen in the next example.
3. The closed unit ball in $\mathbb{R}^n$ is a spectrahedron, defined by the linear matrix polynomial

$$M = \begin{bmatrix} 1 & 0 & \cdots & 0 & x_1 \\ 0 & 1 & \cdots & 0 & x_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & x_n \\ x_1 & x_2 & \cdots & x_n & 1 \end{bmatrix}.$$

   This is a direct computation, to be done in Exercise 2.3. See also Theorem 2.18 for a generalization.
4. For a more interesting example, let $\mathbb{R}[x]_d$ be the vector space of polynomials of degree at most d in $x = (x_1, \ldots, x_n)$, and let $\mathfrak{m} = (p_1, \ldots, p_s)^T$ be an s-tuple of polynomials spanning $\mathbb{R}[x]_d$ (e.g. the monomial basis, with $s = \binom{n+d}{n}$). The map

$$\begin{aligned} \Psi_{\mathfrak{m}} \colon \mathrm{Sym}_s &\to \mathbb{R}[x]_{2d} \\ A &\mapsto \mathfrak{m}^T A \mathfrak{m} \end{aligned}$$

   is linear and surjective. Given $p \in \mathbb{R}[x]_{2d}$, any element of the fibre $\Psi_{\mathfrak{m}}^{-1}(p)$ is called a *Gram matrix* of p, and

$$G_{\mathfrak{m}}(p) = \Psi_{\mathfrak{m}}^{-1}(p) \cap \mathrm{Sym}_s^+$$

   the *Gram spectrahedron* of p (with respect to $\mathfrak{m}$). As the intersection of Sym_s^+ with an affine subspace, $G_{\mathfrak{m}}(p)$ is clearly a spectrahedron. The Gram spectrahedron is nonempty if and only if p is a sum of squares of elements in $\mathbb{R}[x]_d$. For given $A \in G_{\mathfrak{m}}(p)$, we can write $A = B^T B$ with B of size $\mathrm{rk}(A) \times s$ (Exercise 1.1). Then

$$p = \mathfrak{m}^T A \mathfrak{m} = (B\mathfrak{m})^T B \mathfrak{m}$$

is a sum of $\mathrm{rk}(A)$ many squares. Conversely, if $p = \sum_{i=1}^{r} q_i^2$ is a sum of r squares from $\mathbb{R}[x]_d$, we can write $(q_1, \ldots, q_r)^T = B\mathfrak{m}$ for some $B \in \mathrm{Mat}_{r\times s}$, since $\mathfrak{m}$ spans $\mathbb{R}[x]_d$. Then

$$p = (q_1, \ldots, q_r)(q_1, \ldots, q_r)^T = (B\mathfrak{m})^T B\mathfrak{m} = \mathfrak{m}^T (B^T B)\mathfrak{m},$$

so that $B^T B \in G_{\mathfrak{m}}(p)$ and $\mathrm{rk}(B^T B) \leqslant r$. In particular, the shortest sums-of-squares-representations of p correspond to its positive semidefinite Gram matrices of minimal rank.

Moreover, $G_{\mathfrak{m}}(p)$ classifies the representations of p as a sum of squares up to orthogonal equivalence: For if $A \in G_{\mathfrak{m}}(p)$ is split as $A = B^T B = C^T C$ with B and C both of size $r \times s$, where $r = \mathrm{rk}(A)$, then $B = UC$ for some orthogonal matrix U of size $r \times r$ (Exercise 1.1 (2)). Conversely, given such U and a representation $p = \sum_{i=1}^{r} q_i^2$, then $(g_1, \ldots, g_r)^T = U(q_1, \ldots, q_r)^T$ gives another representation $p = \sum_{i=1}^{r} g_i^2$, corresponding to the same Gram matrix.

Exercise 2.3
Verify Example 2.2 (2) and (3) above.

Exercise 2.4
True or false? The spectrahedron $\mathcal{S}(M)$ is a polyhedron if and only if the matrices $M_0, \ldots, M_n$ are pairwise commuting (see also Corollary 2.93).

Exercise 2.5
Let $M(x) = M_0 + \sum_{i=1}^{n} x_i M_i$ be a linear matrix polynomial and put $M'(x) = \sum_{i=1}^{n} x_i M_i$. Show that $\mathcal{S}(M)$ is a cone if and only if $\mathcal{S}(M) = \mathcal{S}(M')$.

Exercise 2.6
Let $S, T \subseteq \mathbb{R}^n$ be spectrahedra. Prove that $S \cap T$ is a spectrahedron.

▶ **Remark 2.7** While the definition of a spectrahedron is simple enough, there are two different ways to think about it, corresponding to somewhat different geometric pictures.

1. We may think of a spectrahedron as a subset of $\mathbb{R}^n$, defined by a linear matrix inequality. This is also the dominant point of view in the dual formulation of a semidefinite program, see Sect. B.3. We can rewrite this as an infinite system of ordinary linear inequalities: If $M(x)$ is a linear matrix polynomial, then $\ell_v(x) = v^T M(x)v$ is a polynomial of degree at most 1 in the variables x, for any $v \in \mathbb{R}^s$, and

$$\mathcal{S}(M) = \left\{u \in \mathbb{R}^n \mid \ell_v(u) \geqslant 0 \text{ for all } v \in \mathbb{R}^s\right\}.$$

We may think of $\mathbb{R}^s$ as a parameter space for the linear inequalities describing the convex set $\mathcal{S}(M)$. Note that *any* closed convex set is described by an infinite family of linear inequalities: Given a closed convex subset S of $\mathbb{R}^n$, let

$$\mathcal{L} = \big\{\ell \in \mathbb{R}[x] \mid \deg(\ell) \leqslant 1, \ell|_S \geqslant 0\big\},$$

then $S = \{u \in \mathbb{R}^n \mid \forall \ell \in \mathcal{L}\colon \ \ell(u) \geqslant 0\}$. This follows from the Bipolar Theorem B.5 for closed convex sets. What makes spectrahedra special is the simple parametrization of $\mathcal{L}$ in terms of a linear matrix polynomial.

2. We may also think of a spectrahedron as a set of matrices, which is the point of view in the primal formulation of a semidefinite program (see again Sect. B.3). Let $M(x) = M_0 + x_1 M_1 + \cdots + x_n M_n$ be a linear matrix polynomial, and

$$\begin{aligned} \Phi_M \colon \mathbb{R}^n &\to \mathrm{Sym}_s \\ u &\mapsto M_0 + u_1 M_1 + \cdots + u_n M_n \end{aligned}$$

the corresponding affine linear map. Consider the set of matrices

$$\mathrm{im}(\Phi_M) \cap \mathrm{Sym}_s^+ .$$

If Φ_M is injective, we can identify this with $\mathcal{S}(M)$. The geometric picture here is that of a convex cone (namely Sym_s^+) sliced by an affine linear subspace. The subspace we are slicing with is the span of $M_1, \ldots, M_n$ shifted by M_0. For instance, this is the way we would think about the Gram spectrahedra in Example 2.2 (4). If Φ_M is not injective, then $\mathcal{S}(M)$ is essentially a cylinder over $\mathrm{im}(\Phi_M) \cap \mathrm{Sym}_s^+$, given in terms of the kernel of Φ_M (see Exercise 2.10 below), so we do not loose too much by passing to the image.
It is instructive to make the analogy with polyhedra again: Let

$$P = \big\{u \in \mathbb{R}^n \mid \ell_1(u) \geqslant 0, \ldots, \ell_s(u) \geqslant 0\big\}$$

and consider the affine linear map

$$\Phi \colon \mathbb{R}^n \to \mathbb{R}^s, u \mapsto (\ell_1(u), \ldots, \ell_s(u)).$$

Then P is the inverse image of the *positive orthant* $\mathbb{R}^s_+$ under Φ. Thus, again up to injectivity of Φ, any polyhedron is a slice of a standard cone $\mathbb{R}^s_+$ for some s, just as in the case of spectrahedra.

The following result provides another class of examples of spectrahedra. It will become very important for the definition of the Lasserre-Parrilo relaxation in Sect. 3.3. We let Σ_{2d} denote the set of sums of squares of polynomials of degree at most d, i.e.,

$$\Sigma_{2d} = \left\{ \sum_{i=1}^{r} p_i^2 \mid r \in \mathbb{N}, p_i \in \mathbb{R}[x], \deg(p_i) \leqslant d \right\}. \tag{2.1}$$

It is a convex cone in $\mathbb{R}[x]_{2d}$, the space of polynomials of degree at most $2d$. For the definition of dual space and cone, see Sect. B.

Proposition 2.8
The dual cone $\Sigma_{2d}^{\vee}$ is a spectrahedron.

Proof By definition, we have

$$\Sigma_{2d}^{\vee} = \left\{L \in \mathbb{R}[x]_{2d}^{*} \mid \forall p \in \mathbb{R}[x]_d \colon L(p^2) \geqslant 0\right\}.$$

Any linear functional $L \in \mathbb{R}[x]_{2d}$ defines a symmetric bilinear form

$$\begin{aligned} b_L \colon \mathbb{R}[x]_d \times \mathbb{R}[x]_d &\to \mathbb{R} \\ (p, q) &\mapsto L(pq) \end{aligned}$$

and from this we obtain a linear map

$$\begin{aligned} \Phi \colon \mathbb{R}[x]_{2d}^{*} &\to \operatorname{Sym}(\mathbb{R}[x]_d) \\ L &\mapsto b_L. \end{aligned}$$

Now $\Sigma_{2d}^{\vee}$ is the spectrahedron $\Phi^{-1}\left(\operatorname{Sym}^{+}(\mathbb{R}[x]_d)\right)$. □

One can make a theory of spectrahedra defined by Hermitian matrices, which may be the natural point of view for certain questions (for example in quantum physics). However, in terms of the class of sets that one obtains, this does not add anything new, essentially due to the following simple observation. We will thus mostly restrict to real symmetric matrix polynomials in the following.

Lemma 2.9
Let M be a Hermitian linear matrix polynomial of size $s \times s$ (i.e., all its coefficient matrices M_i are Hermitian). Then there exists a real symmetric linear matrix polynomial N of size $2s \times 2s$ such that

$$\left\{u \in \mathbb{R}^n \mid M(u) \geqslant 0\right\} = \left\{u \in \mathbb{R}^n \mid N(u) \geqslant 0\right\}$$

and $\det(N) = \det(M)^2$.

Proof Write $M = P + iQ$, where P is real symmetric and Q is real skew-symmetric. Then put

$$N := \begin{bmatrix} P & Q \\ -Q & P \end{bmatrix}.$$

To see that N has the desired property, apply the change of coordinates

$$U^* N U = \begin{bmatrix} P - iQ & 0 \\ 0 & P + iQ \end{bmatrix} \quad \text{where} \quad U = \begin{bmatrix} \frac{1}{\sqrt{2}} \cdot I_s & -\frac{i}{\sqrt{2}} \cdot I_s \\ -\frac{i}{\sqrt{2}} \cdot I_s & \frac{1}{\sqrt{2}} \cdot I_s \end{bmatrix}.$$

In particular, $\det(N) = \det(\overline{M}) \det(M) = \det(M)^2$. □

Exercise 2.10
Let $M = M_0 + \sum_{i=1}^n x_i M_i$ be a linear matrix polynomial. Let $N_1, \ldots, N_m$ be a basis of $\operatorname{span}(M_1, \ldots, M_n)$ and put $N = M_0 + \sum_{i=1}^m x_i N_i$. Show that there exists a bijective linear transformation $\Psi : \mathbb{R}^n \to \mathbb{R}^m \times \mathbb{R}^{n-m}$ such that $\Psi(\mathcal{S}(M)) = \mathcal{S}(N) \times \mathbb{R}^{n-m}$ and $\Phi_M = (\Phi_N \times 0) \circ \Psi$.

While the positive definite matrices form the interior of the cone of positive semidefinite matrices, it may still happen that a linear matrix polynomial M is nowhere positive definite, even if $\mathcal{S}(M)$ has nonempty interior. For a trivial example, we could always artificially enlarge M by adding zeros, since

$$\mathcal{S}(M) = \mathcal{S}\begin{pmatrix} M & 0 \\ 0 & 0 \end{pmatrix}.$$

But this is basically all that can happen, and we can exclude this degenerate case, as we now show.

Definition 2.11

A linear matrix polynomial M of size s is called *monic* if $M(0) = I_s$, the identity matrix.

Lemma 2.12
Let M be a linear matrix polynomial.

1. *If M is monic, the interior of $\mathcal{S}(M)$ is the set*

$$\{u \in \mathbb{R}^n \mid M(u) > 0\}$$

2. *If 0 is an interior point of $\mathcal{S}(M)$, then there exists a monic linear matrix polynomial N of size $\operatorname{rk}(M(0))$ with $\mathcal{S}(M) = \mathcal{S}(N)$.*

Proof (1) is Exercise 2.13. For (2) let $M = M_0 + x_1 M_1 + \cdots + x_1 M_n$. Since $0 \in \mathcal{S}(M)$, we must have $M_0 = M(0) \succcurlyeq 0$. Hence there exists $U \in \mathrm{GL}_s$ such that

$$U^T M_0 U = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

where $r = \mathrm{rk}(M_0)$. Write

$$U^T M U = \begin{bmatrix} N & P \\ P^T & N' \end{bmatrix},$$

where N and N' are linear matrix polynomials of size r resp. $s - r$, and P is a non-symmetric linear matrix polynomial of size $r \times (s - r)$. We claim that $P = 0$ and $N' = 0$. By the choice of U, the constant term of N' is zero, say $N' = x_1 N'_1 + \cdots + x_n N'_n$ with $N'_i \in \mathrm{Sym}_{s-r}$. Since 0 is an interior point of

$$\mathcal{S}(M) \subseteq \mathcal{S}(N'),$$

there is $\varepsilon > 0$ such that $\pm\varepsilon N'_i \succcurlyeq 0$ for $i = 1, \ldots, n$. That is impossible, unless $N'_i = 0$ for $i = 1, \ldots, n$. Now if W is an open neighborhood of 0 with $M(u) \succcurlyeq 0$ for all $u \in W$, then $N'(u) = 0$ implies $P(u) = 0$ for all $u \in W$ (see Exercise 2.14 below). This implies $P = 0$, and the lemma is proved. □

Exercise 2.13
Prove part (1) of Lemma 2.12.

Exercise 2.14
Let

$$M = \begin{bmatrix} N & P \\ P^T & 0 \end{bmatrix}$$

be a real block-matrix with N symmetric. Show that if M is positive semidefinite, then
$P = 0$.

▶ **Remark 2.15** Lemma 2.12 essentially says that it is enough to consider spectrahedra defined by monic linear matrix polynomials. For if S is any spectrahedron, let V be its affine hull. Then S has nonempty relative interior in V, and after a translation we may assume that 0 is in the relative interior of S. We may then change coordinates, replace $\mathbb{R}^n$ by V, and assume that S is given by a monic linear matrix polynomial.
While this argument works fine as a first reduction step in a proof, actually computing a monic representation in large examples can be a difficult and computationally expensive task, which is usually avoided whenever possible.

Corollary 2.16
A spectrahedron S has nonempty interior if and only if there exists a linear matrix polynomial M such that $S = \mathcal{S}(M)$ and M is positive definite in some point of S. In this case,

$$\operatorname{int}(S) = \left\{u \in \mathbb{R}^n \mid M(u) > 0\right\}.$$

Proof If $M(u) > 0$ for some $u \in S$, we may translate and assume $u = 0$. Then $N = M(0)^{-1}$ is positive definite and $M'(x) = \sqrt{N}M(x)\sqrt{N}$ is a monic linear matrix polynomial with $\mathcal{S}(M) = \mathcal{S}(M')$. With this observation, the claim follows from Lemma 2.12. □

We finish this section with a class of spectrahedra constructed in [73], namely convex sets enclosed by ellipsoids.

Definition 2.17

For $u_1, \ldots, u_k \in \mathbb{R}^n$ and $r \geqslant 0$ we define

$$E(u_1, \ldots, u_k, r) := \left\{ u \in \mathbb{R}^n \mid \sum_{j=1}^{k} \|u_j - u\| \leqslant r \right\}$$

as the set of points, whose distances to the u_i sum up to at most r (Fig. 2.2).

Theorem 2.18
For any choice of $k \in \mathbb{N}$, $u_1, \ldots, u_k \in \mathbb{R}^n$ and $r \geqslant 0$, the set

$$E(u_1, \ldots, u_k, r) \subseteq \mathbb{R}^n$$

is a spectrahedron.

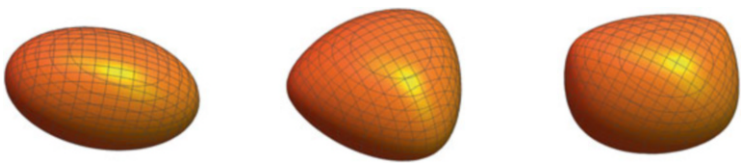


Fig. 2.2 Ellipsoids with $k = 2$, 3 and 4 foci u_j

Proof For matrices $N_1, \ldots, N_k \in \mathrm{Sym}_s$, consider the Kronecker/tensor sum

$$N_1 \boxplus \cdots \boxplus N_k := N_1 \otimes I_s \otimes \cdots \otimes I_s + I_s \otimes N_2 \otimes I_2 \cdots \otimes I_s + \cdots + I_s \otimes \cdots \otimes I_s \otimes N_k.$$

Here, $\otimes$ denotes the Kronecker/tensor product of matrices, so that $N_1 \boxplus \cdots \boxplus N_k$ is a matrix of size s^k. By Exercise 2.19, the eigenvalues of $N_1 \boxplus \cdots \boxplus N_k$ are the sums

$$\lambda_1 + \cdots + \lambda_k$$

where each λ_j is an eigenvalue of N_j. Now for $j = 1, \ldots, k$ consider the symmetric linear matrix polynomial

$$N_j(x) := \begin{bmatrix} 0 & u_{j1} - x_1 & \cdots & u_{jn} - x_n \\ u_{j1} - x_1 & & & \\ \vdots & & 0 & \\ u_{jn} - x_n & & & \end{bmatrix}$$

where $u_j = (u_{j1}, \ldots, u_{jn})$. For $u \in \mathbb{R}^n$, the eigenvalues of $N_j(u)$ are 0 and $\pm\|u_j - u\|$, by an easy argument as probably used in Exercise 2.3 already. So the eigenvalues of

$$(N_1 \boxplus \cdots \boxplus N_k)(u) = N_1(u) \boxplus \cdots \boxplus N_k(u)$$

are all sums of the form

$$\sum_{j \in J} \pm \|u_j - u\|$$

with $J \subseteq \{1, \ldots, k\}$. Each of them is smaller or equal to $\sum_{j=1}^k \|u_j - u\|$, so

$$\left(r \cdot I_{s^k} - N_1 \boxplus \cdots \boxplus N_k\right)(u) \succcurlyeq 0$$

if and only if $\sum_{j=1}^k \|u_j - u\| \leqslant r$, i.e., $u \in E(u_1, \ldots, u_k, r)$. Since $r \cdot I_{s^k} - N_1 \boxplus \cdots \boxplus N_k$ is a symmetric linear matrix polynomial, this proves the claim. □

Exercise 2.19
Let $N_1, \ldots, N_k \in \mathrm{Sym}_s$ be given. Show the following:

1. If each N_j is orthogonally equivalent to some D_j, then $N_1 \boxplus \cdots \boxplus N_k$ is orthogonally equivalent to $D_1 \boxplus \cdots \boxplus D_k$.
2. The eigenvalues of $N_1 \boxplus \cdots \boxplus N_k$ are the sums $\lambda_1 + \cdots + \lambda_k$, where each λ_j is an eigenvalue of N_j.

2.2 First Properties of Spectrahedra

In general it can be very hard to decide whether a given set is a spectrahedron or not. It might even be harder to find an explicit linear matrix polynomial that describes the set. We will thus start with some easy properties that all spectrahedra share, i.e., we will state necessary conditions for a set to be a spectrahedron.

Proposition 2.20
Spectrahedra are convex and basic closed semialgebraic.

Proof Let $S = \mathcal{S}(M)$ for some linear matrix polynomial M of size s. Convexity of S is clear from the definition. We must show that there exist polynomials $p_1, \ldots, p_r \in \mathbb{R}[x]$ such that $S = \mathcal{S}(p_1, \ldots, p_r)$. To say that $M(u)$ is positive semidefinite for $u \in \mathbb{R}^n$ is saying that all its eigenvalues are nonnegative. The eigenvalues are the roots of the characteristic polynomial

$$\chi_{M(u)}(t) = \det(tI_s - M(u)).$$

Thus u is a point in S if and only if $\chi_{M(u)}(-t)$ has no positive roots. By the subsequent lemma, this is the case if and only if all coefficients of $(-1)^s \chi_{M(u)}(-t)$ are greater than or equal to zero. Thus we can take $p_1, \ldots, p_r$ to be the coefficients of

$$(-1)^s \chi_{M(x)}(-t)$$

as a polynomial in t, and even obtain $r = s$. □

Lemma 2.21
Let $p \in \mathbb{R}[t]$ be a monic polynomial in one variable, and assume that all roots of p are real. Then p has no positive roots if and only if all its coefficients are nonnegative.

Proof Clearly, if all coefficients of p are nonnegative, it cannot have positive roots. Conversely, if the roots of p are $-\alpha_1, \ldots, -\alpha_s$ with $\alpha_i \geqslant 0$, then all coefficients of

$$p = (t + \alpha_1) \cdots (t + \alpha_s)$$

are nonnegative. □

▶ **Remark 2.22** If M is a monic linear matrix polynomial of size s, the interior of $\mathcal{S}(M)$ is the basic open set defined by the leading principal minors

of $M(x)$, which are the determinants of $M(x)$ with the last k rows and columns deleted, for $k = s-1, \ldots, 0$. These are s polynomials of ascending degree $1, \ldots, s$. But in general it is not true that the closure of a basic open set

$$\mathcal{U}(g_1, \ldots, g_s) = \{u \in \mathbb{R}^n \mid g_i(u) > 0 \text{ for } i = 1, \ldots, s\}$$

is the basic closed set $\mathcal{S}(g_1, \ldots, g_s)$, and it is indeed not true that a matrix is positive semidefinite if all its leading principal minors are nonnegative (just take the diagonal matrix $\mathrm{Diag}(0, -1)$, all of whose leading principal minors are 0). What is true, however, is that a matrix is positive semidefinite if and only if *all* its principal minors are nonnegative (which are the determinants of the submatrices of M obtained by deleting all rows and columns with indices in some subset of $\{1, \ldots, s\}$), see Exercise 1.1. This gives another proof of Proposition 2.20. However, this description uses $2^s - 1$ inequalities rather than s.

Example 2.23 For

$$M = \begin{bmatrix} 1 & x_1 & x_2 \\ x_1 & 1 & 0 \\ x_2 & 0 & 1 \end{bmatrix}$$

we compute the characteristic polynomial

$$\chi_M(t) = \det(tI_3 - M) = t^3 - 3t^2 + (3 - x_1^2 - x_2^2)t + (x_1^2 + x_2^2 - 1)$$

and the coefficients of $(-1)^3\chi_M(-t)$ are thus $1, 3, 3 - x_1^2 - x_2^3, 1 - x_1^2 - x_2^2$. Only the last one is relevant and necessary to define $\mathcal{S}(M)$ as a basic closed set. The principal minors of M are

$$1, 1, 1, 1 - x_1^2, 1 - x_2^2, 1, 1 - x_1^2 - x_2^2,$$

of which also only the last one is necessary to define $\mathcal{S}(M)$, the planar unit disk.

Exercise 2.24
Let $v_1 = (-1, 0)$ and $v_2 = (1, 0)$ in $\mathbb{R}^2$ and let $S = B_1(v_1) \cup B_1(v_2)$, i.e., the union of two discs of radius 1 about v_1 and v_2. Show that S is basic closed, but the convex hull of S is not. Getting the idea is more important than a rigorous proof. This convex hull is called the (very old-fashioned) *soccer stadium*.



In view of Proposition 2.20, the soccer stadium is not a spectrahedron.

Exercise 2.25
Let $C = \{(b, c) \in \mathbb{R}^2 \mid \text{For all } t \in \mathbb{R}\colon t^4 - bt^2 + (1/4)c \geqslant 0\}$ be an affine section of the cone of univariate nonnegative polynomials. Show that C is closed, convex and semialgebraic, but not basic closed.



So this set is not a spectrahedron, and thus clearly the cone of globally nonnegative univariate polynomials of degree at most 4 is not a spectrahedron. Note that all nonnegative polynomials in one variable are sums of squares of polynomials (see for example [63, 83]).

Another property of spectrahedra is that all their faces are exposed, see Sect. B.2 for the definition and some properties. This was first proven in [86].

Theorem 2.26
All faces of spectrahedra are exposed.

Proof Consider first the cone Sym_s^+ itself. If $A \in \mathrm{Sym}_s^+$ is any positive semidefinite matrix, it is not hard to show (see for example [3, §II.12]) that the unique face of Sym_s^+ containing A in its relative interior is

$$F = \{B \in \mathrm{Sym}_s^+ \mid \ker A \subseteq \ker B\}.$$

To see that such F is an exposed face, let C be a psd matrix with $\operatorname{im} C = \ker A$. Now given $B \in \mathrm{Sym}_s^+$ with $\langle B, C\rangle = 0$, we have $BC = 0$ (since C is also psd) and hence $\ker A = \operatorname{im} C \subseteq \ker B$. This shows $F = \{B \in \mathrm{Sym}_s^+ \mid \langle B, C\rangle = 0\}$, so that F is exposed.

Now if $U \subseteq \mathrm{Sym}_s$ is an affine linear subspace and F a face of the spectrahedron $U \cap \mathrm{Sym}_s^+$, let $A \in \operatorname{relint}(F)$ and let F' be the unique face of Sym_s^+ with $A \in \operatorname{relint}(F')$. Then $F' \cap U$ contains A in its relative interior, hence $F' \cap U = F$. By

what we have just seen, $F' = \{B \in \mathrm{Sym}_s^+ \mid \langle B, C\rangle = 0\}$ for some $C \in \mathrm{Sym}_s^+$, so that $F = \{B \in U \cap \mathrm{Sym}_s^+ \mid \langle B, C\rangle = 0\}$ is also exposed.

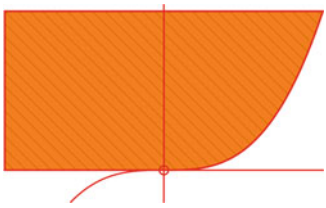
In general, if $\Phi\colon \mathbb{R}^n \to \mathrm{Sym}_s$ is an affine linear map, the faces of $\Phi^{-1}(\mathrm{Sym}_s^+)$ are in bijection with the faces of $S = \mathrm{im}\,\Phi \cap \mathrm{Sym}_s^+$, and if H is a hyperplane in Sym_s exposing a face $H \cap S$, the corresponding face $\Phi^{-1}(H \cap S) = \Phi^{-1}(H) \cap \Phi^{-1}(S)$ of $\Phi^{-1}(S)$ is also exposed. □

Exercise 2.27
Fill in the details in the above proof.

Example 2.28

1. Consider the convex basic closed semialgbraic set

$$S = \mathcal{S}(x_2 - x_1^3, 1 + x_1, 1 - x_1, x_2, 1 - x_2).$$



The origin is a non-exposed face of S, since the only linear polynomial $\ell \in \mathbb{R}[x]_1$ with $\ell|_S \geqslant 0$ and $\ell(0,0) = 0$ is $\ell = x_2$ (up to scaling), which however exposes the larger face $\{(u_1, 0) \in \mathbb{R}^2 \mid -1 \leqslant u_1 \leqslant 0\}$ of S, rather than just the origin. So although S is basic closed and convex, it is not a spectrahedron.

2. Since the set from Exercise 2.25 also has a non-esposed face (the origin), Theorem 2.26 gives another way to see that it is not a spectrahedron.

Finally, there is a very restrictive necessary condition for being a spectrahedron, called *real zero property* (or *hyperbolicity*, in the homogeneous setting). This property comes from the fact that symmetric matrices have real eigenvalues, and it in fact entails all the already mentioned properties of spectrahedra. If $M = I_s + \sum_{i=1}^n x_i M_i$ is a monic linear matrix polynomial of size s, the determinant $p = \det(M)$ is a polynomial (of degree at most s) in $\mathbb{R}[x]$, which vanishes on the boundary of the spectrahedron $\mathcal{S}(M)$. Since all the matrices $\sum u_i M_i$ for $u \in \mathbb{R}^n$ are real symmetric, their characteristic polynomials $\det(t I_s - \sum u_i M_i)$ have only real roots. Since

$$\det\left(t I_s - \sum u_i M_i\right) = t^s \det\left(M\left(t^{-1}u\right)\right) = t^s p\left(t^{-1}u\right),$$

this implies that for all $0 \neq u \in \mathbb{R}^n$ the function $t \mapsto p(tu)$ has only real roots. In other words, the intersection points of the zero set of p with lines through the origin are all real. This can be seen for a polynomial $p \in \mathbb{R}[x_1, x_2]$ of degree 4 in

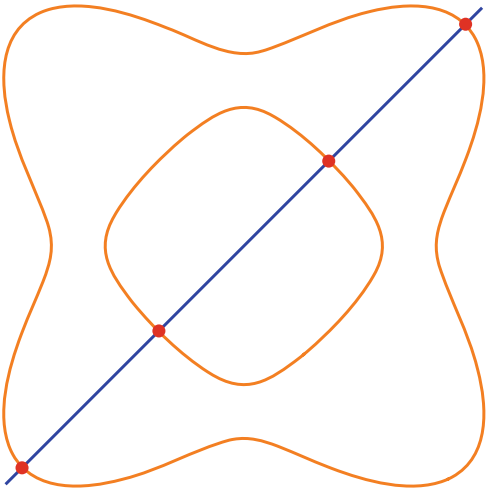


Fig. 2.3 A real zero plane curve of degree 4

Fig. 2.3. An important paper in which such polynomials were studied in connection with spectrahedra is [43]. Many of these results will be discussed in detail in the following sections. For now, the basic point we wish to make is simply that spectrahedra are very special convex sets.

Example 2.29 The set $\{(u_1, u_2) \in \mathbb{R}^2 \mid u_1^4 + u_2^4 \leqslant 1\}$ is known as the (very old-fashioned) *TV screen*.



It is basic closed and convex but not a spectrahedron. The reason is that $p = 1 - x_1^4 - x_2^4$ does not satisfy the hyperbolicity condition above.

Exercise 2.30
Prove that the TV screen is indeed not a spectrahedron.

We will finish this section with a remark on homogenization and dehomogenization of polynomials and convex sets. This will allow us to switch between the homogeneous and non-homogeneous setting in the following, and between arbitrary convex sets and convex cones.

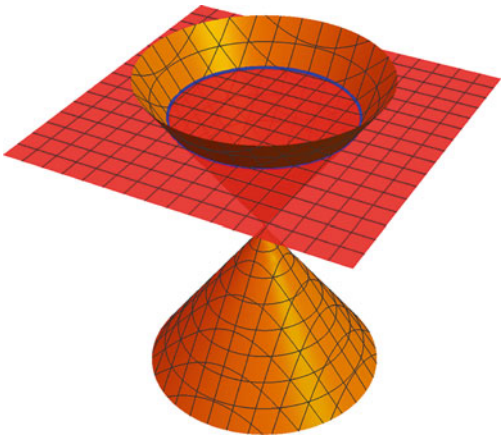


Fig. 2.4 (De-)Homogenization

▶ **Remark 2.31 ((De-)Homogenization)**

1. Let $S \subseteq \mathbb{R}^n$ be a convex set. We embed it into $\mathbb{R}^{n+1}$ as $\{1\} \times S = \{(1, u) \mid u \in S\}$ and take its conic hull

$$C = \operatorname{cone}(\{1\} \times S) \subseteq \mathbb{R}^{n+1}.$$

Thus C is a convex cone by construction, and we recover S by intersecting C with the affine hyperplane defined by $x_0 = 1$ (we use coordinates $x_0, x_1, \ldots, x_n$ on $\mathbb{R}^{n+1}$) (Fig. 2.4). When we intersect the closure $\operatorname{clos}(C)$ with this affine hyperplane, we obtain the closure $\operatorname{clos}(S)$ of S. When we intersect $\operatorname{clos}(C)$ with the hyperplane defined by $x_0 = 0$ instead, we obtain the *recession cone* of S.

If S is a spectrahedron, then $\operatorname{clos}(C)$ is a spectrahedral cone. For if $M_0 + \sum_{i=1}^n x_i M_i$ is a defining matrix polynomial for $S \neq \emptyset$, the homogeneous matrix polynomial $\sum_{i=0}^n x_i M_i$, together with the constraint $x_0 \geqslant 0$, defines $\overline{C}$ (Exercise 2.32). The additional inequality can clearly be added, using Exercise 2.6.

Conversely, if $\operatorname{clos}(C)$ is a spectrahedral cone, then $\operatorname{clos}(S)$ is a spectrahedron. This is easily seen by adding the constraints $\pm(1 - x_0) \geqslant 0$ to the matrix polynomial defining $\operatorname{clos}(C)$.

The cone C can be understood as a homogenization of S. Just as projective geometry often behaves more regular than affine geometry, convex cones often behave more regular than convex sets. The described construction shows how one can often reduce to the conic case. The same construction can clearly be done with any other (affine) hyperplane in $\mathbb{R}^{n+1}$ as well.

2. For a polynomial $p \in \mathbb{R}[x] = \mathbb{R}[x_1, \ldots, x_n]$ of degree d we consider its *homogenization*

$$h = x_0^d \cdot p(x_1/x_0, \ldots, x_n/x_0) \in \mathbb{R}[x_0, x].$$

This is a homogeneous polynomial of degree d in the additional variable x_0, and p is obtained by setting $x_0 = 1$. So the zero set of p is recovered by intersecting the zero set of h with the affine hyperplane defined by $x_0 = 1$, whereas the intersection with the hyperplane $x_0 = 0$ gives the *zeros at infinity* of p.
The real zero property of a polynomial p corresponds to *hyperbolicity* of its homogenization h, a property which we will discuss in more detail in the next section.

Exercise 2.32
Let $M = M_0 + \sum_{i=1}^n x_i M_i$ define the nonempty spectrahedron $S = \mathcal{S}(M)$. Show that the closure $\overline{C}$ of the homogenization of S from Remark 2.31 (1) is defined by the conditions $x_0 M_0 + \cdots + x_n M_n \succcurlyeq 0$ and $x_0 \geqslant 0$.

2.3 Hyperbolic Polynomials

Hyperbolicity is the homogeneous version of the real-zero property introduced above. We will now define and study it in detail. It is closely connected to spectrahedra, but also to partial differential equations [30, 47, 75] and discrete structures [15, 16, 38], and allows for direct convex optimization procedures, called *hyperbolic programming* [4, 37, 89].

Definition 2.33

A homogeneous polynomial $h \in \mathbb{R}[x_0, x_1, \ldots, x_n]$ is called *hyperbolic with respect to a point* $e \in \mathbb{R}^{n+1}$ if $h(e) \neq 0$ and if the univariate polynomial

$$h(u - te) \in \mathbb{R}[t]$$

has only real roots, for every $u \in \mathbb{R}^{n+1}$. It is called *strictly hyperbolic* if it is hyperbolic and the roots of $h(u - te)$ are all distinct, for every $u \in \mathbb{R}^{n+1} \setminus \mathbb{R} \cdot e$.

Exercise 2.34

1. Let $p \in \mathbb{R}[x]$ have the real zero property, i.e., $p(0) \neq 0$ and for all $u \in \mathbb{R}^n$ the function $t \mapsto p(tu)$ has only real zeros. Show that its homogenization
$$h = x_0^{\deg(p)} p(x/x_0) \in \mathbb{R}[x_0, x]$$
is hyperbolic with respect to $e = (1, 0, \ldots, 0)$.
2. Let $h \in \mathbb{R}[x_0, x]$ be homogeneous and hyperbolic with respect to $e = (1, 0, \ldots, 0)$. Show that its dehomogenization $p = h(1, x_1, \ldots, x_n) \in \mathbb{R}[x]$ has the real zero property.

Examples 2.35

1. The polynomial $h = x_0^2 - x_1^2 - x_2^2 \in \mathbb{R}[x_0, x_1, x_2]$ is (strictly) hyperbolic with respect to $e = (1, 0, 0)$, since $h(u - te) = (u_0 - t)^2 - u_1^2 - u_2^2$ has discriminant $4(u_1^2 + u_2^2) > 0$ and therefore two distinct real roots in t, for all $u = (u_0, u_1, u_2) \in \mathbb{R}^3 \setminus \mathbb{R} \cdot e$. This is the homogenization of the polynomial $p = 1 - x_1^2 - x_2^2$ that defines the unit circle. See Fig. 2.5 on the left for an illustration.
2. The polynomial $h = x_0^4 - x_1^4 - x_2^4 \in \mathbb{R}[x_0, x_1, x_2]$ is not hyperbolic with respect to any point in $\mathbb{R}^3$. This is the homogenized version of the polynomial defining the TV screen, see Example 2.29. In particular, for $e = (1, 0, 0)$, one can check that $h(u - te)$ has two real but also a pair of non-real complex-conjugate roots. See Fig. 2.5 in the middle for an illustration.
3. Let h be hyperbolic in direction e. Then
$$\partial_e(h) := \frac{\partial}{\partial t} h(x + te)\Big|_{t=0} = \sum_{i=0}^{n} e_i \frac{\partial}{\partial x_i} h$$

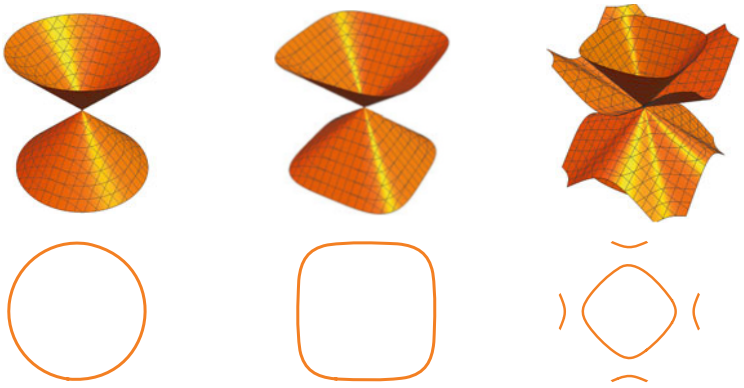


Fig. 2.5 Real zero sets of polynomials from Example 2.35 (1) and (2), and homogenized quartic curve from Fig. 2.3. Dehomogenized curves are shown in the second row

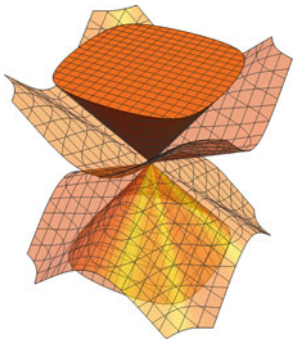


Fig. 2.6 Hyperbolicity cone of a quartic polynomial

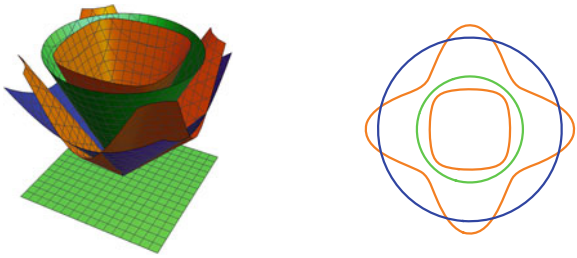


Fig. 2.7 Real zero set of quartic hyperbolic polynomial (orange) with first (green) and second (blue) derivative in direction e, affine section on the right

is again hyperbolic in direction e, by Rolle's Theorem. In fact the zeros of $\partial_e(h)$ interlace the zeros of h, we will discuss this in more detail in Sect. 2.4. These directional derivatives have for example been studied in [89] in the context of hyperbolic programming. See Fig. 2.7 for an illustration of a hyperbolic polynomial and its derivatives.

4. The determinant $\det(X)$ of a general symmetric $s \times s$-matrix $X = (x_{ij})_{1 \leqslant i,j \leqslant s}$, regarded as a polynomial on Sym_s and thus an element of $\mathbb{R}[x_{ij} \mid 1 \leqslant i \leqslant j \leqslant s]$, is hyperbolic with respect to the point $e = I_s$. This is because for any $M \in \mathrm{Sym}_s$, the roots of $\det(M - tI_s)$ are exactly the eigenvalues of M, which are all real. In particular,

$$\mathrm{Sym}_s^+ = \left\{ M \in \mathrm{Sym}_s \mid \det(M - tI_s) \text{ has only nonnegative roots} \right\}.$$

5. Let $M = \sum_{i=0}^n x_i M_i$ be a homogeneous linear matrix polynomial of size $s \times s$ and suppose there exists $e \in \mathbb{R}^{n+1}$ with $M(e) = I_s$. Then $h = \det(M) \in \mathbb{R}[x_0, x_1, \ldots, x_n]$ is homogeneous of degree s and hyperbolic with respect to e. Again, this is because $h(u - te) = \det(M(u) - tI_s)$ is the characteristic polynomial of the symmetric matrix $M(u)$ and therefore has only real roots. More generally, the same remains true if $M(e)$ is just any positive definite matrix, by considering

$$\sqrt{M(e)}^{-1} \cdot M(x) \sqrt{M(e)}^{-1}$$

instead of M. Note that the spectrahedral cone $\mathcal{S}(M)$ can be expressed in terms of h alone, namely as

$$\begin{aligned}\mathcal{S}(M) &= \{u \in \mathbb{R}^{n+1} \mid M(u) \succcurlyeq 0\} \\ &= \{u \in \mathbb{R}^{n+1} \mid h(u - te) \text{ has only nonnegative roots}\}.\end{aligned}$$

In view of these examples, we make the following definition.

Definition 2.36

Let $h \in \mathbb{R}[x_0, \ldots, x_n]$ be hyperbolic with respect to e. The set

$$\overline{\mathcal{C}}_e(h) := \{u \in \mathbb{R}^{n+1} \mid h(u - te) \text{ has only nonnegative roots}\}$$

is called the (closed) *hyperbolicity cone* of h with respect to e.

In this sense, one can think of hyperbolic polynomials as generalized characteristic polynomials, and the hyperbolicity cones as generalized cones of positive semidefinite matrices. But in spite of the name, it is not apparent from the definition that $\overline{\mathcal{C}}_e(h)$ is indeed a convex cone. Also, the name suggests that h should be hyperbolic with respect to any point in the (interior of the) hyperbolicity cone. These statements can be proved directly, as is done for example in [33]. Instead, we will deduce them later from the Helton-Vinnikov Theorem in Sect. 2.5. See Fig. 2.6 for an illustration of a hyperbolicity cone.
In Example 2.35 (5), we have already seen that the hyperbolicity condition is necessary for a cone to be spectrahedral.

Proposition 2.37
If M is a homogeneous linear matrix polynomial with $M(e) \succ 0$ for some $e \in \mathbb{R}^{n+1}$, the spectrahedral cone $\mathcal{S}(M)$ coincides with the hyperbolicity cone $\overline{\mathcal{C}}_e(\det(M))$. □

Thus the following statement is not surprising, though a little additional work is needed for the proof, which we will omit here (see [43, §2]).

Theorem 2.38
Let $C \subseteq \mathbb{R}^{n+1}$ be a basic closed convex cone with nonempty interior. Further, let $h \in \mathbb{R}[x_0, \ldots, x_n] \backslash \{0\}$ be the unique (up to scaling) polynomial of minimal degree vanishing on the boundary of C. Then C is a hyperbolicity cone if and only if h is hyperbolic with respect to some point $e \in \operatorname{int}(C)$, and in this case $C = \overline{C}_e(h)$. □

▶ **Remark 2.39** Theorem 2.38 is useful for checking whether a cone is spectrahedral. The polynomial h is often given with the definition of C, or can easily be computed from it. Now hyperbolicity of h is a necessary condition for being spectrahedral.
For example, it shows that the cone

$$C = \{u \in \mathbb{R}^3 \mid u_0^4 \geqslant u_1^4 + u_2^4, u_0 \geqslant 0\}$$

(the cone over the TV screen) is not spectrahedral, since the polynomial $h = x_0^4 - x_1^4 - x_2^4$ is not hyperbolic (c.f. Example 2.35 (2)). In view of Remark 2.31 this also shows that the TV screen is not a spectrahedron.
Whether hyperbolicity is also sufficient for being spectrahedral is an open question, the so-called *generalized Lax conjecture*, which we will discuss below. Note that checking hyperbolicity of a polynomial will also be hard in general, as hard as checking polynomial nonnegativity [93]. At least, it is clear what needs to be checked, whereas the definition of a spectrahedron requires to produce the defining matrix polynomial, of which not much is known a priori, not even its size. There are also certain approaches to use sums-of-squares techniques for certifying hyperbolicity [22, 44, 69, 85].

Definition 2.40

Given a homogeneous polynomial $h \in \mathbb{R}[x_0, \ldots, x_n]$, and a homogeneous symmetric linear matrix polynomial M such that $h = \det(M)$, we say that M is a *symmetric (linear) determinantal representation* of h. If, in addition, $M(e) > 0$ for some $e \in \mathbb{R}^{n+1}$, then h is hyperbolic with respect to e, and we say that the determinantal representation is *definite*.

Exercise 2.41
Show that every hyperbolic polynomial in two variables x_0, x_1 possesses a definite symmetric determinantal representation.

Unfortunately, not every hyperbolic polynomial possesses a symmetric determinantal representation.

Proposition 2.42
There exist hyperbolic polynomials in $n+1$ variables of degree d that do not possess a symmetric determinantal representation, in all of the following cases:

$$n = 3 \text{ and } d \geqslant 7, \quad n = 4 \text{ and } d \geqslant 3, \quad n \geqslant 5 \text{ and } d \geqslant 2.$$

Proof The set of hyperbolic polynomials of degree d has nonempty interior in the space $V = \mathbb{R}[x_0, \ldots, x_n]_{(d)}$ of homogeneous polynomials of degree d. Indeed, every strictly hyperbolic polynomial is an interior point of that set (see Exercise 2.50), since the roots of a univariate polynomial depend continuously on the coefficients. The dimension of the vector space V is $\binom{n+d}{d}$. On the other hand, if $h \in V$ has a symmetric determinantal representation $h = \det(M)$, then M must be of size $d \times d$, and the space of homogeneous linear matrix polynomials of size d in $n+1$ variables has dimension $(n+1)\binom{d+1}{2}$. The map taking M to $\det(M)$ is polynomial, so the dimension of the image cannot increase (see Theorem A.4 from the Appendix). It follows that if every hyperbolic polynomial is to possess a symmetric determinantal representation, we must have

$$(n+1)\binom{d+1}{2} = \frac{(n+1)(d+1)d}{2} \geqslant \frac{(n+d)!}{d!n!} = \binom{n+d}{d}.$$

This is equivalent to $(n+1)!(d+1)!d \geqslant 2(n+d)!$. Now one can check directly that this inequality fails in all the above cases. □

Exercise 2.43
Examine the inequality in the proof of Proposition 2.42 for all cases that are not stated in the theorem.

Now if we do the count of parameters in the above proof for $n = 2$, we find that the resulting inequality holds for all d. In 1957, it was conjectured by Peter Lax, in connection with the study of hyperbolic PDEs in [60], that every hyperbolic polynomial in three variables possesses a definite determinantal representation. This became known as the *Lax conjecture*. It was proved in [43] through the work of Vinnikov and Helton-Vinnikov.

Theorem 2.44 (Helton-Vinnikov)
Every hyperbolic polynomial in three variables x_0, x_1, x_2 possesses a definite symmetric determinantal representation.

Corollary 2.45
Every three-dimensional hyperbolicity cone is spectrahedral.

We will not give a full proof of the Helton-Vinnikov Theorem. However, we will prove a weaker version in Sect. 2.5 below, namely the existence of definite *Hermitian* determinantal representations. This will still imply Corollary 2.45, using Lemma 2.9.

Since it is clear from Proposition 2.42 that the Helton-Vinnikov Theorem cannot extend to the case $n \geqslant 3$, the search began for a suitable higher dimensional analogue. Various weaker versions have been proposed in recent years, some of which have been disproved (see for example [14, 53, 72]). Perhaps the most natural generalization is simply the statement of Corollary 2.45.

Generalized Lax Conjecture: Every hyperbolicity cone is a spectrahedron.

A few special cases of the conjecture are known, but in general it remains elusive.

Example 2.46 (Quadratic Polynomials) For any hyperbolic polynomial of degree 2, the associated hyperbolicity cone is spectrahedral, independent of the dimension. Note that, in view of Proposition 2.42, this does not mean they all admit a definite determinantal representation. It can however be shown that some high enough power always admits a determinantal representation [72]. Since

$$\overline{\mathcal{C}}_e\left(h^r\right) = \overline{\mathcal{C}}_e\left(h\right)$$

holds for any hyperbolic polynomial h and any $r \geqslant 1$, this is enough to prove spectrahedrality. We will prove it in a slightly different way here. Every quadratic form q can be written as

$$q = (x_0, \ldots, x_n) M (x_0, \ldots, x_n)^T$$

with some $M \in \mathrm{Sym}_{n+1}$. After a change of basis we can assume without loss of generality

$$M = \mathrm{diag}(1, \ldots, 1, -1, \ldots, -1).$$

Now it is easily checked that q can only be hyperbolic (with respect to any direction $e \in \mathbb{R}^{n+1}$) if the signature of M is $(1, n)$ or $(n, 1)$, i.e., we have if $q = \pm(x_0^2 - x_1^2 - \cdots - x_n^2)$, see Exercise 2.47. For this particular polynomial q one immediately checks

$$\det \begin{bmatrix} x_0 & 0 & \cdots & 0 & x_1 \\ 0 & x_0 & \cdots & 0 & x_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & x_0 & x_n \\ x_1 & x_2 & \cdots & x_n & x_0 \end{bmatrix} = x_0^{n-1}\, q.$$

Since this is a definite determinantal representation, we see that

$$\overline{\mathcal{C}}_e\,(q) = \overline{\mathcal{C}}_e\left(x_0^{n-1} q\right)$$

is spectrahedral.

Exercise 2.47
Show that a quadratic form $q \in \mathbb{R}[x_0, \ldots, x_n]_{(2)}$ is only hyperbolic if its signature is $(1, n)$ or $(n, 1)$.

We finish this section by collecting some more facts about hyperbolic polynomials and their hyperbolicity cones.

Proposition 2.48
Hyperbolicity cones are basic closed semialgebraic sets. For example, one has

$$\overline{\mathcal{C}}_e(h) = \mathcal{S}(h, \partial_e h, \partial_e^2 h, \ldots, \partial_e^{\deg(h)-1} h)$$

where $\partial_e(h)$ denotes the directional derivative of h as in Example 2.35 (3).

Proof Exercise 2.49. □

Exercise 2.49
Let h be hyperbolic in direction e. Show the following:

1. $\overline{\mathcal{C}}_e(h) \subseteq \overline{\mathcal{C}}_e(\partial_e h)$.
2. $\overline{\mathcal{C}}_e(h) = \mathcal{S}(h, \partial_e h, \partial_e^2 h, \ldots, \partial_e^{\deg(h)-1} h)$.
   Hint: Reduce to a polynomial in one variable by restricting to a line parallel to e. See Fig. 2.7 for an illustration, and [89] for more information.

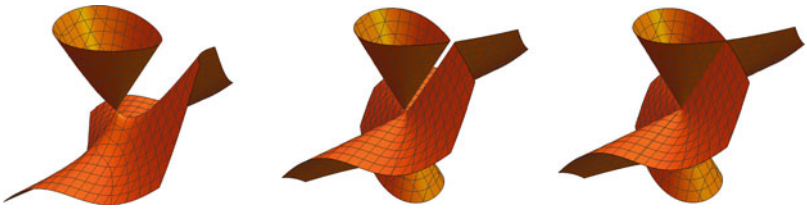


Fig. 2.8 Approximation of hyperbolic polynomial by strictly hyperbolic polynomials

Exercise 2.50
Fix $e \in \mathbb{R}^{n+1}$ and denote by $\mathcal{H}_e \subseteq \mathbb{R}[x_0, \ldots, x_n]_{(d)}$ the set of hyperbolic polynomials of degree d with respect to e.

1. Show that every strictly hyperbolic polynomial of degree d is an interior point of $\mathcal{H}_e$ inside $\mathbb{R}[x_0, \ldots, x_n]_{(d)}$.
2. Show that the strictly hyperbolic polynomials form a dense subset of $\mathcal{H}_e$.
   Hint: If $p \in \mathbb{R}[t]$ has only real roots, examine the roots of $p + \alpha p'$ for $\alpha \in \mathbb{R}$.

In fact more is true: $\mathcal{H}_e$ is connected, simply connected and coincides with the closure of its interior, which is the set of strictly hyperbolic polynomials, see [75] (Fig. 2.8).

2.4 Definite Determinantal Representations and Interlacing

To prove the Helton-Vinnikov Theorem, it is helpful to understand it as a statement in two parts. The first part amounts to the construction of determinantal representations over $\mathbb{R}$ or $\mathbb{C}$, the second to a characterization of those determinantal representations that are definite, and therefore reflect the hyperbolicity. The following notion will be used to address the second part.

Definition 2.51

Let $p, q \in \mathbb{R}[t]$ be univariate polynomials with $\deg(p) = d$ and $\deg(q) = d - 1$, and suppose that p and q have only real roots. Denote the roots of p by

$$\alpha_1 \leqslant \cdots \leqslant \alpha_d$$

and the roots of q by

$$\beta_1 \leqslant \cdots \leqslant \beta_{d-1}.$$

We say that q *interlaces* p if $\alpha_i \leqslant \beta_i \leqslant \alpha_{i+1}$ for all $i = 1, \ldots, d-1$. We say that q *strictly interlaces* p if all these inequalities are strict.

If $h \in \mathbb{R}[x_0, \ldots, x_n]_{(d)}$ is hyperbolic with respect to $e \in \mathbb{R}^{n+1}$, we say that the homogeneous polynomial $g \in \mathbb{R}[x_0, \ldots, x_n]_{(d-1)}$ (strictly) *interlaces h with respect to e*, if $g(u - te)$ (strictly) interlaces $h(u - te)$ in $\mathbb{R}[t]$ for every $u \in \mathbb{R}^{n+1}$, $u \notin \mathbb{R} \cdot e$. Note that this assumes that g is hyperbolic with respect to e as well.

Example 2.52 We have encountered the simplest and most important example already in Example 2.35 (3). If $p \in \mathbb{R}[t]$ is a univariate polynomial with only real roots, then its derivative p' interlaces p, by Rolle's Theorem. If p does not have multiple roots, it is strictly interlaced by p'.

Now more generally, if $h \in \mathbb{R}[x_0, \ldots, x_n]$ is hyperbolic with respect to $e \in \mathbb{R}^{n+1}$, the directional derivative

$$\partial_e(h) = \frac{\partial}{\partial t} h(x + te)\Big|_{t=0} = \sum_{i=0}^{n} e_i \frac{\partial}{\partial x_i} h$$

interlaces h, since $h'(u + te) \in \mathbb{R}[t]$ interlaces $h(u + te) \in \mathbb{R}[t]$ for all $u \in \mathbb{R}^{n+1}$. If h is strictly hyperbolic, then $\partial_e h$ strictly interlaces h.

Lemma 2.53
Suppose that $h \in \mathbb{R}[x_0, \ldots, x_n]_{(d)}$ is irreducible and hyperbolic with respect to e. Fix f, g in $\mathbb{R}[x_0, \ldots, x_n]_{(d-1)}$, where f interlaces h with respect to e. Then g interlaces h with respect to e if and only if fg is nonnegative or nonpositive on the real zero set $\mathcal{V}_{\mathbb{R}}(h)$ of h in $\mathbb{R}^{n+1}$.

Proof It suffices to prove this statement for the restriction to a line $\{u + te \mid t \in \mathbb{R}\}$ for generic $u \in \mathbb{R}^{n+1}$. Since h is irreducible, we may therefore assume that the roots of $h(u + te)$ are distinct from each other and from the roots of $f(u + te) \cdot g(u + te)$ (see Exercise 2.54 below).

Suppose that fg is nonnegative on $\mathcal{V}_{\mathbb{R}}(h)$. By the genericity assumption, the product $f(u + te)g(u + te)$ is positive on all the roots of $h(u + te)$. Between consecutive roots of $h(u + te)$, the polynomial $f(u + te)$ has a single root and thus changes sign. For the product fg to be positive on these roots, $g(u + te)$ must also change sign and have a root between each pair of consecutive roots of $h(u+te)$. Hence g interlaces h with respect to e.

Conversely, suppose that f and g both interlace h. Between any two consecutive roots of $h(u+te)$, both $f(u+te)$ and $g(u+te)$ each have exactly one root, and their product has exactly two. It follows that $f(u + te)g(u + te)$ has the same sign on all the roots of $h(u + te)$. Taking $t \to \infty$ shows this sign to be the sign of $f(e)g(e)$, independent of the choice of u. Hence fg has the same sign at every point of $\mathcal{V}_{\mathbb{R}}(h)$. □

Exercise 2.54
Let $h \in \mathbb{R}[x_0, \ldots, x_n]_{(d)}$, $h \neq 0$, be a square-free homogeneous polynomial and fix $e \in \mathbb{R}^{n+1}$ with $h(e) \neq 0$. Show that there exists a dense open subset $U \subseteq \mathbb{R}^{n+1}$ such that for any $u \in U$ the polynomial $h(u + te)$ has no multiple (real or complex) roots in t. Deduce the assumption made in the first paragraph of the proof above.

Recall that the *adjugate matrix* of a $d \times d$-matrix M (also called *adjoint matrix* or *Cramer matrix*) is the $d \times d$-matrix M^{adj} whose (j, k)-entry is $(-1)^{j+k}$ times the $(d-1) \times (d-1)$-minor of M, obtained by deleting the kth row and jth column and taking the determinant. The fundamental fact about the adjugate matrix is

$$M \cdot M^{\mathrm{adj}} = \det(M) \cdot I_d,$$

which holds for matrices with entries in any commutative ring.

If M is a (symmetric or Hermitian) homogeneous linear matrix polynomial of size $d \times d$, its adjugate M^{adj} is, by definition, a homogeneous matrix polynomial of size $d \times d$ and degree $d - 1$. The relation between M and M^{adj} will play a crucial role in the next section.

Definition 2.55

Let M be a homogeneous Hermitian linear matrix polynomial of size $d \times d$. Then

$$\mathcal{H}(M) \;=\; \left\{\lambda^* M^{\mathrm{adj}} \lambda \;\mid\; \lambda \in \mathbb{C}^d \setminus \{0\}\right\}$$

is a subset of $\mathbb{R}[x]_{(d-1)}$, which we call the *system of hypersurfaces associated with* M.

Here is a useful identity that goes back to the work of Hesse in 1855 [46].

Proposition 2.56
Let M be a homogeneous Hermitian linear matrix polynomial of size d. Then the polynomial

$$(\lambda^* M^{\mathrm{adj}} \lambda)(\mu^* M^{\mathrm{adj}} \mu) \;-\; (\lambda^* M^{\mathrm{adj}} \mu)(\mu^* M^{\mathrm{adj}} \lambda) \tag{2.2}$$

is contained in the ideal generated by $\det(M)$ *in* $\mathbb{R}[x_0, \ldots, x_n]$, *for any* $\lambda, \mu \in \mathbb{C}^d$. *In particular, the polynomial*

$$(\lambda^* M^{\mathrm{adj}} \lambda)(\mu^* M^{\mathrm{adj}} \mu)$$

is nonnegative on the real zero set $\mathcal{V}_{\mathbb{R}}(\det(M)) \subseteq \mathbb{R}^{n+1}$.

Proof Consider a general $d \times d$-matrix of variables $X = (x_{ij})_{i,j}$. At a generic point in $\mathcal{V}_{\mathbb{C}}(\det(X))$, the matrix X has rank $d-1$. The identity $X \cdot X^{\text{adj}} = \det(X) \cdot I_d$ implies that X^{adj} then has rank one at such a point. It follows that the 2×2-matrix

$$\begin{bmatrix} \lambda^* X^{\text{adj}} \lambda & \lambda^* X^{\text{adj}} \mu \\ \mu^* X^{\text{adj}} \lambda & \mu^* X^{\text{adj}} \mu \end{bmatrix}$$

also has rank at most one on the whole of $\mathcal{V}_{\mathbb{C}}(\det(X))$. Since the polynomial $\det(X)$ is irreducible (Exercise 2.57), the determinant of this 2×2 matrix thus lies in the ideal generated by $\det(X)$, using Hilbert's Nullstellensatz (see for example [25]). Restricting to $X = M$ gives the desired identity.

For the claim of nonnegativity, note that $(\mu^* M^{\text{adj}} \lambda) = \overline{(\lambda^* M^{\text{adj}} \mu)}$. So the polynomial

$$(\lambda^* M^{\text{adj}} \lambda)(\mu^* M^{\text{adj}} \mu)$$

is equal to a polynomial times its conjugate modulo $\det(M)$, and is therefore nonnegative on $\mathcal{V}_{\mathbb{R}}(\det(M))$. □

Exercise 2.57
Show that for a matrix $X = (x_{ij})_{i,j}$ of variables, $\det(X)$ is an irreducible polynomial. Show the same for the symmetric case.

We can use the identity from Proposition 2.56 to determine whether a determinantal representation is definite.

Theorem 2.58
Let $h \in \mathbb{R}[x_0, \ldots, x_n]_{(d)}$ be irreducible and hyperbolic with respect to e, and let $h = \det(M)$ be a Hermitian determinantal representation of h. If some polynomial in $\mathcal{H}(M)$ interlaces h with respect to e, every polynomial in $\mathcal{H}(M)$ does, and the matrix $M(e)$ is (positive or negative) definite.

Proof First, suppose that $f = \lambda^* M^{\text{adj}} \lambda$ interlaces h, and let g be another element of $\mathcal{H}(M)$, say $g = \mu^* M^{\text{adj}} \mu$ where $\mu \in \mathbb{C}^d$. From Proposition 2.56 we see that the product fg is nonnegative on $\mathcal{V}_{\mathbb{R}}(h)$. Then, by Lemma 2.53, g interlaces h.

To show that $M(e)$ is definite, we first show that any two elements f, g of $\mathcal{H}(M)$ have the same sign at the point e. Since h is irreducible, the polynomial fg cannot vanish on $\mathcal{V}_{\mathbb{R}}(h)$. Otherwise, by the Real Nullstellensatz (see for example [10] Theorem 4.5.1) h would divide fg and thus one of the two factors. This is impossible for degree reasons.

By Proposition 2.56, the product fg is nonnegative on $\mathcal{V}_{\mathbb{R}}(h)$ and thus strictly positive on a dense subset of $\mathcal{V}_{\mathbb{R}}(h)$. Furthermore, since both f and g interlace h,

the product

$$fg = (\lambda^* M^{\mathrm{adj}} \lambda)(\mu^* M^{\mathrm{adj}} \mu)$$

must be positive at e. Hence $\lambda^* M^{\mathrm{adj}}(e)\lambda \in \mathbb{R}$ has the same sign for all $\lambda \in \mathbb{C}^d$, and the Hermitian matrix $M^{\mathrm{adj}}(e)$ is thus definite. Hence so is $M(e) = h(e)(M^{\mathrm{adj}}(e))^{-1}$. □

▶ **Remark 2.59** The converse of Theorem 2.58 also holds, i.e., if $M(e)$ is definite, then all polynomials from $\mathcal{H}(M)$ interlace h with respect to e, see [81, Theorem 3.3].

We conclude this section with a useful result, showing that the map taking a matrix with linear entries to the determinant is closed when restricted to definite representations, which it need not be in general.

Lemma 2.60
Let $e \in \mathbb{R}^{n+1}$. The set of all homogeneous polynomials $h \in \mathbb{R}[x_0, \ldots, x_n]_{(d)}$ with $h(e) = 1$ that possess a Hermitian determinantal representation $h = \det(M)$ such that $M(e)$ is positive definite is a closed subset of $\mathbb{R}[x_0, \ldots, x_n]_{(d)}$.

Proof First we observe that if $h(e) = 1$ and $h = \det(M)$ with $M(e) > 0$, then h has such a representation M' for which $M'(e)$ is the identity matrix (as seen in the proof of Corollary 2.16).

Now let $h_k \in \mathbb{R}[x_0, \ldots, x_n]_{(d)}$ be a sequence of polynomials converging to h, such that $h_k = \det(M^{(k)})$ with $M^{(k)} = x_0 M_0^{(k)} + \cdots + x_n M_n^{(k)}$ and $M^{(k)}(e) = I_d$. For each j, let e_j denote the j-th unit vector from $\mathbb{R}^d$. Since

$$h_k(te - e_j) = \det\left(t I_d - M^{(k)}(e_j)\right) = \det\left(t I_d - M_j^{(k)}\right)$$

is the characteristic polynomial of $M_j^{(k)}$, the eigenvalues of each $M_j^{(k)}$ converge to the zeros of $h(te - e_j)$. It follows that each sequence

$$\left(M_j^{(k)}\right)_k$$

is bounded. After successively passing to a convergent subsequence of $M_j^{(k)}$ for each $j = 0, \ldots, n$, we may therefore assume that the sequence $M^{(k)}$ is convergent. We can then clearly conclude that $h = \det(\lim_{k\to\infty} M^{(k)})$. □

2.5 Hyperbolic Curves and the Helton-Vinnikov Theorem

The goal of this section is to show that every hyperbolic polynomial in three variables (also called a *hyperbolic plane curve*) possesses a Hermitian determinantal representation. This is a weaker statement than the Helton-Vinnikov theorem (Theorem 2.44), which says that there even exists a symmetric determinantal representation, but it still suffices to characterize three-dimensional spectahedral cones and plane spectrahedra. Our proof follows [81]. An alternative proof for the Hermitian case, using a different approach, can be found in [34].

First of all, we now again speak of curves and convex subsets of the plane, rather than of three-dimensional cones. This is because we consider projective varieties instead of affine cones, which gives a better geometric picture. We will use three variables x, y, z in this section. A homogeneous polynomial $h \in \mathbb{R}[x, y, z]$ of degree d defines the *projective plane curve*

$$\mathcal{Z}_{\mathbb{C}}(h) = \left\{u \in \mathbb{P}^2(\mathbb{C}) \mid h(u) = 0\right\}$$
$$\mathcal{Z}_{\mathbb{R}}(h) = \left\{u \in \mathbb{P}^2(\mathbb{R}) \mid h(u) = 0\right\}$$

where $\mathbb{P}^2(\mathbb{C})$ is the complex projective plane. We use the letter $\mathcal{Z}$ to distinguish the plane projective curve from the affine cone $\mathcal{V}_{\mathbb{C}}(h) \subseteq \mathbb{A}^3(\mathbb{C}) = \mathbb{C}^3$.

Recall how points in the projective plane are denoted in homogeneous coordinates: A point in $\mathbb{P}^2(\mathbb{C})$ is an equivalence class of points in $\mathbb{C}^3 \setminus \{0\}$ defining the same line through the origin. The point in $\mathbb{P}^2(\mathbb{C})$ corresponding to $(a, b, c) \in \mathbb{C}^3 \setminus \{0\}$ is denoted by $(a : b : c)$ and we have

$$(a : b : c) = (\lambda a : \lambda b : \lambda c)$$

for all $\lambda \in \mathbb{C}^*$. In particular, $(a : b : c)$ is real if and only if $\lambda a, \lambda b, \lambda c$ are all real for *some* $\lambda \in \mathbb{C}^*$. Thus $(i : i : i)$ is a real point, while $(1 : 1 : i)$ is not. Complex conjugation acts on $\mathbb{P}^2(\mathbb{C})$ via the rule

$$\overline{(a : b : c)} = (\overline{a} : \overline{b} : \overline{c}),$$

so that $\mathbb{P}^2(\mathbb{R})$ is exactly the set of fixed points of this action.

Let $h \in \mathbb{R}[x, y, z]$ be homogeneous and irreducible of degree d. We wish to find a determinantal representation $h = \det(M)$, where M is a Hermitian matrix of linear forms. We will describe a general method for constructing such a representation. The idea goes back to the work of Hesse in 1855 [46], which was extended by Dixon in 1902 [23].

Construction 2.61 Let $h, g \in \mathbb{R}[x, y, z]$ be homogeneous, with h irreducible, $\deg(h) = d$, and $\deg(g) = d - 1$. Assume that $\mathcal{Z}_{\mathbb{R}}(h) \cap \mathcal{Z}_{\mathbb{R}}(g) = \emptyset$.

1. We split the complex intersection points of the curves $\mathcal{Z}_{\mathbb{C}}(g)$ and $\mathcal{Z}_{\mathbb{C}}(h)$ in two conjugate subsets: Put $S = \mathcal{Z}_{\mathbb{C}}(h) \cap \mathcal{Z}_{\mathbb{C}}(g)$ and let $T \subseteq S$ be such that $S = T \cup \overline{T}$ and $T \cap \overline{T} = \emptyset$.
2. Consider the complex vector space

$$V = \mathcal{I}_{\mathbb{C}}(T) \cap \mathbb{C}[x, y, z]_{(d-1)} = \left\{ p \in \mathbb{C}[x, y, z]_{(d-1)} \mid p|_T = 0 \right\},$$

which is of dimension at least

$$\frac{(d+1)d}{2} - \frac{d(d-1)}{2} = d,$$

which is the dimension of $\mathbb{C}[x, y, z]_{(d-1)}$ minus the number of conditions imposed by the vanishing at the points in T. Put $a_{11} = g$ and extend it to a linearly independent family

$$a_{11}, \ldots, a_{1d} \in V.$$

3. Fix j, k with $2 \leqslant j \leqslant k \leqslant d$. The polynomial $\overline{a}_{1j} a_{1k}$ vanishes on S. If S consists of $d(d-1)$ distinct points, the homogeneous vanishing ideal of S is generated[1] by h and $g = a_{11}$. Thus we obtain polynomials $p, q \in \mathbb{C}[x, y, z]$ such that

$$\overline{a}_{1j} a_{1k} = ph + q a_{11}.$$

Since $\overline{a}_{1j} a_{1k}$, h and a_{11} are all homogeneous, we can assume that p and q are also homogeneous, and we find $\deg(q) = d - 1$.

If S contains fewer than $d(d-1)$ points, we have to take multiplicities into account, but the statement remains true (using Max Noether's Theorem, see for example [29]).

Put $a_{jk} = q$. If $j = k$, then a_{11} and $\overline{a}_{1j} a_{1j}$ are both real and we take a_{jj} to be real as well. Finally put $a_{kj} = \overline{a}_{jk}$ for $j > k$.
4. We denote by A_g the $d \times d$-matrix with entries a_{jk}. By construction, we have

$$a_{11} a_{jk} - a_{1k} a_{j1} \in (h).$$

for all $j < k$. Note that A_g depends not only on h and g, but also on the choice of splitting $S = T \cup \overline{T}$ and the choice of basis of V. We will ignore this and denote by A_g any matrix arising in this way.

Now we are ready for the main result of this section.

[1] By the Nullstellensatz, this is equivalent to showing that the ideal generated by g and h is radical. This can be deduced from Bézout's Theorem for curves, or directly from Max Noether's $AF + BG$ Theorem (see [29]).

Theorem 2.62
Let $h, g \in \mathbb{R}[x, y, z]$ be homogeneous, with h irreducible, $\deg(h) = d$ *and* $\deg(g) = d - 1$. *Assume that $\mathcal{Z}_{\mathbb{R}}(h) \cap \mathcal{Z}_{\mathbb{R}}(g) = \emptyset$ and let A_g be as in Construction 2.61. Then the following holds:*

1. *Every entry of the adjugate matrix $(A_g)^{\mathrm{adj}}$ is divisible by h^{d-2}, and the matrix*

$$M_g := \frac{1}{h^{d-2}} \cdot A_g^{\mathrm{adj}}$$

has linear entries. Furthermore, there exists $\gamma \in \mathbb{R}$ such that

$$\det(M_g) = \gamma h.$$

2. *Assume that h is strictly hyperbolic with respect to a point e, and that g strictly interlaces h. Then $\gamma \neq 0$ and the matrix $M_g(e)$ is (positive or negative) definite.*

The proof will make use of the following simple lemma.

Lemma 2.63
Let A be a $d \times d$-matrix with entries in a factorial ring R. If $h \in R$ is irreducible and divides all 2×2-minors of A, then for every $1 \leqslant k \leqslant d$, the element h^{k-1} divides all $k \times k$-minors of A.

Proof We prove the statement by induction on k. By hypothesis, the claim holds for $k \leqslant 2$. So assume $k > 2$ and suppose that h^{k-2} divides all $(k-1) \times (k-1)$-minors of A. Let B be a submatrix of size $k \times k$ of A. From $B^{\mathrm{adj}} B = \det(B) \cdot I_k$ we conclude

$$\det(B^{\mathrm{adj}}) = \det(B)^{k-1}.$$

Suppose $\det(B) = h^m q$ where h does not divide q. Then

$$\det(B)^{k-1} = h^{m(k-1)} q^{k-1}.$$

By assumption h^{k-2} divides all entries of B^{adj}, hence $h^{k(k-2)}$ divides its determinant $\det(B^{\mathrm{adj}})$, which is $\det(B)^{k-1}$. Since h is irreducible, h does not divide q^{k-1}, so $h^{k(k-2)}$ must divide $h^{m(k-1)}$. Then $k(k-2) \leqslant m(k-1)$, which implies that $k - 1 \leqslant m$, as claimed. □

Proof of Theorem 2.62

(1) By construction, the 2×2 minors of A_g of the form $a_{11}a_{jk} - a_{1k}a_{j1}$ are divisible by h. Therefore, if $u \in \mathcal{Z}_{\mathbb{C}}(h)$ is a point with $a_{11}(u) \neq 0$, we can conclude that every row of $A_g(u)$ is a multiple of the first, so that $A_g(u)$ has rank 1. Since a_{11} is not divisible by h, it follows that $a_{11}(u) \neq 0$ holds on a Zariski-dense subset of $\mathcal{Z}_{\mathbb{C}}(h)$. So all the 2×2 minors of A_g are divisible by h. Since h is irreducible in $\mathbb{C}[x, y, z]$, this implies that all $(d-1) \times (d-1)$-minors of A_g are divisible by h^{d-2}, by Lemma 2.63.

The entries of A_g^{adj} have degree $(d-1)^2$, and h has degree d, so that

$$M_g = \frac{1}{h^{d-2}} \cdot A_g^{\text{adj}}$$

has entries of degree

$$(d-1)^2 - d(d-2) = 1.$$

Furthermore, by Lemma 2.63, $\det(A_g)$ is divisible by h^{d-1}. So $\det(A_g) = qh^{d-1}$ for some $q \in \mathbb{R}[x, y, z]$, and we obtain

$$\begin{aligned}\det(M_g) &= \det(h^{2-d} A_g^{\text{adj}}) = h^{d(2-d)} \det(A_g^{\text{adj}}) = h^{d(2-d)} \det(A_g)^{d-1} \\ &= h^{d(2-d)} q^{d-1} h^{(d-1)^2} = q^{d-1} h.\end{aligned}$$

Since $\det(M_g)$ has degree d, we see that q is a constant and we take $\gamma = q^{d-1}$.

(2) To show that $\det(M_g)$ is not the zero-polynomial, we begin by showing that

$$\lambda^* A_g \lambda$$

is not the zero-polynomial, for any $\lambda \in \mathbb{C}^d \setminus \{0\}$. As argued in the proof of (1), the matrix A_g has rank one at all points of $\mathcal{V}_{\mathbb{C}}(h)$, and for every $\lambda \in \mathbb{C}^d \setminus \{0\}$ we have by construction

$$a_{11} \cdot (\lambda^* A_g \lambda) - (\lambda^* A_g e_1)(\overline{\lambda^* A_g e_1}) \in (h). \tag{2.3}$$

If $\lambda^* A_g \lambda$ is identically zero, we conclude that $\lambda^* A_g e_1$ vanishes on $\mathcal{V}_{\mathbb{C}}(h)$. Since h has degree d and $\lambda^* A_g e_1$ has degree $d-1$, $\lambda^* A_g e_1$ must then vanish identically as well (again using Hilbert's Nullstellensatz). This contradicts the linear independence of the polynomials $\overline{a}_{11}, \ldots, \overline{a}_{1d}$.

Now suppose that the claim is false and that $\det(M_g)$ is identically zero. From the proof of (1) it is clear that $\det(A_g)$ is then zero as well. In particular, the determinant of $A_g(e)$ is zero, so there is some nonzero vector $\lambda \in \mathbb{C}^d \setminus \{0\}$ in its kernel, and $\lambda^* A_g(e)\lambda$ is thus also zero. But we have just shown that the polynomial $\lambda^* A_g \lambda$ is nonzero, and Eq. (2.3) shows that the product $a_{11} \cdot (\lambda^* A_g \lambda)$

is nonnegative on $\mathcal{V}_{\mathbb{R}}(h)$. By Lemma 2.53, $\lambda^* A_g \lambda$ therefore interlaces h. Thus this polynomial cannot vanish at the point e, and so the determinant $\det(M_g)$ is indeed not identically zero.

What remains to show is that $M_g(e)$ is definite. To do this, we show that A_g is the adjugate matrix of M_g. By construction, $M_g = h^{2-d} \cdot A_g^{\mathrm{adj}}$. Taking adjugates, we see that

$$M_g^{\mathrm{adj}} = \frac{1}{h^{(d-2)(d-1)}} \cdot (A_g^{\mathrm{adj}})^{\mathrm{adj}} = \frac{1}{h^{(d-2)(d-1)}} \cdot \det(A_g)^{d-2} \cdot A_g = q^{d-2} A_g,$$

where $\det(A_g) = q h^{d-1}$ as in the proof of (1) above. So a_{11} is a constant multiple of $e_1^* M_g^{\mathrm{adj}} e_1$, and thus belongs to $\mathcal{H}(M_g)$. Since a_{11} interlaces h with respect to e, Theorem 2.58 implies that the matrix $M_g(e)$ is definite. □

Corollary 2.64
Every hyperbolic polynomial in three variables possesses a definite Hermitian determinantal representation.

Proof Suppose $h \in \mathbb{R}[x, y, z]$ is irreducible and strictly hyperbolic with respect to e. Then the directional derivative $\partial_e h$ strictly interlaces h and can be used an input in Construction 2.61. By Theorem 2.62 (2), this will result in a definite Hermitian determinantal representation of h. If h is strictly hyperbolic but not irreducible, then each irreducible factor of h is strictly hyperbolic and we can build a Hermitian determinantal representation of h as a block matrix from the representations of all factors.

In general, if $h \in \mathbb{R}[x, y, z]_{(d)}$ is (not necessarily strictly) hyperbolic with respect to e and $h(e) = 1$, by Exercise 2.50 (2) there exists a sequence of strictly hyperbolic polynomials

$$h_k \in \mathbb{R}[x, y, z]_{(d)}$$

(with respect to e) and $h_k(e) = 1$, converging to h. Now each h_k has a Hermitian determinantal representation, hence so does h by Lemma 2.60. □

Corollary 2.65
Every three-dimensional hyperbolicity cone is spectrahedral.

Proof This follows at once from Corollary 2.64 and Lemma 2.9. □

Let us now use the existence of determinantal representations in dimension three to prove some already announced geometric properties of general hyperbolicity cones. These results were first proven in [30], with a more elementary but also more technical proof.

Corollary 2.66
Every hyperbolicity cone is a convex cone.

Proof Let $h \in \mathbb{R}[x]_{(d)}$ be hyperbolic with respect to e. For $u \in \overline{\mathcal{C}}_e(h)$ and $\alpha > 0$, the roots of $h(\alpha u + te)$ are those of $h(u + \alpha^{-1}te)$, which are just the roots of $h(u - te)$ multiplied with α. So it is clear that $\alpha u \in \overline{\mathcal{C}}_e(h)$.

Given two points $u, v \in \overline{\mathcal{C}}_e(h)$, let V be the three-dimensional subspace spanned by e, u, v. Then $\overline{\mathcal{C}}_e(h) \cap V$ is the hyperbolicity cone of $h|_V$, which is spectrahedral by Corollary 2.65. In particular it is a convex cone, and therefore contains $u + v$. □

Corollary 2.67
If h is hyperbolic with respect to e, then it is hyperbolic with respect to any interior point of $\overline{\mathcal{C}}_e(h)$.

Proof Using arguments as in Corollary 2.16, this is easy to see if h has a definite Hermitian determinantal representation, and by Corollary 2.65 it therefore holds for three-dimensional hyperbolicity cones. The general case can be reduced to the three-dimensional one as in the proof of Corollary 2.66 above. □

Finally, let us see an example. Carrying out Construction 2.61 in practice is not an easy matter. The following example is taken directly from [81, Example 4.11].

Example 2.68 We apply Construction 2.61 to the quartic

$$h = x^4 - 4x^2y^2 + y^4 - 4x^2z^2 - 2y^2z^2 + z^4, \tag{2.4}$$

which is hyperbolic with respect to the point $e = (1 : 0 : 0)$. This curve has two nodes, $(0 : 1 : 1)$ and $(0 : -1 : 1)$, so that h is not strictly hyperbolic. But the construction will still work, and this happens to simplify the explicit computations considerably. Figure 2.9 shows the real curve in the plane $\{x = 1\}$.

We define a_{11} to be the directional derivative $\frac{1}{4}\partial_e h = x^3 - 2xy^2 - 2xz^2$. The curves $\mathcal{V}_{\mathbb{C}}(h)$ and $\mathcal{V}_{\mathbb{C}}(a_{11})$ intersect in the eight points $(2 : \pm\sqrt{3} : \pm i)$, $(2 : \pm i : \pm\sqrt{3})$ and the two nodes, $(0 : \pm 1 : 1)$, each with multiplicity 2, for a total of $4 \cdot 3 = 12$ intersection points, counted with multiplicities. We divide these points into two conjugate sets (making

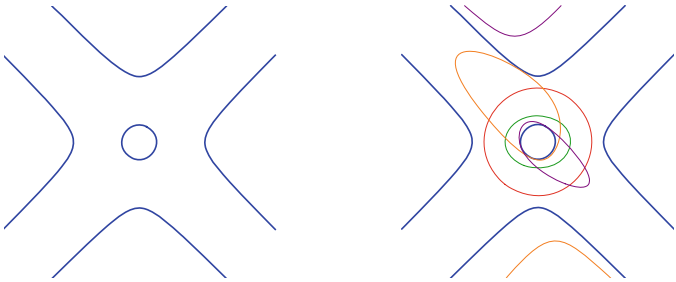


Fig. 2.9 The hyperbolic quartic (2.4) and interlacing cubics

an arbitrary choice) and decompose $S = \mathcal{V}_{\mathbb{C}}(h) \cap \mathcal{V}_{\mathbb{C}}(a_{11})$ into $S = T \cup \overline{T}$ where

$$T = \big\{(0{:}1{:}1),\ (0{:}\,-1:1),\ (2{:}\sqrt{3}{:}i),\ (2{:}\,-\sqrt{3}{:}i),\ (2{:}i{:}\sqrt{3}),\ (2{:}i{:}\,-\sqrt{3})\big\}.$$

The vector space of cubics in $\mathbb{C}[x, y, z]$ vanishing on these six points is four dimensional and we extend a_{11} to a basis $a_{11}, a_{12}, a_{13}, a_{14}$ for this space, where

$$\begin{aligned} a_{12} &= ix^3 + 4ixy^2 - 4x^2z - 4y^2z + 4z^3, \\ a_{13} &= -3ix^3 + 4x^2y + 4ixy^2 - 4y^3 + 4yz^2, \\ a_{14} &= -x^3 - 2ix^2y - 2ix^2z + 4xyz. \end{aligned}$$

Then, to find a_{22} for example, we write $a_{12} \cdot \overline{a}_{12}$ as an element of the ideal (h, a_{11}),

$$a_{12} \cdot \overline{a}_{12} \;=\; (13x^3 - 14xy^2 - 22xz^2) \cdot a_{11} + (16z^2 - 12x^2) \cdot h,$$

and set $a_{22} = 13x^3 - 14xy^2 - 22xz^2$. We proceed similarly for the remaining entries and eventually obtain the output of Construction 2.61, the Hermitian matrix of cubics

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a}_{12} & a_{22} & a_{23} & a_{24} \\ \overline{a}_{13} & \overline{a}_{23} & a_{33} & a_{34} \\ \overline{a}_{14} & \overline{a}_{24} & \overline{a}_{34} & a_{44} \end{bmatrix}.$$

By taking the adjugate of A and dividing by h^2, we find the desired Hermitian determinantal representation,

$$M = \frac{1}{h^2} \cdot A^{\text{adj}} = 2^5 \begin{bmatrix} 14x & 2z & 2ix-2y & 2i(y-z) \\ 2z & x & 0 & -ix+2y \\ -2ix-2y & 0 & x & ix-2z \\ -2i(y-z) & ix+2y & -ix-2z & 4x \end{bmatrix}.$$

The determinant of M is $2^{24} \cdot h$. As promised by Theorems 2.58 and 2.62, the cubics in $\mathcal{H}(M)$ interlace h (see Fig. 2.9), and the matrix M is positive definite at the point $(1, 0, 0)$.

2.6 Hyperbolic Polynomials from Graphs

In this section we will first explain an interesting method to produce hyperbolic polynomials with definite determinantal representations, using the *Matrix-Tree Theorem* by Kirchhoff [51], initially presented already in 1847. Is has been used in the context of hyperbolic polynomials for example in [15, 16]. We will then present Brändén's result [15] on hyperbolicity cones of elementary symmetric polynomials.

Let $G = (V, E)$ be a finite undirected multigraph without loops. Formally, this means that E and V are finite sets, and a map

$$\varphi\colon E \to \binom{V}{2}$$

into the 2-element subsets of V is also given (specifying the two distinct endpoints of each edge). Informally, it means that the finitely many vertices are connected by edges, each edge connecting two different vertices, but there might be (finitely many) multiple edges between any two vertices. The informal approach is sufficient here, and we thus suppress φ from the notation. We will also just say "multigraph" instead of "finite undirected multigraph without loops" from now on.

Definition 2.69

Let $G = (V, E)$ be a multigraph.

1. A subset $T \subseteq E$ is called a *spanning tree* of G if it contains each vertex of G and forms a tree (i.e., a connected subgraph of G without circles).
2. The polynomial

$$h_G := \sum_{\substack{T \subseteq E \\ \text{spanning tree}}} \prod_{e \in T} x_e$$

is called the *spanning tree polynomial* of G. It is a homogeneous polynomial of degree $|V| - 1$ in the variables x_e for $e \in E$.

3. Let $e \in E$ connect the two distinct vertices $v, w \in V$. We denote by

$$J_e := (\delta_v - \delta_w)(\delta_v - \delta_w)^T \in \mathrm{Sym}_{|V|}(\mathbb{R})$$

the matrix (indexed by the vertices of G) with 1 in the (v, v) and (w, w) position, -1 in the (v, w) and (w, v) position, and zeros elsewhere (here, $\delta_v \in \mathbb{R}^{|V|}$ denotes the standard basis vector corresponding to v). We then define the *weighted Laplacian* of G as the homogeneous linear matrix polynomial

$$L_G := \sum_{e \in E} x_e J_e$$

in the variables x_e for $e \in E$. By L_G^v we denote L_G with the row and column indexed by v deleted.

Theorem 2.70 (Matrix-Tree Theorem)
Let G be a connected multigraph. Then h_G is hyperbolic with respect to $e = (1, \ldots, 1)$. For each $v \in V$ we have $h_G = \det\left(L_G^v\right)$, and

$$\overline{C}_e(h_G) = \mathcal{S}\left(L_G^v\right)$$

is a spectrahedron.

Proof In a first step assume that G is a tree. We prove $h_G = \det(L_G^v)$ by induction on the number of vertices of G. For one or two vertices, the statement is obvious. For the induction step, let w be a leaf different from v in G, and denote by G' the tree obtained by deleting w (and its adjacent edge e'). If $\det(L_G^v)$ is computed by Laplace expansion along the row indexed by w, a short calculation shows

$$\det(L_G^v) = x_{e'} \det(L_{G'}^v) = x_{e'} h_{G'} = h_G,$$

where we have used the induction hypothesis for G' for the second equality.

Now assume that G is a general multigraph. Choose an arbitrary orientation of the edges of G, and let I be the weighted incidence matrix of G, by which we mean the $|V| \times |E|$-matrix whose (v, e)-entry is $-\sqrt{x_e}$ if v is the source of e, $\sqrt{x_e}$ if v is the target of e, and 0 else. Then $I \cdot I^T = L_G$ is clear from the construction, and thus

$$L_G^v = I^v \cdot (I^v)^T$$

where I^v is I with the row indexed by v deleted. To compute the determinant we use the Cauchy-Binet Formula (see for example [100]), and obtain

$$\det(L_G^v) = \det\left(I^v (I^v)^T\right) = \sum_{\substack{S \subseteq E \\ |S| = |V|-1}} \det(I_S^v)^2.$$

Here, I_S^v is I^v with only the columns indexed in S remaining.

Now if a subset $S \subseteq E$ with $|S| = |V| - 1$ is not a tree, it defines a subgraph on V that is not connected. So it defines a connected component on some subset of vertices $V' \subseteq V \setminus \{v\}$. Thus the sum of the rows of I_S^v indexed by V' is 0, which implies $\det(I_S^v) = 0$. Conversely, if $\tilde{G} = (V, S)$ is a tree, then

$$\det(I_S^v)^2 = \det\left(I_S^v (I_S^v)^T\right) = \det\left(L_{\tilde{G}}^v\right) = h_{\tilde{G}} = \prod_{e \in S} x_e,$$

where we have used the already proven statement for trees for the third equality. Altogether, this proves $h_G = \det(L_G^v)$.

Since every J_e is positive semidefinite, so is J_e^v, and thus

$$L_G^v(1, \ldots, 1) = \sum_{e \in E} J_e^v \succcurlyeq 0.$$

Furthermore,

$$\det\left(L_G^v(1, \ldots, 1)\right) = h_G(1, \ldots, 1)$$

equals the number of spanning trees of G, which is a positive number, since G is connected. Thus L_G^v provides a definite determinantal representation of h_G, and the remaining claims follow from Proposition 2.37. □

Example 2.71

1. Let G be a tree with $n + 1$ edges. Then

$$h_G = x_0 x_1 \cdots x_n = s_{n+1}(x_0, \ldots, x_n)$$

is the elementary symmetric polynomial of degree $n + 1$. The hyperbolicity cone with respect to $e = (1, \ldots, 1)$ is the positive orthant.

L_G^v is the diagonal determinantal representation of h_G when taking a tree with one root and $n + 1$ leaves, and choosing v as the root.

2. If G has only 2 vertices but $n+1$ edges between them, we obtain the elementary symmetric polynomial of degree 1:

$$h_G = x_0 + \cdots + x_n = s_1(x_0, \ldots, x_n).$$

The hyperbolicity cone is a halfspace.

3. Let G be a circle of length $n+1$. Then

$$h_G = \sum_{i=0}^{n} \prod_{j \neq i} x_j = s_n(x_0, \ldots, x_n)$$

is the elementary symmetric polynomial of degree n. The corresponding matrix polynomial is

$$L_G^v = \begin{bmatrix} x_0+x_n & -x_0 & 0 & & \\ -x_0 & x_0+x_1 & -x_1 & \ddots & \\ 0 & -x_1 & x_1+x_2 & \ddots & 0 \\ & \ddots & \ddots & \ddots & -x_{n-2} \\ & & 0 & -x_{n-2} & x_{n-2}+x_{n-1} \end{bmatrix}.$$

The fact that the hyperbolicity cone of this polynomial is a spectrahedron was first proven in [91].

4. Let G be the complete simple graph on s vertices. For each $1 \leqslant i < j \leqslant s$ we have an edge $x_{i,j}$ and get

$$L_G^s = \begin{bmatrix} x_{1,2}+x_{1,3}+\cdots+x_{1,s} & -x_{1,2} & \cdots & -x_{1,s-1} \\ -x_{1,2} & x_{1,2}+x_{2,3}+\cdots+x_{2,s} & \cdots & -x_{2,s-1} \\ \vdots & \vdots & \ddots & \vdots \\ -x_{1,s-1} & -x_{2,s-1} & \cdots & x_{1,s-1}+\cdots+x_{s-1,s} \end{bmatrix}.$$

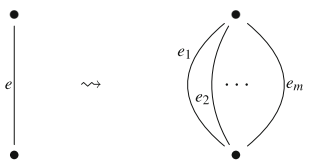
After changing coordinates to

$$y_{i,i} := \sum_{j \neq i} x_{i,j} \quad \text{and} \quad y_{i,j} := -x_{i,j} \ (\text{for } i < j)$$

this becomes $\left(y_{i,j}\right)_{i,j=1}^{n}$. So the hyperbolicity cone of h_G is linearly isomorphic to Sym_s^+.

The following construction is crucial in the proof of Theorem 2.77 below.

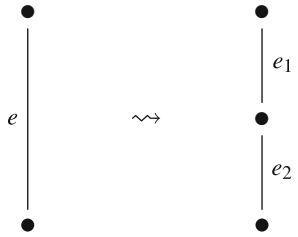
▶ **Remark 2.72** If an edge e in a graph is replaced by m parallel edges $e_1, \ldots, e_m$



the resulting spanning tree polynomial is obtained from the old one by the replacement

$$x_e \rightsquigarrow x_{e_1} + \cdots + x_{e_n}.$$

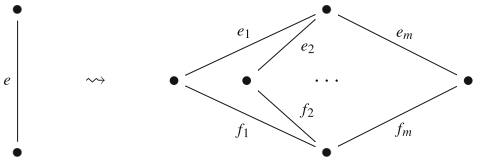
If an edge e is divided into two edges e_1, e_2 by putting a new vertex in the middle



then the resulting spanning tree polynomial is obtained by multiplying the old polynomial with $x_{e_1} + x_{e_2}$ and the replacement

$$x_e \rightsquigarrow \frac{x_{e_1} x_{e_2}}{x_{e_1} + x_{e_2}}.$$

The degree goes up by one with this construction.
Combining these, if a single edge e is replaced by a diamond



then the resulting spanning tree polynomial is obtained by multiplying with $\prod_{i=1}^{m}(x_{e_i} + x_{f_i})$ and the replacement

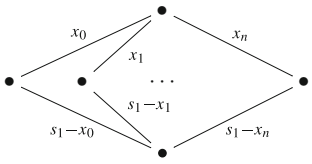
$$x_e \rightsquigarrow \sum_{i=1}^{m} \frac{x_{e_i} x_{f_i}}{x_{e_i} + x_{f_i}}.$$

Exercise 2.73
Let $n \geqslant 2$ be fixed. Show the following:

1. The elementary symmetric polynomial $s_2(x_0, \ldots, x_n)$ of degree 2 is not the spanning tree polynomial of a multigraph.

2. The spanning tree polynomial of the below graph, with edge-variables substituted as indicated in the picture, is

$$2s_1(x_0,\ldots,x_n)^n s_2(x_0,\ldots,x_n):$$



From now on we consider $n \geqslant 0$ fixed, and often suppress it from the notation.

Definition 2.74

For $1 \leqslant k \leqslant n+1$ the *elementary symmetric polynomial of degree k* is

$$s_k := s_k(x_0,\ldots,x_n) := \sum_{0\leqslant i_1<\cdots<i_k\leqslant n} x_{i_1}\cdots x_{i_k}.$$

▶ **Remark 2.75** All elementary symmetric polynomials are hyperbolic in direction $e = (1,\ldots,1)$. This is clearly true for $s_{n+1} = x_0\cdots x_{n+1}$, and the other polynomials are (iterated) directional derivatives of s_{n+1} in direction e. The claim thus follows from Rolle's Theorem, and we also have

$$\mathbb{R}^{n+1}_+ = \overline{\mathcal{C}}_e(s_{n+1}) \subseteq \overline{\mathcal{C}}_e(s_n) \subseteq \cdots \subseteq \overline{\mathcal{C}}_e(s_1),$$

where the largest cone is just a halfspace (see Exercise 2.49) (Fig. 2.10).

We now explain the construction from [15].

Construction 2.76 For $0 \leqslant k \leqslant n$ we construct the *Brändén/Batman graph* B_k, whose spanning tree polynomial will relate to s_{k+1} (see Fig. 2.11 for an example).

First we choose a root α, from wich $n+1$ edges emerge, leading to vertices labeled $0,\ldots,n$. The edge leading to vertex i is labeled $k!x_i$. From each of these

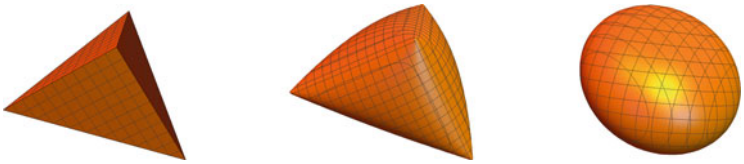


Fig. 2.10 Compact cross-sections of $\overline{\mathcal{C}}_e(s_k)$ for $k = 4, 3$ and 2, with $n = 3$

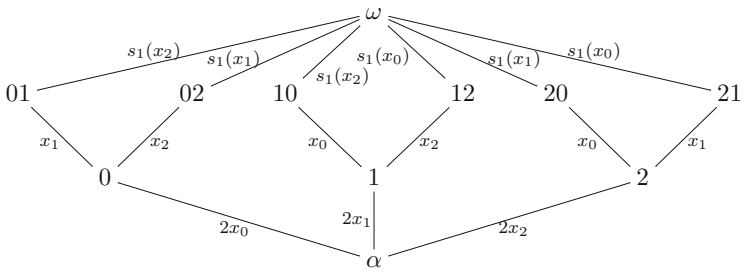


Fig. 2.11 The Brändén/Batman graph B_2 in case of $n = 2$

vertices i, another n edges emerge, leadings to vertices labeled ij with $i \neq j = 0, \ldots, n$. The edge leading from vertex i to vertex ij is labeled $(k-1)!x_j$.

We continue in this way until we have constructed a tree of height k. The vertices at height d (seen from the root α) are labeled with the sequences of length d of numbers between 0 and n, in which no number appears twice. The edge leading to such a vertex (from height $d-1$) is labeled $(k-d+1)!x_j$, when j is the last entry of the vertex label.

Finally, the $(n+1)!/(n+1-k)!$ leaves of the tree are all connected by edges to a final vertex ω. Such an edge is labeled by the first elementary symmetric polynomial s_1, but in only those variables whose indices do *not* appear in the label of the corresponding leaf.

We are now ready to state and prove the main result from [15].

Theorem 2.77
For $e = (1, \ldots, 1)$ and $k = 0, \ldots, n$ we have

$$\overline{C}_e(s_{k+1}) = \overline{C}_e(h_{B_k}),$$

and this cone is thus a spectrahedron.

Proof For $1 \leqslant d \leqslant n+1$ we will use the symmetric rational function

$$q_d := \frac{s_d}{s_{d-1}} \in \mathbb{R}(x_0, \ldots, x_n).$$

For a finite sequence/set v of numbers from $\{0, \ldots, n\}$ we let s_d^v and q_d^v be the functions obtained from s_d and q_d by setting all variables indexed in v to zero, so that these are the corresponding symmetric functions in only the remaining variables.

To compute the spanning tree polynomial of B_k recursively, we define the graph $B_{k,r}$ for $r \geqslant k$ almost as B_k, but with a slightly different labeling of edges. The edges leading from height $d-1$ to d are labeled $(r-d+1)!x_j$, instead of $(k-d+1)!x_j$ as in B_k, and the final edge from a k-sequence v to ω is labeled

$$(r-k+1)!q_{r-k+1}^{v}.$$

Note that clearly $B_{k,k} = B_k$ holds. Now note that $B_{k,r}$ is constructed from $B_{k-1,r}$ by replacing each of the final edges (leading to ω) by a certain edge-labeled diamond, just as in Remark 2.72. On the level of spanning tree polynomials, we thus obtain the one of $B_{k,r}$ from the one of $B_{k-1,r}$ by multiplying with

$$\prod_{j \notin v} (r-k+1)! \left(x_j + q_{r-k+1}^{vj} \right) \tag{2.5}$$

for each vertex v of $B_{k-1,r}$ of height $k-1$, and use instead of $(r-(k-1)+1)!q_{r-(k-1)+1}^{v}$ the rational function

$$(r-k+1)! \sum_{j \notin v} \frac{x_j q_{r-k+1}^{vj}}{x_j + q_{r-k+1}^{vj}}.$$

This last expression is easily seen to be $(r-(k-1)+1)!q_{r-(k-1)+1}^{v}$ (see Exercise 2.78 (3)), so none of these replacements is actually necessary! Thus the spanning tree polynomial of $B_{k,r}$ arises from the one of $B_{k-1,r}$ by just multiplying with all of the factors from (2.5). But up to a positive number, the product in (2.5) equals

$$\prod_{j \notin v} \frac{s_{r-k+1}^{v}}{s_{r-k}^{vj}} \tag{2.6}$$

using Exercise 2.78 (2). The spanning tree polynomial of $B_{0,k}$ is a positive multiple of $q_{k+1} = s_{k+1}/s_k$, and we obtain the spanning tree polynomial h_{B_k} of B_k by multiplying with all the described terms arising in the iterated construction of $B_{1,k}, B_{2,k} \ldots, B_{k,k} = B_k$. In total, we get a product of s_{k+1} and certain $s_{k-|v|}^{v}$ for $v \subseteq \{0, \ldots, n\}$. Indeed, all the denominators from (2.6) cancel with numerators in the following step, in total just some of the numerators survive. Now by Exercise 2.79 we have

$$\overline{\mathcal{C}}_e\,(s_{k+1}) \subseteq \overline{\mathcal{C}}_e \left(s_{k-|v|}^{v} \right)$$

and this clearly implies $\overline{\mathcal{C}}_e(h_{B_k}) = \overline{\mathcal{C}}_e(s_{k+1})$. By Theorem 2.70, this is a spectrahedron. □

Exercise 2.78
Show that for the elementary symmetric polynomials $s_k \in \mathbb{R}[x_0, \dots x_n]$ and the symmetric rational functions $q_k = s_k/s_{k-1} \in \mathbb{R}(x_0, \dots, x_n)$ from the last proof, we have the following recursion formulas:

1. $ks_k = \sum_{j=0}^{n} x_j s_{k-1}^{\{j\}}$.
2. $s_k = s_k^{\{j\}} + x_j s_{k-1}^{\{j\}}$.
3. $\sum_{j=0}^{n} \frac{x_j q_{k-1}^{\{j\}}}{x_j + q_{k-1}^{\{j\}}} = kq_k$.

Exercise 2.79
Show that

$$\overline{\mathcal{C}}_e\left(s_{k+1}\right) \subseteq \overline{\mathcal{C}}_e\left(s_{k-|v|}^v\right)$$

holds for all $1 \leqslant k \leqslant n$ and $v \subseteq \{0, \dots, n\}$ with $|v| \leqslant k$.
Hint: $s_{k-|v|}^v$ is an iterated directional derivative of s_{k+1}. If the derivative direction is only in the boundary of the corresponding hyperbolicity cone, one might perturb it a little bit.

2.7 Derivative Cones

In this section we collect some results on derivative cones of hyperbolicity cones. For a polynomial h which is hyperbolic in direction e, we denote by $\partial_e h$ its derivative in direction e, and call

$$\overline{\mathcal{C}}_e(\partial_e h)$$

the *derivative cone* of $\overline{\mathcal{C}}_e(h)$ (although strictly speaking this is a slight abuse of notation, since the derivative cone depends on h and not only on the cone $\overline{\mathcal{C}}_e(h)$). Of course this process can be repeated and gives rise to the higher order derivative cones

$$\overline{\mathcal{C}}_e(\partial_e^k h).$$

In Exercise 2.49 it is shown that the derivative cone always contains the original cone, so any such cone admits an increasing sequence of higher order derivative cones.

A straightforward consequence of Theorem 2.77 is that all higher order derivative cones of polyhedral cones are spectrahedral:

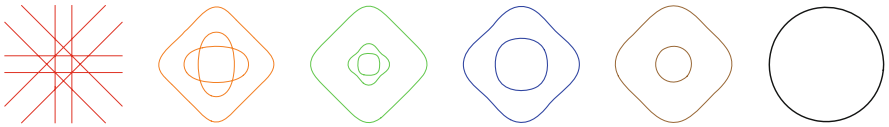


Fig. 2.12 Real zero set of a product of 8 linear forms, and some higher derivatives; affine sections are shown

Corollary 2.80
Let $h = \ell_0 \cdots \ell_d \in \mathbb{R}[x_0, \ldots, x_n]$ be a product of linear forms. Then for any $e \in \mathbb{R}^{n+1}$ with $h(e) \neq 0$ and all $0 \leqslant k \leqslant d$ the derivative cone

$$\overline{\mathcal{C}}_e\left(\partial_e^k h\right)$$

is a spectrahedron.

Proof By Exercise 2.81 we have

$$\partial_e^k h = k! h(e) s_{d-k+1}(\ell_0, \ldots, \ell_d),$$

so the statement is clear from Theorem 2.77, by substituting the ℓ_i for the variables (Fig. 2.12). □

Exercise 2.81
Show that if $h = \ell_0 \cdots \ell_d$ is a product of linear forms, then for any e with $h(e) \neq 0$ and all $0 \leqslant k \leqslant d$ we have $\partial_e^k h = k! h(e) s_{d-k+1}(\ell_0, \ldots, \ell_d)$.

The following result from [92] is a generalization of Corollary 2.80 to arbitrary spectrahedral cones, but only for the first derivative:

Theorem 2.82
Assume the hyperbolic polynomial h admits a definite determinantal representation (w.r.t. direction e). Then $\overline{\mathcal{C}}_e(\partial_e h)$ is a spectrahedron.

Proof It is clearly enough to prove the statement for $h = \det(X)$, where $X = (x_{ij})_{i,j}$ is a symmetric $s \times s$-matrix with variable entries, and $e = I_s$, in which case we have $\overline{\mathcal{C}}_e(h) = \mathrm{Sym}_s^+$. The general case will then easily follow by a linear substitution. We will now show that

$$M \in \overline{\mathcal{C}}_e(\partial_e h) \quad \Leftrightarrow \quad \mathrm{Tr}(YMY) \geqslant 0 \text{ for all } Y \in \mathrm{Sym}_s \text{ with } \mathrm{Tr}(Y) = 0. \tag{2.7}$$

After choosing an arbitrary basis $B_1, \ldots, B_r$ of the space of all symmetric matrices with vanishing trace, the condition on the right in (2.7) is clearly equivalent to

$$\big(\mathrm{Tr}(B_i M B_j)\big)_{i,j} \geqslant 0, \tag{2.8}$$

which will prove that $\overline{\mathcal{C}}_e(\partial_e h)$ is indeed a spectrahedron.

Since the conditions on both sides of (2.7) are invariant under orthogonal conjugations, we can restrict to diagonal matrices $M = \mathrm{diag}(\lambda_1, \ldots, \lambda_s)$ for proving the equivalence. The condition on the left then just means that $\lambda = (\lambda_1, \ldots, \lambda_s)$ belongs to the hyperbolicity cone of the elementary symmetric polynomial of degree $s-1$ (cp. Sect. 2.6). To examine the condition on the right in (2.7) we choose an explicit basis of the space of symmetric matrices with vanishing trace, and look at the matrix polynomial from (2.8). We denote by E_{ij} the matrix with a 1 in the (i, j)-entry and zeros elsewhere. Then the matrices

$$\begin{aligned} S_{ij} &:= E_{ij} + E_{ji} \text{ for } 1 \leqslant i < j \leqslant s \\ D_i &:= E_{ii} - E_{i+1,i+1} \text{ for } 1 \leqslant i < s \end{aligned}$$

form a basis as desired. For $M = \mathrm{diag}(\lambda)$ a straightforward computation shows

$$\mathrm{Tr}(S_{ij} M S_{rt}) = \begin{cases} \lambda_i + \lambda_j : i = r,\, j = t \\ 0 : \text{else} \end{cases}$$

$$\mathrm{Tr}(D_r M S_{ij}) = 0$$

$$\mathrm{Tr}(D_r M D_t) = \begin{cases} \lambda_r + \lambda_{r+1} : r = t \\ -\lambda_t : t = r+1 \\ -\lambda_r : r = t+1 \\ 0 : \text{else.} \end{cases}$$

So for this particular basis, the matrix polynomial from (2.8) has a block-diagonal form, and is thus positive semidefinite if and only if the block indexed by the S_{ij} alone, and the block indexed by the D_r alone, is positive semidefinite. The second block is precisely of the form of Example 2.71 (3), and thus exactly states that λ must be in the hyperbolicity cone of the elementary symmetric polynomial of degree $s-1$, i.e., $M \in \overline{\mathcal{C}}_e(\partial_e h)$. The first block is even diagonal and yields the conditions $\lambda_i + \lambda_j \geqslant 0$ for all $i < j$. But already from the diagonal of the second block we see that $\lambda_i + \lambda_{i+1} \geqslant 0$ holds for all $1 \leqslant i < s$, whenever λ is in the hyperbolicity cone of the elementary symmetric polynomial of degree $s-1$. Since this hyperbolicity cone is invariant under coordinate permutations, this shows that the first block is automatically positive semidefinite, if the second is. This finishes the proof. □

In [54], it has been shown that Theorem 2.82 *does* hold for the higher derivative cones $\overline{C}_e(\partial_e^k h)$, as well. The proof uses Theorem 2.77 and the representation theory of the symmetric group.

2.8 Free Spectrahedra

Some interesting facts about spectrahedra and their defining matrix polynomials can only be learned when looking at their free (that means non-commutative) extensions. They result by evaluating linear matrix polynomials at tuples of Hermitian matrices, instead of just tuples of real numbers. Since matrices do not commute in general (i.e., are free of the burden of commutativity), this accounts for the name *free* spectrahedra. In this section we will introduce the most important concepts and explain some results. We will restrict to convex cones throughout, but it must be pointed out that, while any nonzero number can be rescaled to 1, not every nonzero Hermitian matrix can be rescaled to the identity matrix. This makes passing between homogeneous and non-homogeneous settings a little more difficult. In this section we will also mostly use Hermitian matrices, since the closely related theory of operator systems always works in the complex setting.

Throughout this section, let $M = x_0 M_0 + \cdots + x_n M_n$ be a homogeneous and Hermitian linear matrix polynomial of size s, with

$$M(e) = e_0 M_0 + \cdots + e_n M_n = I_s$$

for some $e \in \mathbb{R}^{n+1}$. Unless otherwise specified, we will also assume that $M_0, \ldots, M_n$ are linearly independent (we have seen in Exercise 2.10 that this is not much of a restriction).

Definition 2.83

For any $k \geqslant 1$, we define

$$\mathcal{S}_k(M) := \left\{ (A_0, \ldots, A_n) \in \mathrm{Her}_k^{n+1} \mid A_0 \otimes M_0 + \cdots + A_n \otimes M_n \succcurlyeq 0 \right\}.$$

Here, Her_k denotes the space of complex Hermitian matrices of size k, and $\otimes$ denotes the Kronecker/tensor product of matrices. So

$$A_0 \otimes M_0 + \cdots + A_n \otimes M_n \in \mathrm{Her}_k \otimes \mathrm{Her}_s \cong \mathrm{Her}_{ks}$$

and this canonical identification is used to define positivity. The (disjoint) union

$$\mathcal{S}_\infty(M) := \bigcup_{k=1}^{\infty} \mathcal{S}_k(M)$$

is called the *free spectrahedron* defined by M, and $\mathcal{S}_k(M)$ is referred to as the *k-th level* of $\mathcal{S}_\infty(M)$.

▶ **Remark 2.84**

1. The first level $\mathcal{S}_1(M)$ coincides with the spectrahedron $\mathcal{S}(M)$ as considered throughout this book so far. This is because $\mathrm{Her}_1 = \mathbb{R}$, and the tensor product is just scalar multiplication in this case.
2. We have defined $\mathcal{S}_\infty(M)$ as a disjoint union, so there is no embedding of matrix spaces involved. Perhaps it would be a cleaner approach to define $\mathcal{S}_\infty(M)$ as the *sequence* of all $\mathcal{S}_k(M)$, but this makes later notation more opaque.
3. One could also add an infinite level by choosing some infinite-dimensional (separable) Hilbert space $\mathcal{H}$ and define

$$\mathcal{S}_\mathcal{H}(M) := \left\{(A_0, \ldots, A_n) \in \mathcal{B}(\mathcal{H})_{\mathrm{sa}}^{n+1} \mid A_0 \otimes M_0 + \cdots + A_n \otimes M_n \succcurlyeq 0\right\},$$

i.e., use bounded self-adjoint operators instead of Hermitian matrices, and work in

$$\mathcal{B}(\mathcal{H})_{\mathrm{sa}} \otimes \mathrm{Her}_s \cong \mathcal{B}(\mathcal{H}^s)_{\mathrm{sa}}.$$

In the general context of free semialgebraic geometry this often makes sense, but for free spectrahedra it does not add anything relevant.

Some important but easy to prove properties of free spectrahedra are collected in the following lemma.

Lemma 2.85
Let M be a matrix polynomial as specified at the beginning of this section.

1. *Each $\mathcal{S}_k(M)$ is a closed, pointed and full-dimensional convex cone in the finite-dimensional real vector space $\mathrm{Her}_k^{n+1} \cong \mathbb{R}^{k^2(n+1)}$. It is also spectrahedral in the classical sense in this space.*
2. *For $A = (A_0, \ldots, A_n) \in \mathcal{S}_k(M)$, $B = (B_0, \ldots, B_n) \in \mathcal{S}_l(M)$ we have*

$$A \oplus B := \left(\begin{bmatrix} A_0 & 0 \\ 0 & B_0 \end{bmatrix}, \ldots, \begin{bmatrix} A_n & 0 \\ 0 & B_n \end{bmatrix}\right) \in \mathcal{S}_{k+l}(M).$$

3. *For $A = (A_0, \ldots, A_n) \in \mathcal{S}_k(M)$ and $V \in \mathrm{Mat}_{k\times l}(\mathbb{C})$ we have*

$$V^*AV := (V^*A_0V, \ldots, V^*A_nV) \in \mathcal{S}_l(M).$$

Proof Exercise 2.86. □

Exercise 2.86
Prove Lemma 2.85.

▶ **Remark 2.87** The properties of Lemma 2.85 can be summarized by saying that $\mathcal{S}_\infty(M)$ is an *abstract operator system* (see for example [79, Chapter 13]). But it can also be understood as a *concrete operator system*. Indeed the space

$$V := \operatorname{span}_{\mathbb{C}}(M_0, \ldots, M_n) \subseteq \operatorname{Mat}_s(\mathbb{C})$$

contains I_s and is closed under taking adjoints (since the M_i are Hermitian). This is already the definition of a concrete operator system. Any such system $V \subseteq \operatorname{Mat}_s(\mathbb{C})$ comes equipped with its set of positive elements of higher level in $\operatorname{Mat}_k(\mathbb{C}) \otimes V \subseteq \operatorname{Mat}_{ks}(\mathbb{C})$, and in our case these coincide exactly with the higher levels $\mathcal{S}_k(M)$.
Every abstract operator system can be realized as a concrete operator system on a possibly infinite-dimensional space of operators, by the Choi-Effros Theorem [79]. But what makes free spectrahedra special (among general abstract operator systems) is that they have a concrete realization in $\operatorname{Mat}_s(\mathbb{C})$, i.e., as operators on a *finite-dimensional space*.
As we will demonstrate in Theorem 2.96 below, they can also be seen as free halfspaces, at least in the sense of separation theory.

Example 2.88 This example is from [40]. It shows that even if two matrix polynomials M, N define the same classical spectrahedron

$$\mathcal{S}_1(M) = \mathcal{S}(M) = \mathcal{S}(N) = \mathcal{S}_1(N),$$

their free spectrahedra might still differ at higher levels. Consider

$$M = \begin{bmatrix} x_0 - x_1 & x_2 \\ x_2 & x_0 + x_1 \end{bmatrix} \text{ and } N = \begin{bmatrix} x_0 & 0 & x_1 \\ 0 & x_0 & x_2 \\ x_1 & x_2 & x_0 \end{bmatrix},$$

the (homogenized) matrix polynomials from Example 2.2 (2) and (3). Both $\mathcal{S}_1(M)$ and $\mathcal{S}_1(N)$ are the standard circular cone in $\mathbb{R}^3$. To be able to visualize the second level, we let x, y, z be real variables and set

$$A_0 = I_2,\ A_1 = \begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix},\ A_2 = \begin{bmatrix} 0 & z \\ z & 0 \end{bmatrix}.$$

In Fig. 2.13 we see the three-dimensional slices of $\mathcal{S}_2(M)$ and $\mathcal{S}_2(N)$ defined by the arising linear matrix inequalities on the A_i. The picture suggests

$$\mathcal{S}_2(M) \subsetneq \mathcal{S}_2(N)$$

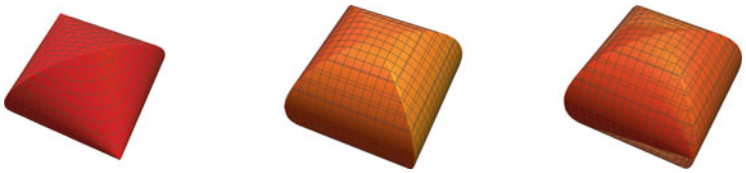


Fig. 2.13 Compact cross-section of $\mathcal{S}_2(M)$ in red, $\mathcal{S}_2(N)$ in orange, and both combined

and one can indeed check by direct computation that

$$\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & \frac{3}{4} \\ \frac{3}{4} & 0 \end{bmatrix}\right) \in \mathcal{S}_2(N) \setminus \mathcal{S}_2(M).$$

The general inclusion $\mathcal{S}_\infty(M) \subseteq \mathcal{S}_\infty(N)$ will be proven in Example 2.90 below.

The following result is from [40], and based on Choi's Representation Theorem for completely positive maps.

Theorem 2.89
Let M, N be homogeneous linear matrix polynomials of size s and t, with $M(e) = I_s$ and $N(e) = I_t$ for some $e \in \mathbb{R}^{n+1}$. Assume the M_i are linearly independent (the N_i not necessarily). Then the following are equivalent:

1. $\mathcal{S}_\infty(M) \subseteq \mathcal{S}_\infty(N)$.
2. $\mathcal{S}_t(M) \subseteq \mathcal{S}_t(N)$.
3. *There exist finitely many* $V_j \in \mathrm{Mat}_{s\times t}(\mathbb{C})$ *with* $\sum_j V_j^* M V_j = N$.

In case this is true, (3) holds with at most $s \cdot t$ many matrices V_j.

Proof That (3) implies (1) is a direct computation: if $A = (A_0, \ldots, A_n) \in \mathcal{S}_k(M)$, then $\sum_{i=0}^n A_i \otimes M_i \succcurlyeq 0$, and thus

$$0 \leqslant \sum_j (I_k \otimes V_j)^* \left(\sum_i A_i \otimes M_i\right)(I_k \otimes V_j) = \sum_i A_i \otimes \sum_j V_j^* M_i V_j = \sum_i A_i \otimes N_i.$$

This implies $A \in \mathcal{S}_k(N)$. (1) trivially implies (2). To show that (2) implies (3) consider

$$V := \mathrm{span}_{\mathbb{C}}\,(M_0, \ldots, M_n) \subseteq \mathrm{Mat}_s(\mathbb{C})$$

and the $\mathbb{C}$-linear map

$$\varphi\colon V \to \mathrm{Mat}_t(\mathbb{C});\ M_i \mapsto N_i.$$

Since the M_i are linearly independent, φ is well-defined. From $M(e) = I_s$ and $N(e) = I_t$ we obtain $\varphi(I_s) = I_t$, i.e., φ is a unital map. Since all M_i and N_i are Hermitian, φ is $*$-linear, i.e., fulfills $\varphi(A^*) = \varphi(A)^*$ for all $A \in V$.

Positive elements in V are precisely the elements from $\mathcal{S}_1(M)$, and from $\mathcal{S}_1(M) \subseteq \mathcal{S}_1(N)$ we deduce that φ maps positive matrices from V to positive matrices in $\mathrm{Mat}_t(\mathbb{C})$, i.e., φ is a positive map. But in fact φ is even *completely positive*, which means the following: For each $k \geqslant 1$ consider the $*$-linear unital map

$$\begin{aligned}\varphi_k\colon \mathrm{Mat}_k(\mathbb{C}) \otimes V &\to \mathrm{Mat}_k(\mathbb{C}) \otimes \mathrm{Mat}_t(\mathbb{C})\\ B \otimes A &\mapsto B \otimes \varphi(A).\end{aligned}$$

For $k = t$, if some element

$$X \in \mathrm{Mat}_t(\mathbb{C}) \otimes V \subseteq \mathrm{Mat}_t(\mathbb{C}) \otimes \mathrm{Mat}_s(\mathbb{C}) = \mathrm{Mat}_{ts}(\mathbb{C})$$

is positive, then $\varphi_t(X)$ is positive in $\mathrm{Mat}_t(\mathbb{C}) \otimes \mathrm{Mat}_t(\mathbb{C}) = \mathrm{Mat}_{t^2}(\mathbb{C})$. This is just a straightforward reformulation of our assumption $\mathcal{S}_t(M) \subseteq \mathcal{S}_t(N)$. So φ_t is a positive map, and by [79], Theorem 6.1, this already implies that *each* φ_k is a positive map (meaning that φ is completely positive, by definition). By Arveson's Extension Theorem (see [79], Theorem 7.5), the map φ extends from V to a completely positive map on the whole of $\mathrm{Mat}_s(\mathbb{C})$. By Choi's Representation Theorem for completely positive maps (see [79], Proposition 4.7), each such map is of the form

$$A \mapsto \sum_j V_j^* A V_j.$$

Since $\sum_j V_j^* M_i V_j = \varphi(M_i) = N_i$ holds for $i = 0, \ldots, n$, this proves (3). □

Example 2.90 Going back to M and N from Example 2.88, one checks

$$V_1^* M V_1 + V_2^* M V_2 = N$$

for

$$V_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \text{ and } V_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

So Theorem 2.89 implies $\mathcal{S}_\infty(M) \subseteq \mathcal{S}_\infty(N)$. In view of $\mathcal{S}_2(N) \not\subseteq \mathcal{S}_2(M)$, the matrix polynomial M does not arise from N as in Theorem 2.89 (3).

▶ **Remark 2.91** Checking containment $\mathcal{S}(M) \subseteq \mathcal{S}(N)$ of classical spectrahedra is an NP-hard problem, already in special cases (see for example [35, 50]). In case that $\mathcal{S}(M)$ is a cube, a condition similar to Theorem 2.89 (3) has been proposed in [6], as a tractable sufficient condition for containment (see also [40]). Theorem 2.89 reveals that this approach in fact detects containment of free spectrahedra. Since containment at level one does in general not imply containment at higher levels (see Example 2.90), it also explains why the approach may fail to detect containment of classical spectrahedra.

There are several cases in which containment of classical spectrahedra *does* imply containment of free spectrahedra (and thus condition (3) from Theorem 2.89), see [28, 50] for more information:

Corollary 2.92
Let M, N be homogeneous linear matrix polynomials with $\mathcal{S}(M) \subseteq \mathcal{S}(N)$. If either $\mathcal{S}(M)$ is a simplex cone, or N is diagonal (so that $\mathcal{S}(N)$ is polyhedral), then

$$\mathcal{S}_\infty(M) \subseteq \mathcal{S}_\infty(N).$$

Proof First, assume that $\mathcal{S}(M)$ is a simplex cone. After a linear transformation we can assume that $\mathcal{S}(M) = \mathbb{R}^{n+1}_+$ is the positive orthant. From $\mathbb{R}^{n+1}_+ = \mathcal{S}(M) \subseteq \mathcal{S}(N)$ we thus obtain $N_i \succcurlyeq 0$ for all i. Now take $A = (A_0, \ldots, A_n) \in \mathcal{S}_k(M)$ and $v \in \mathbb{C}^k$. By Lemma 2.85 (3) we have

$$v^*Av = (v^*A_0v, \ldots, v^*A_nv) \in \mathcal{S}_1(M) = \mathcal{S}(M) = \mathbb{R}^{n+1}_+,$$

which implies $A_i \succcurlyeq 0$ for all i. So $\sum_i A_i \otimes N_i$ is also positive semidefinite, which implies $A \in \mathcal{S}_k(N)$.

Second, assume that $N = \operatorname{diag}(\ell_1, \ldots, \ell_d)$ for linear forms $\ell_j \in \mathbb{R}[x_0, \ldots, x_n]_{(1)}$. Again, for $A \in \mathcal{S}_k(M)$ and $v \in \mathbb{C}^k$ we have

$$v^*Av \in \mathcal{S}_1(M) = \mathcal{S}(M) \subseteq \mathcal{S}(N),$$

so $0 \leqslant \ell_j(v^*Av) = v^*\ell_j(A)v$. This implies $\ell_j(A) \succcurlyeq 0$ for all $j = 1, \ldots, d$, and thus

$$\sum_i A_i \otimes N_i = \begin{bmatrix} \ell_1(A) & & \\ & \ddots & \\ & & \ell_d(A) \end{bmatrix} \succcurlyeq 0.$$

So $A \in \mathcal{S}_k(N)$. □

We can use the theory of free spectrahedra to recover one of the main results of [88] and [7], saying that spectrahedra are polyhedra only in the obvious cases:

Corollary 2.93
Let M be a homogeneous linear matrix polynomial of size s, as specified at the beginning of this section. Assume that $\ell \in \mathbb{R}[x_0, \ldots, x_n]_{(1)}$ is a linear form that exposes a face of $\mathcal{S}(M)$ of codimension 1. Then there is some invertible $A \in \mathrm{GL}_s(\mathbb{C})$, and a linear matrix polynomial M' of size $s-1$, with

$$A^* M A = \begin{bmatrix} \ell & 0 \\ 0 & M' \end{bmatrix}.$$

Hence if $\mathcal{S}(M)$ is polyhedral, there is some invertible $A \in \mathrm{GL}_s(\mathbb{C})$ with

$$A^* M A = \begin{bmatrix} D & 0 \\ 0 & M'' \end{bmatrix}$$

where D is diagonal and $\mathcal{S}(M) = \mathcal{S}(D)$.

Proof By Theorem 2.89 (2)⇒(3), the inclusion $\mathcal{S}(M) \subseteq \mathcal{S}(\ell)$ implies the existence of $v_j \in \mathbb{C}^k$ with $\sum_j v_j^* M v_j = \ell$ (note that we can assume $\ell(e) = 1$). We can assume $v_1^* M v_1 \neq 0$, and after an invertible linear transformation also that v_1 is the first standard basis vector. So

$$M_{11} + \sum_{j \geqslant 2} v_j^* M v_j = \ell.$$

Since ℓ vanishes along a face of codimension 1 of $\mathcal{S}(M)$, $0 \neq M_{11}$ must vanish there as well, which implies $M_{11} = \ell$, without loss of generality. Looking at all 2×2 minors of M that involve M_{11}, we see that all $M_{1j} = \overline{M_{j1}}$ must also vanish on that face, and they are thus also scalar multiples of ℓ. Using M_{11}, one can now eliminate all the M_{1j} and M_{j1} symmetrically, and the resulting matrix polynomial is of the desired form. The second claim follows by simply iterating this process. □

Note that the first step in the previous proof could also be done with a more elementary argument, for example as used in the proof of Theorem 3.5 (8) below. Using results on free spectrahedra is much less elementary, but avoids a lot of technical arguments.

One would like to completely classify linear matrix polynomials that define the same spectrahedron. As we have seen already, this is harder than one might suspect, since the free spectrahedra impose important conditions that cannot be seen at level one alone. However, classifying linear matrix polynomials defining the same free spectrahedron is easier. The following notion and result was also proven in [40].

Definition 2.94

A matrix polynomial M of size s is *minimally free defining* if it is *not* possible to find a unitary $A \in \mathrm{U}_s(\mathbb{C})$, such that

$$A^* M A = \begin{bmatrix} M_1 & 0 \\ 0 & M_2 \end{bmatrix}$$

with M_1 of size $< s$, and $\mathcal{S}_\infty(M_1) = \mathcal{S}_\infty(M)$.

Theorem 2.95

Let M, N be homogeneous linear matrix polynomials as specified at the beginning of this section, and assume both are minimally free defining. If $\mathcal{S}_\infty(M) = \mathcal{S}_\infty(N)$, then M and N are of the same size and unitarily equivalent.

Proof We only give a rough idea of the proof and refer to [40] for details.

We let $V = \mathrm{span}_{\mathbb{C}}(M_0, \ldots, M_n) \subseteq \mathrm{Mat}_s(\mathbb{C})$, $W = \mathrm{span}_{\mathbb{C}}(N_0, \ldots, N_n) \subseteq \mathrm{Mat}_t(\mathbb{C})$, and consider the unital $*$-linear map

$$\varphi\colon V \to W;\ M_i \mapsto N_i.$$

Then φ is an isomorphism, and, as in the proof of Theorem 2.89, we see that both φ and φ^{-1} are completely positive. After checking the technial assumptions, we can apply [2], Theorem 2.2.5, and extend φ to a $*$-homomorphism $\pi\colon \mathrm{Mat}_s(\mathbb{C}) \to \mathrm{Mat}_t(\mathbb{C})$. This implies $s \leqslant t$, and by symmetry thus $s = t$. It is well-known that each unital $*$-homomorphism on $\mathrm{Mat}_s(\mathbb{C})$ is of the form $A \mapsto U^* A U$ for some unitary $U \in \mathrm{U}_s(\mathbb{C})$, which finishes the proof. □

We finish this section with a version of the free separation theorem of [24]. It shows that free spectrahedra play the same role in free convexity as halfspaces do in classical convexity theory.

Theorem 2.96 (Free Separation Theorem)

For each $k \geqslant 1$ let $C_k \subseteq \mathrm{Her}_k^{n+1}$ be a convex cone, such that the union $C_\infty := \bigcup_{k \geqslant 1} C_k$ fulfills condition (3) from Lemma 2.85. If $A \in \mathrm{Her}_k^{n+1} \setminus \mathrm{clos}(C_k)$, then there exists a homogeneous linear matrix polynomial M of size k, such that

$$C_\infty \subseteq \mathcal{S}_\infty(M) \ \text{ and } \ A \notin \mathcal{S}_k(M).$$

Proof Since $A \notin \operatorname{clos}(C_k)$, we can use the classical Separation Theorem B.4 to obtain a linear functional $\varphi\colon \operatorname{Her}_k^{n+1} \to \mathbb{R}$ with $\varphi_{|C_k} \geqslant 0$ and $\varphi(A) < 0$. As a linear form on $\operatorname{Her}_k^{n+1}$, φ is of the form

$$\varphi\colon (B_0, \ldots, B_n) \mapsto \sum_i \operatorname{Tr}(B_i M_i)$$

for certain $M_i \in \operatorname{Her}_k(\mathbb{C})$. We claim that the linear matrix polynomial

$$M = x_0 \overline{M}_0 + \cdots + x_n \overline{M}_n,$$

where $\overline{M}_i$ is the entrywise complex conjugate of M_i, has the desired property. First, with e_j denoting the j-th standard basis vector of $\mathbb{C}^k$, we directly compute

$$\left(\sum_{j=1}^k e_j \otimes e_j\right)^* \left(\sum_i A_i \otimes \overline{M}_i\right) \left(\sum_{j=1}^k e_j \otimes e_j\right) = \sum_i \operatorname{Tr}(A_i M_i) = \varphi(A) < 0.$$

So $\sum_i A_i \otimes \overline{M}_i \not\succcurlyeq 0$ and thus $A \notin \mathcal{S}_k(M)$. Further, for any $t \geqslant 1$, $B \in C_t$ and $v_1, \ldots, v_k \in \mathbb{C}^t$ we have

$$\left(\sum_j v_j \otimes e_j\right)^* \left(\sum_i B_i \otimes \overline{M}_i\right) \left(\sum_j v_j \otimes e_j\right) = \sum_i \operatorname{Tr}(V^* B_i V M_i) = \varphi(V^* B V),$$

where V is the $t \times k$-matrix with columns v_j. Since $V^* B V \in C_k$ by condition (3), and φ is nonnegative on C_k, this number is nonnegative. Furthermore, since every vector in $\mathbb{C}^t \otimes \mathbb{C}^k$ is of the form $\sum_j v_j \otimes e_j$, this proves $\sum_i B_i \otimes \overline{M}_i \succcurlyeq 0$, and thus $B \in \mathcal{S}_t(M)$. □

Example 2.97 Separation with classical functionals is not possible in the free setup in general, i.e., linear matrix polynomials of size $k > 1$ are necessary. For example, consider

$$A = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & \frac{3}{4} \\ \frac{3}{4} & 0 \end{bmatrix}\right) \in \mathcal{S}_2(N) \setminus \mathcal{S}_2(M)$$

as in Example 2.88. Whenever ℓ is a classical linear functional with $\ell_{|\mathcal{S}(M)} \geqslant 0$, then the inclusion $\mathcal{S}_1(N) = \mathcal{S}_1(M) \subseteq \mathcal{S}_1(\ell)$ implies $\mathcal{S}_\infty(N) \subseteq \mathcal{S}_\infty(\ell)$, by Theorem 2.89 (2)⇒(1). So $A \in \mathcal{S}_2(N) \subseteq \mathcal{S}_2(\ell)$ is not separated from $\mathcal{S}_2(M)$ by ℓ.



Spectrahedral Shadows

3

In this chapter we introduce the notion of a spectrahedral shadow, and examine these sets thoroughly. They are still feasible sets for semidefinite programming, and in contrast to spectrahedra, many of the standard constructions from convexity theory can be applied to them. We explain how general semialgebraic sets can be approximated by spectrahedral shadows and how these methods provide criteria for being a spectrahedral shadow for many sets. While all convex semialgebraic sets in the plane are spectrahedral shadows, we also provide examples of convex semialgebraic sets that cannot be realized as a projection of a spectrahedron.

3.1 Spectrahedral Shadows

A *spectrahedral shadow* or *projected spectrahedron* is the image of a spectrahedron under an affine-linear map. Such a shadow is clearly again convex, and also semialgebraic (by the Projection Theorem 1.2) but need not be a spectrahedron. One obvious reason is that a spectrahedral shadow is not necessarily closed, since linear maps are not closed. For example, the projection of the spectrahedron

$$\left\{(u_1, u_2) \in \mathbb{R}^2 \;\middle|\; \begin{bmatrix} u_1 & 1 \\ 1 & u_2 \end{bmatrix} \succcurlyeq 0\right\}$$

onto the first factor is the open interval $(0, \infty)$, see Fig. 3.1. But there is much more to it, as we will (see also Fig. 3.2).

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Fig. 3.1 Non-closed shadow (blue) of a spectrahedron (orange)

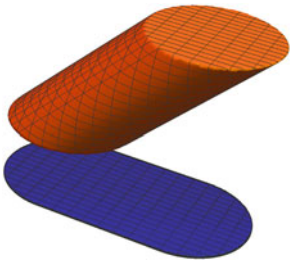


Fig. 3.2 A compact spectrahedron (orange) and its shadow (blue)

Example 3.1

1. The TV-screen $\left\{(u_1, u_2) \in \mathbb{R}^2 \mid u_1^4 + u_2^4 \leqslant 1\right\}$ is not a spectrahedron (Example 2.29). But the spectrahedron

$$S = \left\{ (u_1, u_2, a, b) \in \mathbb{R}^4 \;\middle|\; \begin{bmatrix} 1+a & b & & & \\ b & 1-a & & & \\ & & 1 & u_1 & \\ & & u_1 & a & \\ & & & & 1 \quad u_2 \\ & & & & u_2 \quad b \end{bmatrix} \succcurlyeq 0 \right\}$$

   in $\mathbb{R}^4$ maps onto the TV-screen under the projection $(u_1, u_2, a, b) \mapsto (u_1, u_2)$. In fact, an element $(u_1, u_2, a, b) \in S$ satisfies $a^2 + b^2 \leqslant 1$ and $u_1^2 \leqslant a$, $u_2^2 \leqslant b$, hence

$$u_1^4 + u_2^4 \leqslant a^2 + b^2 \leqslant 1.$$

   Conversely, any point (u_1, u_2) in the TV screen lifts to the point $(u_1, u_2, u_1^2, u_2^2) \in S$.

2. Let $\mathbb{R}[x]_d$ be the vector space of polynomials in $x = (x_1, \ldots, x_n)$ of degree at most d and let

$$\Sigma_{2d} = \left\{ p_1^2 + \cdots + p_r^2 \mid p_1, \ldots, p_r \in \mathbb{R}[x]_d,\ r \in \mathbb{N} \right\}$$

   be the cone of sums of squares (*sos-cone*). The sos-cone is not a spectrahedron. In fact, it is closed and convex, but it is not basic closed if $d > 1$. In fact, in Exercise 2.25 we identified an affine-linear slice that is not basic closed. It is however a spectrahedral shadow, namely

it is the image of Sym_s^+ under the Gram map

$$\begin{aligned}\Psi_{\mathfrak{m}}\colon\ \mathrm{Sym}_s &\to \mathbb{R}[x]_{2d}\\ A &\mapsto \mathfrak{m}^T A\mathfrak{m}\end{aligned}$$

where $\mathfrak{m}$ is a vector of polynomials spanning $\mathbb{R}[x]_d$, as we have seen in Example 2.2 (4).

Exercise 3.2
Represent the football stadium from Exercise 2.24 as a spectrahedral shadow.

Exercise 3.3
Show that Σ_{2d} is a closed convex cone in the finite-dimensional space $\mathbb{R}[x]_{2d}$.

Note first that we speak of *projected* spectrahedra, because every affine-linear map can be factored into an injective one, followed by a projection. This simple fact translates into the following statement:

Lemma 3.4
Let $P \subseteq \mathbb{R}^n$ be a spectrahedral shadow. Then there exists a linear matrix polynomial $M(x, y)$ in variables $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$, for some m, with

$$P = \left\{u \in \mathbb{R}^n \mid \exists v \in \mathbb{R}^m\colon M(u, v) \succcurlyeq 0\right\}.$$

Proof Let $P = \varphi(S)$ where $S = \mathcal{S}(N)$ is a spectrahedron in $\mathbb{R}^m$ and $\varphi\colon \mathbb{R}^m \to \mathbb{R}^n$ an affine-linear map. Let $\Gamma\colon \mathbb{R}^m \to \mathbb{R}^n \times \mathbb{R}^m$, $v \mapsto (\varphi(v), v)$ be the graph map. Since Γ is injective, $\Gamma(S)$ is a spectrahedron. Namely, $\Gamma(S) = \mathcal{S}(M)$ for

$$M(x, y) = \begin{bmatrix} N(y) & 0 & 0\\ 0 & \varphi(y) - x & 0\\ 0 & 0 & x - \varphi(y)\end{bmatrix},$$

where the entries $\varphi(y) - x$ and $x - \varphi(y)$ are diagonal blocks. This gives rise to the desired representation of P. □

A representation of P as in Lemma 3.4 is also called a *lifted linear matrix inequality representation*, an *extended formulation* or simply a *semidefinite representation*. If $M(x, y)$ is a linear matrix polynomial as above, we will often denote the projection $\mathbb{R}^{n+m} \to \mathbb{R}^n$ onto the first coordinates by π_x. Thus the lifted representation will be denoted by

$$P = \pi_x\mathcal{S}(M(x, y)) \quad \text{or just} \quad P = \pi_x\mathcal{S}(M).$$

While spectrahedra are very special convex sets, it is still unclear which distinguishing features spectrahedral shadows have in general (beyond the obvious of being convex and semialgebraic). It was asked in [65] whether every convex semialgebraic set is a spectrahedral shadow. This also became known as the Helton-Nie conjecture [41]. This conjecture was proven by Scheiderer for two-dimensional sets [98], but later disproven in higher dimensions [99]. Helton and Nie [41, 42] proved a series of results which can be summed up as saying that any convex semialgebraic set with sufficiently regular boundary is a spectrahedral shadow. Understanding and proving some of these results will be one of our main goals in the following sections.

We will start by establishing some basic properties of spectrahedral shadows. In particular, we will show that many of the usual operations preserving convexity also preserve the property of being a spectrahedral shadow.

3.2 Operations on Spectrahedral Shadows

The following theorem contains many of the constructions that can be performed with convex semialgebraic sets, and states that all these operations preserve the property of being a spectrahedral shadow. Item (6) was first proven in [41], see also [64, 71]. Items (8), (9) and thus (10) are already implicitly contained in the duality theory of semidefinite programming (see for example [87]), and (11) is from [67].

Theorem 3.5
Let $P, Q \subseteq \mathbb{R}^n$, $R \subseteq \mathbb{R}^m$ be spectrahedral shadows. Then the following sets are again spectrahedral shadows:

1. *Intersection:* $P \cap Q$.
2. *Cartesian product:* $P \times R$.
3. *Affine-linear images and preimages of* P.
4. *Minkowski sum:* $P + Q$.
5. *Conic hull:* $\operatorname{cone}(P)$.
6. *Convex hull:* $\operatorname{conv}(P \cup Q)$.
7. *Projective transformations of* P.
8. *Polar:* $P^\circ = \left\{v \in \mathbb{R}^n \mid \forall u \in P : u^T v \geqslant -1\right\}$.
9. *Dual cone:* $P^\vee = \left\{v \in \mathbb{R}^n \mid \forall u \in P : u^T v \geqslant 0\right\}$.
10. *Closure:* $\operatorname{clos}(P)$.
11. *Relative interior:* $\operatorname{relint}(P)$.

Proof Throughout the proof let

$$P = \pi_x \mathcal{S}(M) \text{ and } Q = \pi_x \mathcal{S}(N)$$

with linear matrix polynomials $M(x, y), N(x, z)$ in variables $x = (x_1, \ldots, x_n)$, $y = (y_1, \ldots, y_r), z = (z_1, \ldots, z_s)$.

(1) We have $P \cap Q = \pi_x \mathcal{S}(L)$ where

$$L(x, y, z) := \begin{bmatrix} M(x, y) & 0 \\ 0 & N(x, z) \end{bmatrix}.$$

(2) Clearly, $P \times \mathbb{R}^m = \{(u, v) \in \mathbb{R}^{n+m} \mid \exists w \in \mathbb{R}^r \colon M(u, w) \succcurlyeq 0\}$ is a spectrahedral shadow, and so is $\mathbb{R}^n \times R$. Therefore $P \times R = (P \times \mathbb{R}^m) \cap (\mathbb{R}^n \times R)$ is a spectrahedral shadow, using (1).

(3) For affine-linear images, this is clear by definition. For the preimage of P under the affine-linear map $\varphi \colon \mathbb{R}^l \to \mathbb{R}^n$, consider the set

$$\{(u, v) \in \mathbb{R}^l \times P \mid \varphi(u) = v\},$$

which is a is a spectrahedral shadow, using (1) and (2). Clearly $\varphi^{-1}(P)$ is the projection onto the first factor, so the claim follows.

(4) Let $\sigma \colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ be the linear map $(u, v) \mapsto u + v$. Then $P + Q = \sigma(P \times Q)$, so the result follows from (2) and (3).

(5) For $i = 1, \ldots, n$ let

$$L_i(x, y, s, t) := \begin{bmatrix} s & x_i \\ x_i & t \end{bmatrix}$$

so that

$$\mathcal{S}(L_i) = \left\{(u, v, \lambda, \mu) \in \mathbb{R}^n \times \mathbb{R}^r \times \mathbb{R} \times \mathbb{R} \mid \lambda, \mu \geqslant 0, \lambda\mu \geqslant u_i^2, i = 1, \ldots, n\right\}.$$

Further let $\widetilde{M}(x, y, s, t) := s \cdot M(x/s, y/s)$ be the homogenization of M by the variable s. We consider $\widetilde{M}$ as a matrix polynomial with an additional variable t, although it does not appear. We claim that

$$\operatorname{cone}(P) = \pi_x \left(\mathcal{S}(\widetilde{M}) \cap \mathcal{S}(L_1) \cap \cdots \cap \mathcal{S}(L_n)\right), \tag{3.1}$$

which will prove the statement, using (1) and (3) (or directly Exercise 2.6). To see "$\subseteq$" in (3.1), take $u \in \operatorname{cone}(P)$. If $u = 0$, it clearly belongs to the set on the right, so assume $u \neq 0$. Since P is already convex, there exists $\lambda > 0$ and $u' \in P$ with $u = \lambda u'$. There further exists $v' \in \mathbb{R}^r$ with $M(u', v') \succcurlyeq 0$, since $u' \in P$. Multiplying this with λ proves $(u, \lambda v', \lambda, \mu) \in \mathcal{S}(\widetilde{M})$, for all $\mu \in \mathbb{R}$.

Clearly μ can be chosen large enough to fulfill $\lambda\mu \geqslant u_i^2$ for all i, thus also $(u, \lambda v', \lambda, \mu) \in \mathcal{S}(L_i)$ for all i and such μ. This proves that u belongs to the set on the right in (3.1).

For "$\supseteq$" let (u, v, λ, μ) belong to $\mathcal{S}(\widetilde{M}) \cap \mathcal{S}(L_1) \cap \cdots \cap \mathcal{S}(L_n)$. If $\lambda > 0$, we clearly obtain $M(u/\lambda, v/\lambda) \succcurlyeq 0$, so $u/\lambda \in \pi_x \mathcal{S}(M) = P$ and thus $u \in \operatorname{cone}(P)$. If $\lambda = 0$, from the conditions in $\mathcal{S}(L_i)$ it follows that $u = 0$, which also belongs to $\operatorname{cone}(P)$.

(6) Let $\widetilde{P}, \widetilde{Q} \subseteq \mathbb{R}^{n+1}$ be the conic homogenizations of P and Q, as described in Remark 2.31 (1). By (5) both are spectrahedral shadows, and so is $C := \widetilde{P} + \widetilde{Q}$, using (4). A direct computation shows

$$\operatorname{conv}(P \cup Q) = \left\{u \in \mathbb{R}^n \mid (u, 1) \in C\right\},$$

which proves the statement, using (3).

(7) A projective transformation of P can be realized by embedding $\mathbb{R}^n$ as an affine subspace into $\mathbb{R}^{n+1}$, taking the conic hull of P there, applying a linear transformation of $\mathbb{R}^{n+1}$, and finally taking the inverse image with respect to the embedding of $\mathbb{R}^n$. So the claim follows from (3) and (5).

(8) We first assume that $P = \mathcal{S}(L)$ is a spectrahedron, defined by the monic linear matrix polynomial $L = I_k + \sum_{i=1}^n x_i L_i$. Then for any $A \in \operatorname{Sym}_k^+$ with $\operatorname{Tr}(A) \leqslant 1$ and $u \in P$ we have

$$0 \leqslant \operatorname{Tr}(AL(u)) \leqslant 1 + \sum_{i=1}^n u_i \operatorname{Tr}(AL_i), \tag{3.2}$$

using (the first part of) self-duality of Sym_k^+ with respect to the trace inner product (Proposition B.6). Clearly, the set of positive semidefinite matrices of trace at most one is a compact spectrahedron. So, as a linear image of this set,

$$S := \left\{(\operatorname{Tr}(AL_1), \ldots, \operatorname{Tr}(AL_n)) \mid A \in \operatorname{Sym}_k^+, \operatorname{Tr}(A) \leqslant 1\right\}$$

is also a compact spectrahedral shadow, and from (3.2) we see that S is contained in P°. To prove equality of the two sets, assume $u \notin S$. By the Bipolar Theorem B.5 applied to S, there exists $v \in S^\circ$ with $v^T u < -1$. From $v \in S^\circ$ we obtain

$$0 \leqslant 1 + \sum_{i=1}^n v_i \operatorname{Tr}(AL_i) = \operatorname{Tr}(AL(v))$$

for all $A \in \operatorname{Sym}_k^+$ with $\operatorname{Tr}(A) = 1$. Again by self-duality of Sym_k^+ (this time the other part, and using that $\operatorname{Tr}(A) = 1$ can be achieved by scaling any nonzero psd matrix) we get $L(v) \in \operatorname{Sym}_k^+$, which means $v \in P$ and thus $u \notin P^\circ$. From all of this we conclude that $P^\circ = S$ is a spectrahedral shadow.

In the next step assume that P is an arbitrary spectrahedron. Since a translation of P leads to a projective transformation of P°, we can assume that P contains the origin in its relative interior, using (7). Furthermore, P° is simply a cylinder over the polar of P in its affine hull, so the statement follows from what we have just proven, using Lemma 2.12 and (2).

Finally, in the general case $P = \pi_x(S)$ for some spectrahedron S, we have

$$v \in P^\circ \Leftrightarrow (v, 0) \in S^\circ$$

and the result follows from what we have already proven and (3).

(9) follows from $P^\vee = \mathrm{cone}(P)^\vee = \mathrm{cone}(P)^\circ$, using (5) and (8).

(10) follows from (8), since we may assume $0 \in P$ and thus $\mathrm{clos}(P) = (P^\circ)^\circ$ by the Bipolar Theorem B.5.

(11) Since $\pi_x(\mathrm{relint}(\mathcal{S}(M))) = \mathrm{relint}(\pi_x(\mathcal{S}(M))) = \mathrm{relint}(P)$ (see Exercise 3.7), it suffices to prove the statement for spectrahedra. We thus assume that L is a linear matrix polynomial with $P = \mathcal{S}(L)$. For a fixed point $u_0 \in \mathrm{relint}(P)$ and $u \in \mathbb{R}^n$, we have $u \in \mathrm{relint}(P)$ if and only if there exists $\varepsilon > 0$ such that $u + \varepsilon(u - u_0) \in P$, i.e., $L(u + \varepsilon(u - u_0)) \succcurlyeq 0$ (see Exercise 3.6). Now we just write out $L(x) = L_0 + L'(x)$ with L' homogeneous, put $\delta = \frac{1}{1+\varepsilon}$, compute

$$\delta L(u + \varepsilon(u - u_0)) = \delta L_0 + L'(u) - \delta\varepsilon L'(u_0)$$

and conclude that

$$\mathrm{relint}(P) = \left\{u \in \mathbb{R}^n \mid \exists \delta \in (0, 1):\ \delta L_0 + L'(u) + (\delta - 1)L'(u_0) \succcurlyeq 0\right\}.$$

Combining this with a representation of the open unit interval

$$(0, 1) = \left\{\delta \in \mathbb{R} \;\middle|\; \exists \lambda:\ \begin{bmatrix} \lambda & 1 \\ 1 & \delta \end{bmatrix} \succcurlyeq 0 \text{ and } \begin{bmatrix} \lambda & 1 \\ 1 & 1 - \delta \end{bmatrix} \succcurlyeq 0\right\},$$

we obtain a representation of $\mathrm{relint}(P)$ as a spectrahedral shadow (Fig. 3.3). □

Exercise 3.6

Let S be a convex subset of a finite-dimensional real vector space V, and let U be the affine hull of S. Recall that the *relative interior* of S is the set

$$\mathrm{relint}(S) := \left\{u \in S \mid \exists \varepsilon > 0:\ B_\varepsilon(u) \cap U \subseteq S\right\}.$$

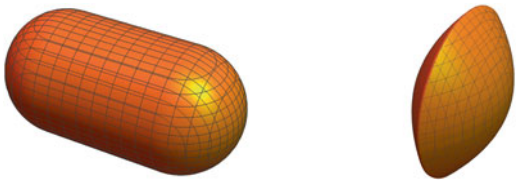


Fig. 3.3 Convex hull of two balls, and its polar

1. For any fixed point $u_0 \in \operatorname{relint}(S)$, show

$$\operatorname{relint}(S) = \{u \in V \mid \exists \varepsilon > 0 \colon\ u + \varepsilon(u - u_0) \in S\}.$$

2. Show that $\operatorname{relint}(S)$ is convex.

Exercise 3.7
Let $S \subseteq \mathbb{R}^n$ be a convex set and $\varphi \colon \mathbb{R}^n \to \mathbb{R}^m$ a linear map. Show that

$$\varphi(\operatorname{relint}(S)) = \operatorname{relint}(\varphi(S)).$$

Exercise 3.8
To what extent do the results in Theorem 3.5 also hold for spectrahedra? More precisely, if P, Q, R are spectrahedra, which of the sets constructed in (1)–(10) are again spectrahedra?

Note that the proof of Theorem 3.5 is completely constructive, so given concrete representations of P, Q, R as spectrahedral shadows, the steps in the proof can be turned into concrete representations of the resulting convex sets.

In Sect. 3.4 below we will need the following refinement of Theorem 3.5 (11) on interiors of spectrahedral shadows, from [67]:

Theorem 3.9
Let S, T be spectrahedral shadows in $\mathbb{R}^n$ with $T \subseteq S$ and S closed. Let $\mathcal{F}_T$ be the set of all faces F of S with $F \cap T \neq \emptyset$, and set

$$T \hookleftarrow S := \bigcup_{F \in \mathcal{F}_T} \operatorname{relint}(F).$$

Then $T \hookleftarrow S$ is a spectrahedral shadow.

Proof First, assume that S is a spectrahedron, and let $M(x)$ be a linear matrix polynomial of size s defining S. For $u \in S$ we denote by F_u the unique face of S that contains u in its relative interior. Now, for fixed $v \in T$, we have

$$\bigcup_{F \in \mathcal{F}_T, v \in F} \operatorname{relint}(F) = \{u \in S \mid v \in F_u\}$$
$$= \left\{u \in \mathbb{R}^n \mid M(u) \succcurlyeq 0, \ker M(u) \subseteq \ker M(v)\right\},$$

where we use the arguments from the proof of Theorem 2.26 for the second equation. Using Exercise 3.10 we thus have

$$T \leftarrow\!\!\!\rho\ S = \left\{u \in \mathbb{R}^n \mid \exists v \in T,\ \exists \lambda\colon \left[\begin{array}{c|c} M(u) & M(v) \\ \hline M(v) & \lambda \cdot I_s \end{array}\right] \succcurlyeq 0\right\},$$

which proves the claim.

Now let S be a spectrahedral shadow. Hence there is a spectrahedron $\widetilde{S}$ in some $\mathbb{R}^{n+r}$ that projects onto S via the projection map $\pi_x\colon \mathbb{R}^{n+r} \to \mathbb{R}^n$. Define

$$\widetilde{T} := \pi^{-1}(T) \cap \widetilde{S} = \left\{(u, v) \in \mathbb{R}^{n+r} \mid (u, v) \in \widetilde{S}, u \in T\right\},$$

which is clearly a spectrahedral shadow, and also a subset of $\widetilde{S}$. We thus already know that $\widetilde{T} \leftarrow\!\!\!\rho\ \widetilde{S}$ is a spectrahedral shadow, so we finish the proof by showing

$$\pi_x\left(\widetilde{T} \leftarrow\!\!\!\rho\ \widetilde{S}\right) = T \leftarrow\!\!\!\rho\ S.$$

For the first inclusion, let $(u, v) \in \widetilde{T} \leftarrow\!\!\!\rho\ \widetilde{S}$ be given. Thus there is $(a, b) \in \widetilde{T}$ and some face $\widetilde{F}$ of $\widetilde{S}$ containing (a, b), such that $(u, v) \in \operatorname{relint}(\widetilde{F})$. So there is some $\varepsilon > 0$ such that

$$(u, v) + \varepsilon\left((u, v) - (a, b)\right) \in \widetilde{F}.$$

It follows that $u + \varepsilon(u - a) \in \pi_x(\widetilde{F}) \subseteq S$, which implies $a \in F_u$. Since $a \in T$ and $u \in \operatorname{relint}(F_u)$, this proves $u \in T \leftarrow\!\!\!\rho\ S$.

For the other inclusion, let F be a face of S that contains some element from T. Then

$$\widetilde{F} := \pi_x^{-1}(F) \cap \widetilde{S}$$

is a face of $\widetilde{S}$ that contains some element from $\widetilde{T}$. By Exercise 3.7 we find

$$\pi_x\left(\operatorname{relint}\left(\widetilde{F}\right)\right) = \operatorname{relint}\left(\pi_x\left(\widetilde{F}\right)\right) = \operatorname{relint}(F),$$

which proves the desired inclusion. □

Exercise 3.10
Let $A \in \mathrm{Sym}_s$ and $B \in \mathrm{Mat}_{t\times s}$. Show that the following are equivalent:

1. $\exists \lambda \in \mathbb{R}$ s.t. $\left[\begin{array}{c|c} A & B^t \\ \hline B & \lambda \cdot I_t \end{array}\right] \succcurlyeq 0$.
2. $A \succcurlyeq 0$ and $\ker A \subseteq \ker B$.

Exercise 3.11
Show the following (directly or using Theorem 3.9): If S is a spectrahedral shadow and u an extreme point of S, then $S \setminus \{u\}$ is a spectrahedral shadow.
Hint: Consider first the case in which u is an exposed face of S.

Exercise 3.12
Let $D \subseteq \mathbb{R}^2$ be the unit disk. Show that for any semialgebraic subset S of the boundary of D, the set $D \setminus S$ is a spectrahedral shadow.

3.3 Positive Polynomials and the Lasserre-Parrilo Relaxation

In this section we introduce an important method to systematically and explicitly construct spectrahedal shadows, approximating a given semialgebraic set. It is described in [57], similar ideas can be found in [76], and we thus call it the *Lasserre-Parrilo relaxation method*. In many cases this method can be used to show that the convex hull of the initial semialgebraic set is a spectrahedral shadow, as we will demonstrate in the following sections. The basic building blocks for Lasserre-Parrilo relaxations are truncated quadratic modules and their duals. We briefly review the definitions here, and refer to Sect. A in the appendix for more detailed explanations.

Throughout, let $p = (p_1, \ldots, p_r)$ be a tuple of polynomials from $\mathbb{R}[x]$. We set $p_0 = 1$ and denote by

$$\mathrm{QM}(p) := \left\{ \sum_{i=0}^{r} \sigma_i p_i \mid \sigma_i \in \Sigma \text{ for } i = 0, \ldots, r \right\}$$

the *quadratic module generated by* p. Here, Σ denotes the set of sums of squares of polynomials in $\mathbb{R}[x]$. Just as ideals contain polynomials that obviously vanish on the zero set of its generators, a quadratic module contains polynomials that are obviously nonnegative on the basic closed semialgebraic set defined by the generators $p_1, \ldots, p_r$.

For $d \geqslant 0$ we set

$$\mathrm{QM}_d(p) := \mathrm{QM}(p) \cap \mathbb{R}[x]_d$$

and

$$\mathrm{QM}_{[d]}(p) := \left\{ \sum_{i=0}^{r} \sigma_i \, p_i \mid \forall i : \ \deg(\sigma_i \, p_i) \leqslant d \right\}.$$

Note that clearly

$$\mathrm{QM}_{[d]}(p) \subseteq \mathrm{QM}_d(p)$$

holds, but this inclusion can be strict, since degree cancelling might occur when summing the terms $\sigma_i \, p_i$.

Exercise 3.13
Check carefully which of the following two equalities is trivial by definition, and prove the other one:

$$\mathrm{QM}_{[2d]}(\emptyset) = \mathrm{QM}_{2d}(\emptyset) = \Sigma_{2d}.$$

See (2.1) for the definition of Σ_{2d}.

By $\mathrm{QM}_{[d]}(p)^\vee$ we denote the dual cone of $\mathrm{QM}_{[d]}(p)$ in $\mathbb{R}[x]_d^*$, and we further define

$$\mathrm{QM}_{[d]}(p)' := \left\{ L \in \mathrm{QM}_{[d]}(p)^\vee \mid L(1) = 1 \right\}.$$

These dual cones are all spectrahedra, by the following direct generalization of Proposition 2.8.

Proposition 3.14
For all $d \geqslant 0$, the cones $\mathrm{QM}_{[d]}(p)^\vee$ and $\mathrm{QM}_{[d]}(p)'$ are spectrahedra, and the cone $\mathrm{QM}_{[d]}(p)$ is a spectrahedral shadow.

Proof Consider the finite-dimensional vector space

$$V = \left\{ (f_0, \ldots, f_r) \in \mathbb{R}[x]^{r+1} \mid \forall i : 2\deg(f_i) + \deg(p_i) \leqslant d \right\}.$$

Similar as in the proof of Proposition 2.8 we associate with $L \in \mathbb{R}[x]_d^*$ the bilinear form

$$b_L : V \times V \to \mathbb{R}$$

$$((f_0, \ldots, f_r), (g_0, \ldots, g_r)) \mapsto \sum_{i=0}^{r} L(f_i \, g_i \, p_i)$$

and obtain a linear map

$$\Phi\colon \mathbb{R}[x]_d^* \to \mathrm{Sym}(V)$$
$$L \mapsto b_L$$

with $\mathrm{QM}_{[d]}(p)^\vee = \Phi^{-1}(\mathrm{Sym}^+(V))$. So $\mathrm{QM}_{[d]}(p)^\vee$ and thus clearly also $\mathrm{QM}_{[d]}(p)'$ is a spectrahedron. The statement about $\mathrm{QM}_{[d]}(p)$ follows from a slight modification of the argument in Example 3.1 (2), see Exercise 3.15. □

Exercise 3.15
Prove that each $\mathrm{QM}_{[d]}(p)$ is a spectrahedral shadow.

Proposition 3.16
Let $S = \mathcal{S}(p)$ be a basic closed semialgebraic subset of $\mathbb{R}^n$, and consider the projection

$$\pi : \mathbb{R}[x]_d^* \to \mathbb{R}^n$$
$$L \mapsto \big(L(x_1), \ldots, L(x_n)\big).$$

Then $\mathrm{conv}(S) \subseteq \pi(\mathrm{QM}_{[d]}(p)')$.

Proof For $u \in S$ consider the evaluation map

$$L_u\colon \mathbb{R}[x]_d \to \mathbb{R}$$
$$q \mapsto q(u)$$

as an element in the dual space $\mathbb{R}[x]_d^*$. Since $L_u(1) = 1$ and all polynomials in $\mathrm{QM}_{[d]}(p)$ are nonnegative on S, we have $L_u \in \mathrm{QM}_{[d]}(p)'$. From

$$\pi(L_u) = u$$

we obtain $S \subseteq \pi(\mathrm{QM}_{[d]}(p)')$, and convexity of $\pi(\mathrm{QM}_{[d]}(p)')$ thus proves the claim. □

Definition 3.17

With notations as above, we define the following:

1. For $d \geqslant 0$, the spectrahedral shadow

$$\mathcal{R}_{[d]}(p) := \pi\left(\mathrm{QM}_{[d]}(p)'\right)$$

in $\mathbb{R}^n$ is called the *Lasserre-Parrilo relaxation* of degree d of $\operatorname{conv}(S)$ with respect to p.

2. The Lasserre-Parrilo relaxation is *exact* in degree d if

$$\mathcal{R}_{[d]}(p) = \operatorname{conv}(S),$$

and *exact up to closure* if these two sets have the same closure.

3. The sequence of Lasserre-Parrilo relaxations $(\mathcal{R}_{[d]}(p))_{d\in\mathbb{N}}$ *converges* to a subset $C \subseteq \mathbb{R}^n$ if

$$\operatorname{clos}(C) = \operatorname{clos}\left(\bigcap_{d\geqslant 0} \mathcal{R}_{[d]}(p)\right).$$

Note that the chain $\mathrm{QM}_{[1]}(p) \subseteq \mathrm{QM}_{[2]}(p) \subseteq \cdots$ is ascending by definition, therefore the chains $\mathrm{QM}_{[1]}(p)' \supseteq \mathrm{QM}_{[2]}(p)' \supseteq \cdots$ and

$$\mathcal{R}_{[1]}(p) \supseteq \mathcal{R}_{[2]}(p) \supseteq \cdots \supseteq \operatorname{conv}(\mathcal{S}(p))$$

are descending. Our first goal now is to characterize exactness in terms of nonnegative polynomials and sums of squares with degree bounds. It will allow us to use Positivstellensätze to prove that certain sets are spectrahedral shadows.

Proposition 3.18
Let $p = (p_1, \ldots, p_r) \in \mathbb{R}[x]^r$ and $S = \mathcal{S}(p)$. For $d \geqslant 0$ consider the following statements:

1. *The Lasserre-Parrilo relaxation is exact up to closure in degree d, i.e.,*

$$\operatorname{conv}(S) \subseteq \mathcal{R}_{[d]}(p) \subseteq \operatorname{clos}(\operatorname{conv}(S)).$$

2. *Every $\ell \in \mathbb{R}[x]$ with $\deg(\ell) = 1$ and $\ell|_S \geqslant 0$ is contained in $\mathrm{QM}_{[d]}(p)$.*

Then (2) *implies* (1). *The converse also holds if S has non-empty interior.*

Proof (2)⇒(1): Suppose that (1) does not hold, and set $C = \operatorname{clos}(\operatorname{conv}(S))$. Since $\operatorname{conv}(S) \subseteq \mathcal{R}_{[d]}(p)$ by Proposition 3.16, there must exist $u \in \mathcal{R}_{[d]}(p) \setminus C$. Hence, by the Separation Theorem B.4, there is a polynomial $\ell \in \mathbb{R}[x]_1$ with $\ell|_S \geqslant 0$

and $\ell(u) < 0$. Since $u \in \mathcal{R}_{[d]}(p)$, there exists $L \in \mathrm{QM}_{[d]}(p)'$ such that $u = \big(L(x_1), \ldots, L(x_n)\big)$. Now since $\deg(\ell) = 1$ and $L(1) = 1$, we have

$$L(\ell) = \ell\big(L(x_1), \ldots, L(x_n)\big) = L(u) < 0,$$

which implies $\ell \notin \mathrm{QM}_{[d]}(p)$, so also (2) fails.

(1)⇒(2): Let $\ell \in \mathbb{R}[x]$ with $\deg(\ell) = 1$ and $\ell_{|_S} \geqslant 0$. Since S has non-empty interior, and polynomials from $\mathrm{QM}(p)$ are nonnegative on S, we must have

$$\mathrm{QM}(p) \cap (-\mathrm{QM}(p)) = \{0\}.$$

By Proposition A.19 this implies that $\mathrm{QM}_{[d]}(p)$ is closed. So if ℓ is not contained in $\mathrm{QM}_{[d]}(p)$, there exists

$$L \in \mathrm{QM}_{[d]}(p)^\vee \text{ with } L(\ell) < 0,$$

by the Biduality Theorem B.5. We claim that there also exists such L with $L(1) = 1$. If $L(1) \neq 0$, then $L(1) > 0$ since $1 \in \mathrm{QM}_{[d]}(p)$, so we can just rescale. If $L(1) = 0$, take any $L_1 \in \mathrm{QM}_{[d]}(p)^\vee$ with $L_1(1) = 1$ (for example a point evaluation) and let $L' = \alpha L + L_1$. Then L' has the desired property when α is sufficiently large.

It follows that $u := \big(L(x_1), \ldots, L(x_n)\big)$ is a point in $\mathcal{R}_{[d]}(p) \subseteq \mathrm{clos}(\mathrm{conv}(S))$. On the other hand,

$$\ell(u) = \ell\big(L(x_1), \ldots, L(x_n)\big) = L(\ell) < 0$$

yields a contradiction, since ℓ is nonnegative on S and thus on $\mathrm{clos}(\mathrm{conv}(S))$. □

Definition 3.19

Let $p = (p_1, \ldots, p_r) \in \mathbb{R}[x]^r$ be a tuple of polynomials, and $S \subseteq \mathbb{R}^n$ a subset. We say that p *is an Archimedean description of* S if $S = \mathcal{S}(p)$ and the quadratic module $\mathrm{QM}(p)$ is Archimedean (see Definition A.12).

Note that if p is an Archimedean description of S, then S must be bounded (and thus compact). In case that $\mathcal{S}(p)$ is bounded, but the description by p is not Archimedean, one can redress this in principle by adding $\lambda - \sum_i x_i^2$ to the generators, for some sufficiently large real λ. Also note that if $\mathrm{QM}(p)$ is closed under products, i.e., a *preordering*, then boundedness of $S = \mathcal{S}(p)$ already implies that p is an Archimedean description of S, by Schmüdgen's Positivstellensatz Theorem A.11.

A direct consequence of the Positivstellensätze by Schmüdgen Theorem A.11 and Putinar Theorem A.13 is the following important convergence and approximation result:

Theorem 3.20
Let p be an Archimedean description of S. Then the Lasserre-Parrilo relaxations of S with respect to p converge to $\mathrm{conv}(S)$.

Proof We will show

$$\mathrm{conv}(S) = \bigcap\nolimits_{d \geqslant 0} \mathcal{R}_{[d]}(p)$$

where the inclusion from left to right is clear by Proposition 3.16. Now if $u \notin \mathrm{conv}(S)$, then there exists $\ell \in \mathbb{R}[x]_1$ with $\ell|_S > 0$ and $\ell(u) < 0$, by the Separation Theorem B.4 (note again that S and thus $\mathrm{conv}(S)$ are compact, and thus closed). By Putinar's Positivstellensatz A.13 we have $\ell \in \mathrm{QM}(p)$ and hence $\ell \in \mathrm{QM}_{[d]}(p)$ for some $d \geqslant 0$ (depending on ℓ). This implies $u \notin \mathcal{R}_{[d]}(p)$, by the same argument as in Proposition 3.18. □

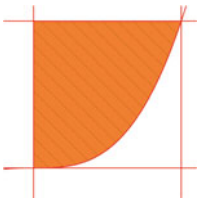
Example 3.21
To illustrate the Lasserre-Parrilo relaxation method, we discuss an example in detail. Consider the polynomials

$$p_1 = x_2 - x_1^3,\ p_2 = x_1,\ p_3 = 1 - x_1,\ p_4 = x_2,\ p_5 = 1 - x_2$$

in the variables x_1, x_2. Then $S = \mathcal{S}(p)$ is already convex, and we claim that

$$S = \mathcal{R}_{[3]}(p).$$

In particular, S is a spectrahedral shadow. We use Proposition 3.18 and show that $\mathrm{QM}_{[3]}(p)$ contains all $\ell \in \mathbb{R}[x]_1$ with $\ell|_S \geqslant 0$.



Let $\ell \in \mathbb{R}[x]_1$ be such a polynomial. If $\ell(u) > 0$ for all $u \in S$, then ℓ will assume its minimum ε in some point of S, since S is compact. Since $\varepsilon \in \mathrm{QM}_{[3]}(p)$ it suffices to show $\ell - \varepsilon \in \mathrm{QM}_{[3]}(p)$. Also, if ℓ is already nonnegative on the box $[0, 1] \times [0, 1]$, we can use Farkas's Lemma B.23 and conclude that ℓ is contained in

$$\mathrm{cone}(x_1, 1 - x_1, x_2, 1 - x_2) \subseteq \mathrm{QM}_{[3]}(p).$$

Thus we are left with the case that ℓ describes a tangent to the cubic curve $x_2 = x_1^3$. The tangent at a point (a, a^3) with $a \in (0, 1)$ is given by the polynomial

$$\ell_a(x_1, x_2) = x_2 - 3a^2x_1 + 2a^3.$$

Direct computation now shows

$$\begin{aligned}\ell_a &= x_2 - 3a^2x_1 + 2a^3 \\ &= x_1^3 - 3a^2x_1 + 2a^3 + (x_2 - x_1^3) \\ &= (x_1 - a)^2x_1 + 2a\big(x_1 - a\big)^2 + (x_2 - x_1^3) \\ &= 2a\big(x_1 - a\big)^2 + p_1 + (x_1 - a)^2p_2 \in \mathrm{QM}_{[3]}(p).\end{aligned}$$

Exercise 3.22
Compute the first and second Lasserre-Parrilo relaxation in Example 3.21.

▶ **Remark 3.23** Sometimes a slightly different version of the Lasserre-Parrilo relaxation is used, especially in the context of theta bodies (see Remark 3.25 below). First recall that $\mathrm{QM}_{[d]}(p)$ (and thus also $\mathrm{clos}(\mathrm{QM}_{[d]}(p))$) is a spectrahedral shadow, by Proposition 3.14 and Theorem 3.5 (10). So the sets

$$\mathbb{R}[x]_1 \cap \mathrm{QM}_{[d]}(p) \quad \text{and} \quad \mathbb{R}[x]_1 \cap \mathrm{clos}(\mathrm{QM}_{[d]}(p))$$

of linear polynomials from $\mathrm{QM}_{[d]}(p)$ and its closure are also spectrahedral shadows. Since, by Proposition A.19, the truncated quadratic module is often closed anyway, we will now just work with the closure (but of course everything works with $\mathrm{QM}_{[d]}(p)$, as well). The set

$$\widetilde{\mathcal{R}}_{[d]}(p) := \big\{u \in \mathbb{R}^n \mid \ell(u) \geqslant 0 \,\forall \ell \in \mathbb{R}[x]_1 \cap \mathrm{clos}(\mathrm{QM}_{[d]}(p))\big\} \tag{3.3}$$

is easily checked to be a spectrahedral shadow again. In fact, it is the inverse image of the dual of $\mathbb{R}[x]_1 \cap \mathrm{clos}(\mathrm{QM}_{[d]}(p))$, under the canonical affine embedding of $\mathbb{R}^n$ into the dual space of $\mathbb{R}[x]_1$.
One advantage of this approach is that $\widetilde{\mathcal{R}}_{[d]}(p)$ is defined by a family of linear inequalities. In particular, it is always closed. On the other hand, our earlier defined $\mathcal{R}_{[d]}(p)$ is, by definition, an explicit projection of a spectrahedron, whereas for $\widetilde{\mathcal{R}}_{[d]}(p)$ the semidefinite description is much less explicitly given. The relation between the two notions is discussed in Exercise 3.24.

Exercise 3.24
Show that $\widetilde{\mathcal{R}}_{[d]}(p)$, as defined in (3.3), coincides with the closure of $\mathcal{R}_{[d]}(p)$.

▶ **Remark 3.25 (Theta Bodies)** A construction closely related to the Lasserre-Parrilo relaxation is the *theta body* construction. A first version was used by Lovasz already in 1979 [62], a general treatment can be found in [32]. Theta bodies are mostly used for semidefinite relaxations of hard combinatorial optimization problems. It turns out that they are equivalent to Lasserre-Parrilo relaxations for real varieties, under some mild assumptions. This is what we will now explain.
Let $I \subseteq \mathbb{R}[x]$ be an ideal and

$$V = \mathcal{V}_{\mathbb{R}}(I) = \{u \in \mathbb{R}^n \mid p(u) = 0 \; \forall p \in I\}$$

the set of real points of the associated affine variety. For $d \geqslant 0$, we define

$$\mathrm{SOS}_{[2d]}(I) := \Sigma_{2d} + I \subseteq \mathbb{R}[x]$$

as the set of those polynomials that are sums of squares of degree $2d$ modulo the ideal I. The *d-th theta body of* I is then defined as

$$\Theta_{[d]}(I) := \left\{u \in \mathbb{R}^n \mid \ell(u) \geqslant 0 \; \forall \ell \in \mathbb{R}[x]_1 \cap \mathrm{SOS}_{[2d]}(I)\right\}.$$

The $\Theta_{[d]}(I)$ thus provide a decreasing outer approximation of the closed convex hull of V by closed sets. To compare theta bodies with Lasserre-Parrilo relaxations, we first note that V is also a basic closed semialgebraic set. If $I = (p_1, \ldots, p_r)$, then clearly

$$V = \mathcal{S}(p_1, -p_1, \ldots, p_r, -p_r).$$

Since every polynomial q is a difference of two sums of squares (even of degree at most $\deg(q) + 1$, see Exercise 3.27), we obtain

$$\mathrm{QM}(p_1, -p_1, \ldots, p_r, -p_r) = \Sigma + I,$$

where Σ denotes the set of all sums of squares of polynomials. However, there is a slight difference between $\mathrm{QM}_{[2d]}(p_1, -p_1, \ldots, p_r, -p_r)$ and $\mathrm{SOS}_{[2d]}(I)$. But for the right choice of generators $p_1, \ldots, p_r$ we indeed have

$$\begin{aligned} \mathrm{QM}_{[2d]}(p_1, -p_1, \ldots, p_r, -p_r) &\subseteq \mathrm{SOS}_{[2d]}(I) \cap \mathbb{R}[x]_{2d} \\ &\subseteq \mathrm{QM}_{[2d+1]}(p_1, -p_1, \ldots, p_r, -p_r), \end{aligned}$$

where the first inclusion is obvious. For the second assume that

$$p = \sigma + q$$

for some $\sigma \in \Sigma_{2d}$, $q \in I$, and $\deg(p) \leqslant 2d$. We get $q \in I \cap \mathbb{R}[x]_{2d}$ and can find a representation

$$q = \sum_i q_i p_i$$

with certain degree bounds on the terms $q_i p_i$, depending only on d and $p_1, \ldots, p_r$. This is due to the fact that $I \cap \mathbb{R}[x]_{2d}$ is a finite-dimensional vector space, and thus has a finite basis. But by choosing the generators of I well, one can in fact get explicit control over these degrees. For example, if $p_1, \ldots, p_r$ is a Gröbner basis of I with respect to some degree-compatible monomial order, we obtain a representation with $\deg(q_i p_i) \leqslant 2d$ for all i. Now writing each q_i as a difference of two sums of squares as in Exercise 3.27, the representation $p = \sigma + q$ turns into a representation from $\mathrm{QM}_{[2d+1]}(p_1, -p_1, \ldots, p_r, -p_r)$. We thus immediately obtain

$$\widetilde{\mathcal{R}}_{[2d+1]}(p_1, -p_1, \ldots, p_r, -p_r) \subseteq \Theta_{[d]}(I) \subseteq \widetilde{\mathcal{R}}_{[2d]}(p_1, -p_1, \ldots, p_r, -p_r)$$

where we use the Lasserre-Parrilo relaxation described in Remark 3.23. To be precise, the inclusion on the right is only valid if $\mathrm{QM}_{[2d]}(p_1, -p_1, \ldots, p_r, -p_r)$ or $\mathrm{SOS}_{[2d]}(I) \cap \mathbb{R}[x]_{2d}$ is closed, since we have used the closure in Construction (3.3) from Remark 3.23. If I is a real radical ideal, then $\mathrm{SOS}_{[2d]}(I) \cap \mathbb{R}[x]_{2d}$ is closed, see for example [82], Proposition 2.6 (b).
Summarizing, if $p_1, \ldots, p_r$ form a Gröbner basis (with respect to a degree compatible monomial order) of the real radical ideal I, then theta bodies and Lasserre-Parrilo relaxations are nested in between each other, and can therefore be considered equivalent relaxations of the set $\mathrm{conv}(\mathcal{V}_{\mathbb{R}}(I))$.

Example 3.26
Consider the full 0/1-hypercube

$$C := \{0, 1\}^n \subseteq \mathbb{R}^n.$$

We have $C = \mathcal{V}_{\mathbb{R}}(I)$ with $I = (x_1 - x_1^2, \ldots, x_n - x_n^2)$. From

$$x_i = x_i^2 + (x_i - x_i^2) \text{ and } (1 - x_i) = (1 - x_i)^2 + (x_i - x_i^2)$$

we obtain $x_i, 1 - x_i \in \mathrm{SOS}_2(I)$ and thus $\Theta_1(I) = \mathrm{conv}(C) = [0, 1]^n$.

Exercise 3.27
Show $\mathbb{R}[x]_{2d} = \Sigma_{2d} - \Sigma_{2d}$, i.e., every polynomial can be written as a difference of two sums of squares of the same degree if the degree is even, and one more if the degree is odd.
Hint: Show that $\Sigma_{2d} - \Sigma_{2d}$ is a vector space, then work with monomials.

3.4 Convex Hulls of Curves

In this chapter we discuss Scheiderer's positive solution of the Helton-Nie conjecture in dimension two [98]: Every convex semialgebraic subset of the plane is a spectrahedral shadow. This is obtained as a consequence of a stronger result, namely the stability of quadratic modules defining 1-dimensional bounded sets.

We recall the most important notions and results on affine varieties: To an affine variety

$$V = \mathcal{V}_{\mathbb{C}}(I) \subseteq \mathbb{C}^n$$

defined over $\mathbb{R}$ by an ideal $I \subseteq \mathbb{R}[x]$, there corresponds the coordinate ring

$$\mathbb{R}[V] := \mathbb{R}[x]/\mathcal{I}(V)$$

where $\mathcal{I}(V) = \sqrt{I}$ is the vanishing ideal. The coordinate ring is a finitely-generated and reduced $\mathbb{R}$-algebra. The variety V is often treated as an abstract object encoded in $\mathbb{R}[V]$, independent of the choice of coordinates, i.e., the choice of the surjection $\mathbb{R}[x] \to \mathbb{R}[V]$. For example, V can be recovered as the set of all algebra homomorphisms from $\mathbb{R}[V]$ to $\mathbb{C}$. Recall also that a *morphism*

$$\varphi\colon V \to W$$

between two affine varieties V and W over $\mathbb{R}$ is simply a real polynomial map from V to W, with respect to some embedding of V and W into affine space. Such a morphism induces a homomorphism

$$\varphi^*\colon \mathbb{R}[W] \to \mathbb{R}[V]$$

of $\mathbb{R}$-algebras, given by $p \mapsto p \circ \varphi$, and each algebra homomorphism comes from a morphism of varieties.

If Q is a finitely generated quadratic module in $\mathbb{R}[V]$, we want to make sense of the notion of stability for Q (see Sect. A.2 for details on stability). With the notation from the last section, stability of a quadratic module means that for every d there exists some e such that

$$\mathrm{QM}_d(p) \subseteq \mathrm{QM}_{[e]}(p).$$

Informally, stability expresses the fact that in sums-of-squares representation of elements we have degree-bounds on the single terms in the representation. The problem here is that we have no well-defined notion of degree for elements in $\mathbb{R}[V]$. There are two solutions: Fixing coordinates, choosing $p_1, \ldots, p_r \in \mathbb{R}[x]$ with $Q = \mathrm{QM}(\overline{p}_1, \ldots, \overline{p}_r)$, and generators $\mathcal{I}(V) = (q_1, \ldots, q_s)$ of the vanishing ideal, we can consider the quadratic module

$$Q_0 = \mathrm{QM}(p_1, \ldots, p_r, \pm q_1, \ldots, \pm q_s)$$

in $\mathbb{R}[x]$, which is just the preimage of Q in $\mathbb{R}[x]$ under the residue map $\mathbb{R}[x] \to \mathbb{R}[x]/\mathcal{I}(V)$ (c.f. proof of Corollary A.26). Then we say that Q is stable if and only if Q_0 is. We would have to show that this does not depend on the choice of coordinates.

More elegantly, we can just eliminate the notion of degree from the definition of stability: Let A be any commutative $\mathbb{R}$-algebra and $Q = \mathrm{QM}(p)$ a finitely generated quadratic module in A. Given a linear subspace W of A, we write

$$\mathrm{QM}_{[W]}(p) = \left\{ \sum_{i=0}^{r} \sigma_i p_i \mid \sigma_i = \sum_j g_{ij}^2 \text{ with } g_{ij} \in W \text{ for all } i, j \right\}$$

and say that Q is *stable* if for every finite-dimensional subspace U of A there exists a finite-dimensional subspace W of A, such that

$$\mathrm{QM}(p) \cap U \subseteq \mathrm{QM}_{[W]}(p).$$

Note that the earlier definition of $\mathrm{QM}_{[d]}(p)$ and the new one of $\mathrm{QM}_{[W]}(p)$ are slightly different in flavour, since the first imposes some degree-restriction on the terms $\sigma_i p_i$, whereas the latter imposes it on the single g_{ij}. However, if the degree of a sum

$$\sigma_i p_i = \left(\sum_j g_{ij}^2 \right) p_i$$

in the polynomial ring is bounded by some number d, we immediately get that the degree of each g_{ij} is bounded by $\frac{1}{2}(d - \deg(p_i))$ (compare to Exercise 3.13). So the two notions of truncations are indeed equivalent. In particular, the new definition of stability agrees with the above one for the polynomial ring, and it is independent of the choice of generators.

Exercise 3.28
Show that the notion of stability of a quadratic module $\mathrm{QM}(p)$ in a commutative $\mathbb{R}$-algebra A, as just defined, does not depend on the choice of generators p.

Now let V be an affine $\mathbb{R}$-variety. We write $V(\mathbb{R})$ for the set of real points of V. If $V = \mathcal{V}_{\mathbb{C}}(I)$ is an explicit embedding of V, then clearly $V(\mathbb{R}) = \mathcal{V}_{\mathbb{R}}(I)$ is the set of real zeros of the elements from I (otherwise $V(\mathbb{R})$ can for example be defined as the set of real-valued algebra homomorphisms on $\mathbb{R}[V]$). Given a subset S of $V(\mathbb{R})$, the preordering

$$\mathcal{P}(S) := \{q \in \mathbb{R}[V] \mid q_{|S} \geq 0\}$$

is called the *saturated preordering of* S. A finitely generated quadratic module $Q = \mathrm{QM}(p)$, with $p = (p_1, \ldots, p_r)$ and all $p_i \in \mathbb{R}[V]$, is called *saturated* if $Q = \mathcal{P}(\mathcal{S}(p))$, where

$$\mathcal{S}(p) := \{u \in V(\mathbb{R}) \mid p_1(u) \geqslant 0, \ldots, p_r(u) \geqslant 0\}$$

is the basic closed semialgebraic set defined by p within $V(\mathbb{R})$. As explained in Sect. A.2, the saturated preordering $\mathcal{P}(S)$ of a semialgebraic set S is never finitely generated if $\dim(S) \geqslant 3$ (i.e., never of the form $\mathrm{QM}(p)$ for some finite tuple p of generators). On the other hand, it is finitely generated for any subset of the line (see Example A.9). It turns out that this is also true for compact subsets of smooth algebraic curves. Recall that a variety is called *smooth* if it does not possess any singular points, neither real nor complex. For an affine hypersurface $\mathcal{V}_{\mathbb{C}}(q)$, with $q \in \mathbb{R}[x]$ irreducible, this just means $\nabla q(u) \neq 0$ for all $u \in \mathcal{V}_{\mathbb{C}}(q)$. The affine variety V is called an affine *curve* if all of its irreducible components are one-dimensional (which means that every prime ideal in the coordinate ring is either minimal or maximal).

The following result by Scheiderer [94] is the first main ingredient for the result on planar spectrahedral shadows.

Theorem 3.29
Let Z be a smooth affine curve over $\mathbb{R}$, and $S \subseteq Z(\mathbb{R})$ a compact semialgebraic subset. Then the saturated preordering $\mathcal{P}(S) \subseteq \mathbb{R}[Z]$ is finitely generated (as a quadratic module).

Since the saturated preordering $\mathcal{P}(S)$ is finitely generated, it makes sense to ask whether it is stable. To show that it is, we will consider its behavior under real closed extensions of $\mathbb{R}$, see Sect. A.4 for a detailed explanation of this connection, and for example [83] for more information on general real closed fields and some model theory. Let $\mathrm{R}/\mathbb{R}$ be a real closed field extension and let $\mathrm{C} = \mathrm{R}[\sqrt{-1}]$ be its algebraic closure. If

$$V = \mathcal{V}_{\mathbb{C}}(q_1, \ldots, q_s) \subseteq \mathbb{C}^n$$

is an affine variety defined over $\mathbb{R}$ by $q_1, \ldots, q_s \in \mathbb{R}[x]$ with coordinate ring

$$\mathbb{R}[V] = \mathbb{R}[x]/\sqrt{(q_1, \ldots, q_s)},$$

then we can regard

$$\mathcal{V}_{\mathrm{C}}(q_1, \ldots, q_s) \subseteq \mathrm{C}^n$$

as an affine variety defined over R with coordinate ring $\mathrm{R}[V] = \mathrm{R}[x]/\sqrt{(q_1, \ldots, q_s)}$. One can show that there is a canonical isomorphism

$$\mathrm{R}[V] \cong \mathbb{R}[V] \otimes_{\mathbb{R}} \mathrm{R},$$

which yields a more intrinsic description of $\mathrm{R}[V]$. The main technical result of Scheiderer in [98] is the following.

Theorem 3.30
Let Z be a smooth affine curve over $\mathbb{R}$, let $S \subseteq Z(\mathbb{R})$ be a compact semialgebraic subset and let $P = \mathcal{P}(S) \subseteq \mathbb{R}[Z]$ be the saturated preordering of S. Then for any real closed field extension $\mathrm{R}/\mathbb{R}$, the preordering P_{R} generated by P in $\mathrm{R}[Z]$ is again saturated.

Note that if P is generated by the tuple p as a quadratic module in $\mathbb{R}[Z]$, then P_R is generated by p as a quadratic module in $\mathrm{R}[Z]$. Also note that the semialgebraic subset $\mathcal{S}(P_{\mathrm{R}})$ of $Z(\mathrm{R})$ is the base extension $S(\mathrm{R})$ of S (i.e., the set defined by the same equations as S, for example by p), hence the theorem says exactly that

$$\mathrm{QM}(p) = \mathcal{P}(S(\mathrm{R})) \subseteq \mathrm{R}[Z].$$

We will discuss this result and its proof in Sect. A.5 of the appendix. For now, we will just apply it to show stability of $\mathcal{P}(S)$.

Corollary 3.31
For Z and S as above, the saturated preordering $\mathcal{P}(S)$ in $\mathbb{R}[Z]$ is finitely generated (as a quadratic module) and stable.

Proof We only need to show stability of P. This comes as an application of Proposition A.39, which also holds for stability of quadratic modules in general $\mathbb{R}$-algebras, with the same proof. That P_{R} is saturated implies that the intersection of P_{R} with any finite-dimensional subspace of $\mathrm{R}[Z] = \mathbb{R}[Z] \otimes_{\mathbb{R}} \mathrm{R}$ is semialgebraic

over R (the proof is completely analogues to that of Proposition A.17). This holds for any real closed R/$\mathbb{R}$, so P is stable by Proposition A.39. □

We now apply this result in the context of the Lasserre-Parrilo relaxation. Simply speaking, we would like to show that the convex hull of a compact 1-dimensional set in $\mathbb{R}^n$ possesses an exact Lasserre-Parrilo relaxation. This is easy to deduce from Corollary 3.31, but only for subsets of smooth curves. To avoid this assumption, we need a few more preparations. Since stability means that we have degree bounds for *all* non-negative polynomials, not only linear ones, we have more flexibility and coordinate-independence, which can be exploited as follows.

Proposition 3.32
Let V be an affine $\mathbb{R}$-variety, and $S \subseteq V(\mathbb{R})$ be a semialgebraic set whose saturated preordering $\mathcal{P}(S)$ in $\mathbb{R}[V]$ is finitely generated and stable. Then for any morphism of varieties

$$\varphi\colon V(\mathbb{R}) \to \mathbb{R}^n$$

the set clos (conv($\varphi(S)$)) *is a spectrahedral shadow.*

Proof Let $\varphi^*\colon \mathbb{R}[x] \to \mathbb{R}[V]$ be the homomorphism of $\mathbb{R}$-algebras induced by φ, and fix generators $p_1, \ldots, p_r \in \mathbb{R}[V]$ of the saturated preordering of S:

$$\mathcal{P}(S) = \mathrm{QM}(p).$$

Since $\mathcal{P}(S)$ is stable, there exists a finite-dimensional subspace W of $\mathbb{R}[V]$ such that

$$\varphi^*(\mathbb{R}[x]_1) \cap \mathcal{P}(S) \subseteq \mathrm{QM}_{[W]}(p).$$

Choose another finite-dimensional subspace U of $\mathbb{R}[V]$, which contains both $\mathrm{QM}_{[W]}(p)$ and the space $\varphi^*(\mathbb{R}[x]_1)$. Now just as in Proposition 3.14, the convex set

$$\mathrm{QM}_{[W]}(p)' := \{L \in U^* \mid L|_{\mathrm{QM}_{[W]}(p)} \geqslant 0 \text{ and } L(1) = 1\}$$

is a spectrahedron, and we can thus define the generalized Lasserre-Parrilo relaxation $\mathcal{R}_{[W]} \subseteq \mathbb{R}^n$ as the image of $\mathrm{QM}_{[W]}(p)'$ under the linear map

$$\pi : L \mapsto \big(L(\varphi^*(x_1)), \ldots, L(\varphi^*(x_n))\big).$$

For $u \in S$, the functional $L_u \in \mathbb{R}[V]^*$ given by evaluation in u is contained in $\mathrm{QM}_{[W]}(p)'$, which implies

$$\mathrm{conv}(\varphi(S)) \subseteq \mathcal{R}_{[W]}.$$

As in the proof of Proposition 3.18, suppose we had $u \in \mathcal{R}_{[W]}$, say $u = \pi(L)$ for $L \in \mathrm{QM}_{[W]}(p)'$, but $u \notin \mathrm{clos}\big(\mathrm{conv}(\varphi(S))\big)$. Then we can pick $\ell \in \mathbb{R}[x]_1$ with $\ell|_{\varphi(S)} \geqslant 0$ and $\ell(u) < 0$, using the Separation Theorem B.4. We conclude $L(\varphi^*(\ell)) = \ell(u) < 0$, hence $\varphi^*(\ell) \notin \mathrm{QM}_{[W]}(p)$. On the other hand, from $\ell \geqslant 0$ on $\varphi(S)$ we obtain $\varphi^*(\ell) \geqslant 0$ on S, i.e.,

$$\varphi^*(\ell) \in \varphi^*(\mathbb{R}[x]_1) \cap \mathcal{P}(S) \subseteq \mathrm{QM}_{[W]}(p),$$

a contradiction. Thus $\mathrm{clos}(\mathrm{conv}(\varphi(S))) = \mathrm{clos}(\mathcal{R}_{[W]})$ is a spectrahedral shadow (using Proposition 3.5 (10)). □

▶ **Remark 3.33** In spite of its apparent generality, the hypotheses of this proposition can only be satisfied if V has dimension at most 1, by the main result of [95].

Now if Z is any affine curve over $\mathbb{R}$, possibly singular, we can get rid of the singularities by passing to the *normalisation* of Z. If Z is irreducible, this is the curve corresponding to the integral closure of the domain $\mathbb{R}[Z]$ in its quotient field $\mathbb{R}(Z)$. This integral closure is again a finitely generated $\mathbb{R}$-algebra, and therefore corresponds to an affine curve $\widetilde{Z}$ over $\mathbb{R}$. The inclusion $\mathbb{R}[Z] \subseteq \mathbb{R}[\widetilde{Z}]$ corresponds to a morphism $\widetilde{Z} \to Z$ of curves, which is an isomorphism everywhere except over the singular points of Z. Since $\mathbb{R}[\widetilde{Z}]$ is integrally closed, the curve $\widetilde{Z}$ is smooth.[1] The map $\widetilde{Z} \to Z$ is surjective, but its restriction $\widetilde{Z}(\mathbb{R}) \to Z(\mathbb{R})$ to real points may be non-surjective. In fact, a point $u \in Z(\mathbb{R})$ is outside the image of $\widetilde{Z}(\mathbb{R})$ if and only if it is an isolated singularity of Z, in which case it is the image of (potentially several) pairs of complex-conjugate points in $\widetilde{Z}$. The map $\widetilde{Z}(\mathbb{R}) \to Z(\mathbb{R})$ is however proper in the euclidean topology, i.e., it is closed with compact fibres.

If Z has several irreducible components, say $Z = Z_1 \cup \cdots \cup Z_k$, we can apply the normalisation separately to each component and obtain the normalisation $\widetilde{Z} = \widetilde{Z}_1 \sqcup \cdots \sqcup \widetilde{Z}_k$ of Z (where $\sqcup$ denotes the disjoint union) with coordinate ring

$$\mathbb{R}[\widetilde{Z}] = \mathbb{R}[\widetilde{Z}_1] \times \cdots \times \mathbb{R}[\widetilde{Z}_k].$$

With all of this we are ready for the first main result on curves and spectrahedral shadows (see Fig. 3.4 for an example).

[1] In general, the singular locus of an irreducible affine variety V with integrally closed coordinate ring has codimension at least 2 in V. Thus if V is a curve, it must be smooth.



Fig. 3.4 Convex hull of moment curve $r \mapsto (r, r^2, r^4)$ with $r \in [0, 1]$

Theorem 3.34
Every compact convex semialgebraic set whose set of extreme points has dimension at most 1 *is a spectrahedral shadow.*

Proof Denote the compact convex set by C. Let S be the closure of the set of extreme points of C, a compact semialgebraic set of dimension at most 1 by hypothesis. Let Z be the Zariski-closure of S, an affine $\mathbb{R}$-variety of dimension at most 1. Let $Z_1, \ldots, Z_k$ be the irreducible components of Z, let $\widetilde{Z}_i \to Z_i$ be the normalisation of each, and let $\widetilde{Z} \to Z$ be the normalisation of Z, given by $\widetilde{Z} = \widetilde{Z}_1 \sqcup \cdots \sqcup \widetilde{Z}_k$.

Now let $u_1, \ldots, u_l \in S$ be the isolated singularities of Z contained in S. As explained above, $u_1, \ldots, u_l$ do not lie in the image of $\widetilde{Z}(\mathbb{R}) \to Z(\mathbb{R})$. Consider the abstract variety

$$Y = \widetilde{Z} \sqcup p_1 \sqcup \cdots \sqcup p_l$$

where each p_i is a real point. Its real coordinate ring is the $\mathbb{R}$-algebra

$$\mathbb{R}[Y] = \mathbb{R}[\widetilde{Z}_1] \times \cdots \times \mathbb{R}[\widetilde{Z}_k] \times \underbrace{\mathbb{R} \times \cdots \times \mathbb{R}}_{l \text{ times}}.$$

The variety Y comes with a natural morphism $\varphi\colon Y \to Z$ which is the normalisation on $\widetilde{Z}$ and sends each p_i to u_i. Let $\widetilde{S}$ be the preimage of S in $Y(\mathbb{R})$ under φ. Then $\widetilde{S}$ is again compact, and by Corollary 3.31 the saturated preordering $\mathcal{P}(\widetilde{S}) \subseteq \mathbb{R}[Y]$ is finitely generated and stable. By Proposition 3.32, this implies that

$$\operatorname{clos}(\operatorname{conv}(\varphi(\widetilde{S}))) = \operatorname{clos}(\operatorname{conv}(S))$$

is a spectrahedral shadow, and it coincides with C by the Krein-Milman Theorem (see for example [3], Theorem 3.3). □

Corollary 3.35
Every compact convex semialgebraic subset of $\mathbb{R}^2$ is a spectrahedral shadow.

Proof For a convex semialgebraic set in the plane, the extreme points form a semialgebraic set of dimension at most 1, so Theorem 3.34 applies. □

Finally, we show how to deduce the full two-dimensional Helton-Nie conjecture from the above result. This amounts to dealing first with closed but not necessarily bounded semialgebraic subsets of the plane, and then performing some surgery on the boundary in the general case.

Theorem 3.36
Every convex semialgebraic subset of $\mathbb{R}^2$ is a spectrahedral shadow.

Proof Let S be such a set. Suppose first that S is closed, and consider the homogenization

$$C := \operatorname{clos}(\operatorname{cone}(S \times \{1\})) \subseteq \mathbb{R}^3.$$

As usual, since $S \times \{1\} = C \cap \left(\mathbb{R}^2 \times \{1\}\right)$, and by using Theorem 3.5 (3), it is enough to show that C is a spectrahedral shadow. Let U be the lineality space of C (c.f. Exercise B.3). Then

$$C = (C \cap U^{\perp}) + U$$

and $C \cap U^{\perp}$ is pointed, so using Theorem 3.5 (4) we may assume that C is pointed. In that case, by Proposition B.1, there exists an affine hyperplane $H \subseteq \mathbb{R}^3$, such that $C = \operatorname{cone}(C \cap H)$ and $C \cap H$ is compact. Now $C \cap H$ is a spectrahedral shadow by Corollary 3.35, hence so is C by Theorem 3.5 (5).

For the general case, we may assume as usual that $\operatorname{int}(S) \neq \emptyset$. We know that $\operatorname{clos}(S)$ is a spectrahedral shadow and $B = \operatorname{clos}(S) \setminus S$ is a certain one-dimensional semialgebraic subset of the boundary of $\operatorname{clos}(S)$. Let $\mathcal{F}$ be the finite set of one-dimensional faces of $\operatorname{clos}(S)$. We decompose B as follows:

$$B_0 = \text{the relative interior of } B \cap \operatorname{Ex}(\operatorname{clos}(S)) \text{ inside } \partial(S)$$
$$B_F = F \cap B \text{ for } F \in \mathcal{F}$$

Then B is the union of $B_0, \bigcup_{F \in \mathcal{F}} B_F$, and finitely-many extreme points $u_1, \ldots, u_k$ of $\operatorname{clos}(S)$. For $F \in \mathcal{F}$, let H_F be the open halfplane with $\operatorname{int}(S) \subseteq H_F$, $F \cap H_F = \emptyset$,

so that $F \subseteq \partial H_F$. Put

$$\begin{aligned} S_0 &= \operatorname{clos}(S) \setminus B_0 \\ S_F &= H_F \cup (F \cap S) = H_F \cup (\operatorname{clos}(H_F) \cap S) \\ S_u &= \operatorname{clos}(S) \setminus \{u\} \text{ for } u \in \operatorname{Ex}(\operatorname{clos}(S)). \end{aligned}$$

By construction, we now have

$$S = S_0 \cap \bigcap_{F \in \mathcal{F}} S_F \cap S_{u_1} \cap \cdots \cap S_{u_k}.$$

It therefore suffices to show that each of the finitely many sets appearing in this intersection is a spectrahedral shadow. The sets S_F are the union of an open halfplane and an interval and are therefore spectrahedral shadows (either by a direct argument, or using Theorem 3.5 (6)). For S_{u_i}, see Exercise 3.11. To deal with S_0, let

$$T = \operatorname{clos}\big(\operatorname{conv}(\partial S \setminus B_0)\big).$$

Since T is closed, convex and semialgebraic, we already know that T is a spectrahedral shadow. Now let $\mathcal{F}_T$ be the set of all faces F of $\operatorname{clos}(S)$ such that $F \cap T \neq \emptyset$ and consider

$$T \leftarrow\!\!\!\rho \operatorname{clos}(S) = \bigcup_{F \in \mathcal{F}_T} \operatorname{relint}(F).$$

We claim that $T \leftarrow\!\!\!\rho \operatorname{clos}(S) = S_0$, showing that S_0 is a spectrahedral shadow, by Theorem 3.9. To see this let $u \in S_0$. If u is an interior point of $\operatorname{clos}(S)$, then clearly $u \in T \leftarrow\!\!\!\rho \operatorname{clos}(S)$ (unless $T = \emptyset$, which is a trivial case, since it implies that S is the interior of $\operatorname{clos}(S)$). If $u_0 \in \partial S$, then it is not in B_0, hence it lies in T and therefore in $T \leftarrow\!\!\!\rho \operatorname{clos}(S)$. Conversely, given u in $\operatorname{clos}(S)$ but not in S_0, then it is a point of B_0. By the definition of B_0, this implies that $\{u\}$ is the unique face of $\operatorname{clos}(S)$ containing u in its relative interior. Since $u \notin T$, this means $u \notin T \leftarrow\!\!\!\rho \operatorname{clos}(S)$. This completes the proof. □

▶ **Remark 3.37** Using some more convex geometry, one can refine the argument in the first part of the proof of Theorem 3.36 to show that the closure of the convex hull of any semialgebraic set of dimension at most 1 is a spectrahedral shadow, see [98] Theorem 6.1.

3.5 General Exactness Results: The Helton-Nie Theorems

After having settled the two-dimensional case completely, we now turn to positive results in higher dimensions. As we will see in Sect. 3.8, not every convex semialgebraic set is a spectrahedral shadow. However, the results of Helton and

Nie, which we present in this section, state sufficient conditions for semialgebraic sets to be spectrahedral shadows. These conditions can be applied to sets of arbitrary dimension, and they even imply the exactness of a Lasserre-Parrilo relaxation. The results are all from [41] and [42]. We begin with the easiest case.

Theorem 3.38
Let $S = \mathcal{S}(p)$ be a bounded convex set with non-empty interior. Suppose that the following condition is satisfied:

[SOS-Concavity] *The Hessian matrices $-\nabla^2 p_i \in \mathrm{Sym}_n \mathbb{R}[x]$ are sums of squares for $i = 1, \ldots, r$.*

Then S possesses an exact Lasserre-Parrilo relaxation with respect to p, in fact

$$S = \mathcal{R}_{[d]}(p)$$

where $d = \max\{\deg(p_i) \mid i = 1, \ldots, r\}$. In particular, S is a spectrahedral shadow.

For the proof we need the following important lemma:

Lemma 3.39
Let $F \in \mathrm{Sym}_k \mathbb{R}[x]$ be a matrix polynomial which is a sum of squares, and fix $u \in \mathbb{R}^n$. Then the matrix polynomial

$$G_u(x) := \int_0^1 \int_0^t F(u + s(x - u))\mathrm{d}s\mathrm{d}t$$

(where the integration is carried out entry- and coefficientwise) is again a sum of squares in $\mathrm{Sym}_k \mathbb{R}[x]$.

Proof By Exercise A.29, a matrix polynomial $G \in \mathrm{Sym}_k \mathbb{R}[x]$ is a sum of squares if and only if the polynomial

$$yGy^T \in \mathbb{R}[x, y],$$

which is homogeneous of degree 2 in $y = (y_1, \ldots, y_k)$, is a sum of squares. Thus, by hypothesis, the polynomial

$$p := yF(u + s(x - u))y^T \in \mathbb{R}[x, y, s]$$

is a sum of squares, and therefore possesses a positive semidefinite Gram matrix (recall Example 2.2 (4) for the definition). This means

$$p = (B\mathfrak{m})^T B\mathfrak{m}$$

where $\mathfrak{m}$ is a column vector of monomials in x, y, s, and B is a suitable real matrix. Now write $\mathfrak{m} = U(s) \cdot \overline{\mathfrak{m}}$, where $\overline{\mathfrak{m}}$ is a vector of monomials in x, y and U is a univariate matrix polynomial in the variable s of appropriate size.[2] So putting

$$A(s) := \big(BU(s)\big)^T BU(s)$$

we find that $p = \overline{\mathfrak{m}}^T A(s)\overline{\mathfrak{m}}$, and hence

$$y^T G_u y = \int_0^1\int_0^t p(x, y, s)\mathrm{d}s\mathrm{d}t = \overline{\mathfrak{m}}^T \underbrace{\left(\int_0^1\int_0^t A(s)\mathrm{d}s\mathrm{d}t\right)}_{\succcurlyeq 0} \overline{\mathfrak{m}},$$

showing that G_u is a sum of squares, as claimed. □

Proof of Theorem 3.38 Let $\ell \in \mathbb{R}[x]_1$ with $\ell|_S \geqslant 0$, and let $u \in S$ be a minimizer of ℓ. Since the matrix polynomials $-\nabla^2 p_i$ are sums of squares, the polynomials p_i are concave (Exercise B.24). Then Corollary B.28 guarantees the existence of Lagrange multipliers for ℓ in u, so that we have

$$\begin{aligned} \nabla\ell(u) &= \sum_{i=1}^r \lambda_i \nabla p_i(u) \\ \lambda_i p_i(u) &= 0 \qquad\qquad \text{for all } i = 1, \ldots, r \end{aligned}$$

for certain $\lambda_1, \ldots, \lambda_r \geqslant 0$. It follows that the function

$$f_\ell := \ell - \ell(u) - \sum_{i=1}^r \lambda_i p_i$$

[2] To see that this is possible, let $\overline{\mathfrak{m}}$ contain all monomials in x, y up to degree d. Let $U(s)$ be the rectangular matrix $[I_N, sI_N, \cdots, s^d I_N]^T \in \mathrm{Mat}_{(d+1)N\times N}\,\mathbb{R}[s]$, where $N = \mathrm{length}(\overline{\mathfrak{m}})$. Then $U(s) \cdot \overline{\mathfrak{m}}$ contains all monomials up to degree d (and more) in x, y, s.

and its gradient both vanish at u, so the fundamental theorem of calculus implies

$$f_\ell = \int_0^1 \frac{\partial}{\partial t} f_\ell(u + t(x-u))\mathrm{d}t = \iint_{00}^{1t} \frac{\partial^2}{\partial s^2} f_\ell(u + s(x-u))\mathrm{d}s\mathrm{d}t = \tag{3.4}$$

$$= \sum_{i=1}^{r} \lambda_i \cdot (x-u)^T \Bigg(\underbrace{\iint_{00}^{1t} -\nabla^2 p_i(u + s(x-u))\mathrm{d}s\mathrm{d}t}_{=:A_u^{(i)}(x)} \Bigg)(x-u).$$

By Lemma 3.39, the matrix polynomials $A_u^{(i)}$ are sums of squares, and therefore

$$\ell = \ell(u) + \sum_{i=1}^{r} \lambda_i p_i + \sum_{i=1}^{r} \lambda_i (x-u)^T A_u^{(i)}(x-u)$$

is a represention of ℓ in $\mathrm{QM}_{[d]}(p)$ where $d = \max\{\deg(p_i) \mid i = 1, \ldots, r\}$. Note that the degree of the sum of squares on the very right is indeed at most d, which can either be seen by close inspection of its construction, or directly from the fact that ℓ has degree 1. So now the claim follows from Proposition 3.18. □

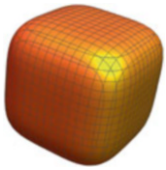
A polynomial p whose Hessian is a sum of squares is called *sos-convex*. By Exercise B.24, any such polynomial is convex. But as usual, the converse does not hold in general.

Exercise 3.40
Let $h \in \mathbb{R}[x_0, \ldots, x_n]$ be a homogeneous polynomial.

(a) Show that if h is convex, then h is nonnegative on $\mathbb{R}^{n+1}$.
(b) Show that if h is sos-convex, then h is a sum of squares.
(c) Give an example of a convex polynomial $p \in \mathbb{R}[x]$, such that its homogenization $x_0^{\deg p} p(x/x_0)$ is not convex.

For an example of a convex sum of squares that is not sos-convex, see [1]. There also exist convex homogeneous polynomials that are not sums of squares, even though not a single explicit example of such a polynomial is known (see [8]).

Example 3.41
We consider the n-dimensional TV-screen $S = \mathcal{S}(p) \subseteq \mathbb{R}^n$, defined by the polynomial $p = 1 - x_1^4 - \cdots - x_n^4$.



The Hessian of p is

$$\nabla^2 p = \begin{pmatrix} -12x_1^2 & & \\ & \ddots & \\ & & -12x_n^2 \end{pmatrix}$$

so p is obviously sos-concave. Thus Theorem 3.38 implies that the Lasserre-Parrilo relaxation of S with respect to p becomes exact in degree 4.

Exercise 3.42
Work out the construction in the proof of Theorem 3.38 for the TV-screen. In which degree does the Lasserre-Parrilo relaxation really become exact?

Next, we present a more sophisticated version of Theorem 3.38, in which the defining polynomials are not required to be sos-concave. Instead, we will assume that the concavity is strict, at least along the extreme part of the boundary, with a uniform lower bound on the curvature. The proof makes use of Putinar's Theorem for matrix polynomials, as proven in the appendix.

Theorem 3.43
Let p be an Archimedean description of the convex set S with non-empty interior. Suppose that for each $i = 1, \ldots, r$ the following condition is satisfied:

[Strict Concavity] *The function p_i is concave on S, and if $u \in S$ is in the closure of the set of extreme points of S at with p_i vanishes, then $\nabla^2 p_i(u)$ is negative definite.*

Then S possesses an exact Lasserre relaxation with respect to p. In particular, S is a spectrahedral shadow.

Proof For any $1 \leqslant i \leqslant r$, let Z_i be the closure of the set of extreme points of S at which p_i vanishes. For $u \in \mathbb{R}^n$ define

$$A_u^{(i)}(x) := \int_0^1 \int_0^t -\nabla^2 p_i(u + s(x-u))\mathrm{d}s\mathrm{d}t \in \mathrm{Sym}_n \mathbb{R}[x].$$

Since $-\nabla^2 p_i(u) > 0$ for all $u \in Z_i$, and $-\nabla^2 p_i(v) \succcurlyeq 0$ for all $v \in S$ by hypothesis, it follows that

$$A_u^{(i)}(v) > 0$$

for all $v \in S$ and $u \in Z_i$. Thus by compactness of S and Z_i, there exists $\delta > 0$ with

$$A_u^{(i)}(v) \succcurlyeq \delta I_n$$

for all $u \in Z_i$, $v \in S$. We may therefore apply Putinar's Theorem for matrix polynomials (Theorem A.35) and obtain representations

$$A_u^{(i)}(x) = \sum_{j=0}^{r} p_j S_{j,u}^{(i)}$$

where each $S_{j,u}^{(i)} \in \mathrm{Sym}_n \mathbb{R}[x]$ is a sum of squares of degree bounded by

$$\deg\left(S_{j,u}^{(i)}\right) \leqslant D(p, \deg(A_u^{(i)}), \|A_u^{(i)}\|, \delta).$$

Now we again use compactness of Z_i to make the bound independent of u. By taking $\|A^{(i)}\| = \max\{\|A_u^{(i)}\| \mid u \in Z_i\}$ and noting that $\deg(A_u^{(i)}) \leqslant \deg(p_i)$, we have

$$\deg\left(S_{j,u}^{(i)}\right) \leqslant D(p, \max\{\deg(p_i) \mid i = 1, \ldots, r\}, \|A^{(i)}\|, \delta).$$

Now, just as in the proof of Theorem 3.38, let $\ell \in \mathbb{R}[x]_1$ with $\ell|_S \geqslant 0$ and let $u \in S$ be a minimizer of ℓ, which we may assume to be an extreme point of S (see Exercise B.26). Again fix Lagrange multipliers $\lambda_1, \ldots, \lambda_r \geqslant 0$ for ℓ at u (Corollary B.28), so that

$$\nabla \ell(u) = \sum_{i=1}^{r} \lambda_i \nabla p_i(u)$$

$$\lambda_i p_i(u) = 0 \qquad \text{for all } i = 1, \ldots, r.$$

Proceeding as in the proof of Theorem 3.38, using identity (3.4) again, we obtain a representation

$$\ell = \ell(u) + \sum_{i=1}^{r} \lambda_i p_i + \sum_{i=1}^{r} \sum_{j=0}^{r} \left(\lambda_i (x-u)^T S_{j,u}^{(i)} (x-u)\right) p_j$$

in which the degrees are independent of u, and hence of ℓ. In view of Proposition 3.18, this finishes the proof. □

Since not every strictly concave polynomial is sos-concave, it is clear that Theorem 3.43 can be applied to examples in which Theorem 3.38 fails. In fact, one can even show that *most* concave polynomials (in a suitable sense) are not sos-concave (see [8]). But explicit examples of such polynomials (as in [1]) are not easy to come by.

On the other hand, the assumption that the defining polynomials p_i be strictly concave on $\mathcal{S}(p)$ is still quite restrictive. We can further weaken the hypotheses if we make use of our freedom in choosing the defining polynomials p of the basic closed set $\mathcal{S}(p)$. This needs some technical preliminaries.

Definition 3.44

A twice continuously differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is called *strictly quasi-concave* at a point $u \in \mathbb{R}^n$, if the Hessian $\nabla^2 f(u)$ is negative definite on the algebraic tangent space $\{v \in \mathbb{R}^n \mid \nabla f(u)^T v = 0\}$, i.e., if

$$\text{For all } v \in \mathbb{R}^n \setminus \{0\}: \quad \nabla f(u)^T v = 0 \implies v^T \left(\nabla^2 f(u)\right) v < 0.$$

The definition is simple enough, but not very intuitive. The more natural (but weaker condition) definition of non-strict quasi-concavity is treated in the following exercise.

Exercise 3.45

Let $S \subseteq \mathbb{R}^n$ be convex. A function $f : \mathbb{R}^n \to \mathbb{R}$ is called *quasi-concave* on S if all its superlevel sets

$$S_a = \{u \in S \mid f(u) \geqslant a\},$$

for $a \in \mathbb{R}$, are convex. Show that a strictly quasi-concave function on S, as defined above, is quasi-concave on S.

Example 3.46
The polynomial $p = x_1x_2$ is strictly quasi-concave on the open quadrant $(0, \infty) \times (0, \infty)$. Indeed, we compute $\nabla p = (x_2, x_1)^T$ and

$$\nabla^2 p = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Then given $0 \neq (u_1, u_2) \in \mathbb{R}^2$, we have $\nabla p(u_1, u_2)^\perp = \text{span}(-u_1, u_2)$, and the restriction of $\nabla^2 p$ to that line is $-2u_1u_2$, which is negative for $u_1, u_2 > 0$. On the other hand, $\nabla^2 p$ is constant and indefinite, so that p is not concave anywhere.

Lemma 3.47
A twice continuously differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is strictly quasi-concave at a point $u \in \mathbb{R}^n$, if and only if there exists $\mu \geqslant 0$ such that

$$\nabla^2 f(u) - \mu \cdot \nabla f(u) \nabla f(u)^T \prec 0.$$

Proof Put $v_0 = \nabla f(u)$, $A = \nabla^2 f(u)$, and suppose there exists μ as above. Then given $v \in \mathbb{R}^n$ with $v_0^T v = 0$, we have

$$v^T A v = v^T A v - \mu v^T v_0 v_0^T v < 0.$$

Conversely, suppose that f is strictly quasi-concave at u. If $v_0 = 0$, quasi-concavity implies $A \prec 0$ and there is nothing to show, so assume $v_0 \neq 0$. Since A is negative definite and hence non-degenerate as a bilinear form on the subspace

$$V := \text{span}(v_0)^\perp,$$

it admits an orthogonal complement, i.e., there exists $w \in \mathbb{R}^n$ such that

$$\mathbb{R}^n = V \oplus \text{span}(w) \text{ and } v^T A w = w^T A v = 0 \text{ for all } v \in V.$$

Then for any vector $v + \lambda w \in \mathbb{R}^n$, with $0 \neq v \in V$ and $\lambda \in \mathbb{R}$, and for any $\mu \in \mathbb{R}$, we compute

$$(v + \lambda w)^T (A - \mu v_0 v_0^T)(v + \lambda w) = \underbrace{v^T A v}_{<0} + \lambda^2 (w^T A w - \mu (w^T v_0)^2).$$

So we choose $\mu \geqslant 0$ such that $\mu (w^T v_0)^2 > w^T A w$ (note that $w^T v_0 \neq 0$, since $w \notin V$). □

Lemma 3.48
For any $\mu \geqslant 0$ there exists a univariate polynomial $g \in \mathbb{R}[t]$ that is a sum of squares and satisfies the following for all $a \in [-1, 1]$:

1. $g(a) > 0$
2. $g(a) + g'(a)a > 0$
3. $\frac{2g'(a)+g''(a)a}{g(a)+g'(a)a} \leqslant -\mu$.

Proof Exercise 3.49. □

Exercise 3.49
Give a proof of Lemma 3.48.
Hint: Show first that

$$f(a) := \frac{1 - e^{-(\mu+1)a}}{(\mu + 1)a}$$

defines a twice continuously differentiable function with the desired properties.

The next result explains how strictly quasi-concave defining polynomials can be changed to strictly concave polynomials (to which then Theorem 3.43 can be applied).

Proposition 3.50
Let $p = (p_1, \ldots, p_r)$ be an Archimedean description of S, and assume that the polynomials p_i are strictly quasi-concave on S. Then there exist $q_0, \ldots, q_r \in \mathrm{QM}(p)$, which are strictly concave on S (see Theorem 3.43), and which also provide an Archimedean description of S.

Proof Choose $\rho > 1$ with $S \subseteq B_{\rho-1}(0)$. After rescaling, we may assume $p_i(B_\rho(0)) \subseteq [-1, 1]$ for $i = 1, \ldots, r$. Now for $\mu \geqslant 0$ let $g \in \mathbb{R}[t]$ be as in Lemma 3.48, and put

$$q_0 := \rho^2 - \sum_{i=1}^n x_i^2 \qquad \text{and} \qquad q_i := p_i \cdot g(p_i) \text{ for } i = 1, \ldots, r.$$

Then $S \subseteq \mathcal{S}(q)$ is clear. Conversely, let $u \in \mathbb{R}^n \setminus S$. If $u \notin B_\rho(0)$, then $q_0(u) < 0$, hence $u \notin \mathcal{S}(q)$. If $u \in B_\rho(0) \setminus S$, then $-1 \leqslant p_i(u) < 0$ for some i, and hence

$$(g(p_i))(u) = g(p_i(u)) > 0,$$

which implies $q_i(u) < 0$. Thus we have shown $S = \mathcal{S}(q)$, and by the way q_0 is defined, q is again an Archimedean description of S. Also, since g is a sum of squares, so is $g(p_i)$, which implies $q_1, \ldots, q_r \in \mathrm{QM}(p)$. Since $\mathrm{QM}(p)$ is Archimedean, we also have $q_0 \in \mathrm{QM}(p)$, using Putinar's Positivstellensatz A.13.

Now we need to make sure that $q_0, \ldots, q_r$ are strictly concave on S. The polynomial q_0 is everywhere strictly concave, since $\nabla^2 q_0 = -2I_n$. For the others we compute

$$\begin{aligned}\nabla^2 q_i &= \big(g(p_i) + g'(p_i)p_i\big)\nabla^2 p_i + \big(2g'(p_i) + g''(p_i)p_i\big)\nabla p_i \cdot (\nabla p_i)^T \\ &= \big(g(p_i) + g'(p_i)p_i\big)\left(\nabla^2 p_i + \frac{2g'(p_i) + g''(p_i)p_i}{g(p_i) + g'(p_i)p_i}\nabla p_i \cdot (\nabla p_i)^T\right) \\ &\leqslant \big(g(p_i) + g'(p_i)p_i\big)\big(\nabla^2 p_i - \mu \cdot \nabla p_i \cdot (\nabla p_i)^T\big).\end{aligned}$$

Thus using property (2) of g and applying Lemma 3.47 to p_i, we can make all $\nabla^2 q_i$ negative definite at all points of S, by choosing μ sufficiently large (using that S is compact). □

We now sum up the results of this chapter, combining the various conditions into a single statement.

Corollary 3.51
Let p be an Archimedean description of a convex set S with non-empty interior, and suppose that for each $i = 1, \ldots, r$ one of the following two conditions is satisfied:

[SOS-Concavity]	*The matrix $-\nabla^2 p_i \in \mathrm{Sym}_n \mathbb{R}[x]$ is a sum of squares.*
[Strict quasi-concavity]	*The function p_i is strictly quasi-concave on S.*

Then S possesses an exact Lasserre-Parrilo relaxation with respect to p.

Proof Suppose that $p_1, \ldots, p_m$ are sos-concave, and $p_{m+1}, \ldots, p_r$ are strictly quasi-concave. Since $\mathrm{QM}(p)$ is Archimedean, we have $\rho - \sum_{i=1}^n x_i^2 \in \mathrm{QM}(p)$ for some $\rho > 0$, by Putinar's Positivstellensatz A.13. By Proposition 3.50 we can

replace

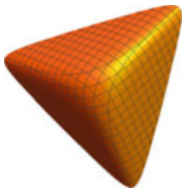
$$\rho - \sum_{i=1}^{n} x_i^2,\ p_{m+1}, \ldots, p_r$$

by strictly concave polynomials in $\mathrm{QM}(p)$, defining the same set, and obtain a new description $S = \mathcal{S}(q_1, \ldots, q_s)$ with $q_1, \ldots, q_s \in \mathrm{QM}(p)$ such that each q_i for $i = 1, \ldots, s$ is either sos-concave or strictly concave on S. Now the arguments in the proofs of Theorems 3.38 and 3.43 can be combined by a simple case distinction between the sos-concave and strictly concave defining polynomials to show that S has an exact Lasserre-Parrilo relaxation with respect to q. Since each $\mathrm{QM}_{[d]}(q)$ is contained in some $\mathrm{QM}_{[e]}(p)$, also some relaxation with respect to p is exact. □

Example 3.52
For some $\lambda > n$ let

$$p_1 = x_1,\ \ldots,\ p_n = x_n,\ p_{n+1} = \lambda - x_1 - \cdots - x_n$$

and $p_{n+2} = x_1 \cdots x_n - 1$. It is not hard to check that $\mathcal{S}(p)$ is bounded and convex, and that $\mathrm{QM}(p)$ is Archimedean.



One can show that p_{n+2} is strictly quasi-concave on $(0, \infty)^n$ (Exercise 3.53), and thus on $\mathcal{S}(p)$. Since $p_1, \ldots, p_{n+1}$ are linear, they are clearly sos-concave. So $\mathcal{S}(p)$ possesses an exact Lasserre-Parrilo relaxation with respect to p.

Exercise 3.53
Show that the polynomial

$$p = x_1 \cdots x_n - 1$$

is strictly quasi-concave at each point $u = (u_1, \ldots, u_n) \in \mathbb{R}^n$ with $u_1 \cdots u_n > 0$.

Example 3.54
The statement in Corollary 3.51 is still not the best possible. Consider the polynomials

$$p_1 = x_2 - x_1^3,\ p_2 = x_1,\ p_3 = 1 - x_1,\ p_4 = x_2,\ p_5 = 1 - x_2$$

in the variables x_1, x_2. We saw in Example 3.21 that the third Lasserre relaxation of $S = \mathcal{S}(p)$ with respect to p is exact. But the polynomial p_1 has Hessian

$$\nabla^2 p_1 = \begin{bmatrix} -6x_1 & 0 \\ 0 & 0 \end{bmatrix}$$

and is therefore not strictly quasi-concave at the origin. Therefore, the exactness of the Lasserre relaxation does not follow from Corollary 3.51.

3.6 Hyperbolicity Cones as Spectrahedral Shadows

Following [70], we can use the exactness results of Helton and Nie to study representations of hyperbolicity cones as spectrahedral shadows. For a hyperbolic polynomial h with smooth hyperbolicity cone, this boils down to verifying quasi-concavity of h on the boundary. But this requires some care, since the concavity cannot be strict along lines through the origin, since h is homogeneous. Thus we have to take a suitable dehomogenization. First, we will need the following basic lemma (for a proof avoiding the use of the Helton-Vinnikov Theorem see [81], Lemma 2.4).

Lemma 3.55
Let $h \in \mathbb{R}[x_0, \ldots, x_n]$ be hyperbolic with respect to e. If $u \in \mathbb{R}^{n+1}$ is a point with $h(u) = 0$ and $\nabla h(u) \neq 0$, then $\partial_e h(u) \neq 0$.

Proof If $\nabla h(u) \neq 0$, then

$$\partial_v h(u) = \frac{\partial}{\partial s} h(u + sv)|_{s=0} \neq 0$$

for generic $v \in \mathbb{R}^{n+1}$. Fix such $v \in \operatorname{int}(\overline{\mathcal{C}}_e(h))$ and consider the hyperbolic polynomial

$$h(ru + sv + te)$$

in three variables r, s, t. By Corollary 2.64, this polynomial has a definite Hermitian determinantal representation

$$h(ru + sv + te) = \det(rA + sB + tC)$$

where B and C are positive definite, hence factor as $B = UU^*$ and $C = VV^*$ with invertible matrices U, V. Now $s = 0$ is a simple root of

$$h(u + sv) = \det(A + sB),$$

which means that $U^{-1}A(U^*)^{-1}$ has one-dimensional kernel. But then so does A and thus also $V^{-1}A(V^*)^{-1}$, hence $t = 0$ is also a simple root of $\det(A+tC) = h(u+te)$. This implies $\partial_e h(u) \neq 0$. □

The following lemma contains the crucial statement of strict quasi-concavity of smooth hyperbolic polynomials. A proof avoiding the Helton-Vinnikov Theorem can for example be found in [89], Theorem 10.

Lemma 3.56
Let $h \in \mathbb{R}[x_0, \ldots, x_n]$ be hyperbolic with respect to e with $h(e) > 0$, and suppose that $\overline{\mathcal{C}}_e(h)$ is pointed (i.e., does not contain a full line). Let H be any affine hyperplane with $e \in H$, $0 \notin H$. Then $h|_H$ is strictly quasi-concave at any point $u \in \overline{\mathcal{C}}_e(h) \cap H$ with $h(u) \neq 0$ or with $h(u) = 0$ and $\nabla h(u) \neq 0$.

Proof Let $H' := H - e$, a linear hyperplane in $\mathbb{R}^{n+1}$. Note that showing strict quasi-concavity of $h|_H$ at u amounts to the following: If $v \in H' \setminus \{0\}$ is such that $\partial_v h(u) = 0$, then $\partial_v^2 h(u) < 0$ (see Exercise 3.57 below).

(1) Let $u \in \overline{\mathcal{C}}_e(h) \cap H$ with $h(u) \neq 0$. Then $h(u) > 0$, so we may rescale and assume $h(u) = 1$. By Corollary 2.67, h is hyperbolic with respect to u. Thus for any $v \in H' \setminus \{0\}$, the univariate polynomial $h(v + tu)$ has only real roots. Since h is homogeneous, the same is true for $h(u + tv) = t^d h(t^{-1}u + v)$, where $d = \deg(h)$. Since $h(u) \neq 0$, all these roots are different from 0, hence we may write

$$h(u + tv) = (1 + \lambda_1 t) \cdots (1 + \lambda_k t)$$

where $k = \deg_t(h(u + tv)) \leqslant d$ and $-(1/\lambda_1), \ldots, -(1/\lambda_k)$ are the roots of $h(u + tv)$. Since $\overline{\mathcal{C}}_e(h)$ is pointed, $h(u + tv)$ cannot be constant for any $v \in H' \setminus \{0\}$, and we must have $k > 0$. In particular, the coefficient of t in $h(u + tv)$ is

$$a_1 = \sum_{i=1}^{k} \lambda_i$$

and the coefficient of t^2 is

$$a_2 = \frac{1}{2}\left(a_1^2 - \sum_{i=1}^{k} \lambda_i^2\right).$$

So if $a_1 = 0$, then $a_2 < 0$, showing that $h|_H$ is strictly quasi-concave at u.

(2) Let $u \in \overline{\mathcal{C}}_e(h)$ with $h(u) = 0$ and $\nabla h(u) \neq 0$. For any $v \in H' \setminus \{0\}$, the polynomial

$$h\,(re + s(u - e) + tv) \in \mathbb{R}[r, s, t]$$

is the restriction of h to the subspace spanned by e, u, v, a hyperbolic polynomial of degree $d = \deg(h)$ in three variables r, s, t. Applying Corollary 2.64 and assuming $h(e) = 1$, we can find Hermitian matrices A, B of size $d \times d$ such that

$$h\,(re + s(u - e) + tv) = \det\,(r\,I_d + s\,A + t\,B)\,.$$

Evaluating at $r = s = 1, t = 0$ gives $h(u) = \det(I_d + A)$, and since $\nabla h(u) \neq 0$, we must have $\operatorname{rk}(I_d + A) = d - 1$ by Lemma 3.55. After a change of coordinates, we may assume $I_d + A = \operatorname{Diag}(1, \ldots, 1, 0)$. Write $B = (b_{jk})_{j,k \leqslant d}$ and expand

$$h(u + tv) = a_1 t + \cdots + a_k t^k.$$

Comparing coefficients on both sides of $h(u + tv) = \det(I_d + A + tB)$, we find

$$a_1 = b_{dd} \qquad \text{and} \qquad a_2 = \sum_{j=1}^{d-1} b_{jj} b_{dd} - b_{jd}\overline{b}_{jd}.$$

So if $a_1 = 0$, we cannot have $a_2 \geqslant 0$ unless $b_{jd} = 0$ for $j = 1, \ldots, d - 1$. But that would imply $h(u + tv) = 0$ for all t, which is impossible under the assumption that $\overline{\mathcal{C}}_e(h)$ is pointed. □

Exercise 3.57
Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice continuously differentiable function. Show that f is strictly quasi-concave at a point $u \in \mathbb{R}^n$ if and only if $\partial_v f(u) = 0$ implies $\partial_v^2 f(u) < 0$ for all $v \in \mathbb{R}^n \setminus \{0\}$.

Definition 3.58

Let $h \in \mathbb{R}[x_0, \ldots, x_n]$ be hyperbolic with respect to $e \in \mathbb{R}^{n+1}$. We say that $\overline{\mathcal{C}}_e(h)$ has *smooth boundary* if $\nabla h(u) \neq 0$ for all $u \neq 0$ in the boundary of $\overline{\mathcal{C}}_e(h)$.

This notion of smoothness reflects the usual one from algebraic geometry, but indeed coincides with other possible notions from convexity theory or differential geometry (at least if h is square-free), as for example explained in [89]. The following is the main result of [70].

Theorem 3.59
Every hyperbolicity cone with smooth boundary is a spectrahedral shadow.

Proof Let $h \in \mathbb{R}[x]$ be hyperbolic with respect to e with $h(e) > 0$, and assume that $\overline{\mathcal{C}}_e(h)$ has smooth boundary. Let $V \subseteq \mathbb{R}^{n+1}$ be the lineality space of $\overline{\mathcal{C}}_e(h)$ (see Exercise B.3). Then

$$\overline{\mathcal{C}}_e(h) = \left(\overline{\mathcal{C}}_e(h) \cap V^\perp\right) + V$$

by Exercise B.3 (2). If $\overline{\mathcal{C}}_e(h) \cap V^\perp$ is a spectrahedral shadow, then so is $\overline{\mathcal{C}}_e(h)$, by Theorem 3.5 (4). Thus, with a suitable choice of coordinates and using Exercise B.3 (4), we may assume that $\overline{\mathcal{C}}_e(h)$ is pointed. This implies that there exists an affine hyperplane $H \subseteq \mathbb{R}^{n+1}$ with $e \in H, 0 \notin H$ such that

$$\overline{\mathcal{C}}_e(h) = \operatorname{cone}(\overline{\mathcal{C}}_e(h) \cap H)$$

and such that $\overline{\mathcal{C}}_e(h) \cap H$ is compact (c.f. Proposition B.1).

Now Lemma 3.56 says that $h|_H$ is strictly quasi-concave at all points of $\overline{\mathcal{C}}_e(h) \cap H$. Furthermore, given $u \in \overline{\mathcal{C}}_e(h) \setminus \{0\}$ with $h(u) = 0$, there exists a radius ε such that

$$\overline{B}_\varepsilon(u) \cap \mathcal{S}(h) = \overline{B}_\varepsilon \cap \overline{\mathcal{C}}_e(h),$$

because $t = 0$ is a simple root of $h(u+te)$ by Lemma 3.55. In other words, h locally describes the hyperbolicity cone by nonnegativity. Since the defining polynomial

$$\varepsilon^2 - \sum_{i=1}^{n+1}(x_i - u_i)^2$$

of the closed ball $\overline{B}_\varepsilon(u)$ is everywhere strictly quasi-concave, we may apply Corollary 3.51 and conclude that

$$\overline{B}_\varepsilon \cap \overline{\mathcal{C}}_e(h) \cap H$$

possesses an exact Lasserre-Parrilo relaxation, and this thus a spectrahedral shadow. Using the compactness of the boundary of $\overline{\mathcal{C}}_e(h) \cap H$ in H, we can write $\overline{\mathcal{C}}_e(h) \cap H$

as the convex hull of finitely many spectrahedral shadows. Applying Theorem 3.5 (6), we see that

$$\overline{\mathcal{C}}_e(h) \cap H$$

is a spectrahedral shadow. Hence so is $\overline{\mathcal{C}}_e(h) = \operatorname{cone}(\overline{\mathcal{C}}_e(h) \cap H)$, by Theorem 3.5 (5). □

3.7 Necessary Conditions for Exactness

So far we have only seen sufficient conditions for the exactness of the Lasserre-Parrilo relaxations, but not a single example in which it demonstrably fails to become exact. In this chapter, we will fill this gap. Note however that the results in this section do *not* imply that the corresponding sets are not spectrahedral shadows! They just say that the very convenient relaxation method of Lasserre and Parrilo fails to prove this property (see Examples 3.62 and 3.65).

The principal obstruction against exactness is the following result from [31].

Proposition 3.60
Let $S = \mathcal{S}(p) \subseteq \mathbb{R}^n$ and let $Z \subseteq \mathbb{R}^n$ be a line, such that $S \cap Z$ has non-empty interior in Z. Assume that there exists a point $u_0 \in S$, in the relative boundary of $\operatorname{clos}(\operatorname{conv}(S)) \cap Z$, such that the gradients $\nabla p_i(u_0)$ are orthogonal to Z, whenever $p_i(u_0) = 0$. Then

$$\operatorname{clos}(\operatorname{conv}(S)) \subsetneq \mathcal{R}_{[d]}(p)$$

for all $d \geqslant 1$, i.e., no Lasserre-Parrilo relaxation with respect to p is exact.

Proof Let Z and u_0 be as in the hypothesis. After a change of coordinates, we may assume $u_0 = 0$ and $Z = \{u \in \mathbb{R}^n \mid u_2 = \cdots = u_n = 0\}$. We may further assume that $u_1 \geqslant 0$ holds for all $u \in \operatorname{clos}(\operatorname{conv}(S)) \cap Z$. Let $q_1, \ldots, q_r \in \mathbb{R}[x_1]$ be the polynomials obtained from $p_1, \ldots, p_r$ by setting the last $n - 1$ variables to zero. We claim that

$$\mathcal{R}_{[d]}(q) \subseteq \mathcal{R}_{[d]}(p) \cap Z$$

holds for any $d \geqslant 1$. To see this, let $u \in \mathcal{R}_{[d]}(q)$, i.e., $u = (L(x_1), 0, \ldots, 0)$ for some

$$L \in \mathrm{QM}_{[d]}(q)' \subseteq \mathbb{R}[x_1]^*.$$

Then L extends to $L_0 \in \mathbb{R}[x]^*$ via $L_0(g) := L(g(x_1, 0, \ldots, 0))$ for $g \in \mathbb{R}[x]$. We have $L_0 \in \mathrm{QM}_{[d]}(p)'$, since $g(x_1, 0, \ldots, 0) \in \mathrm{QM}_{[d]}(q)$ for any $g \in \mathrm{QM}_{[d]}(p)$. Thus

$$u = (L_0(x_1), \ldots, L_0(x_n)) \in \mathcal{R}_{[d]}(p).$$

Now set $S' := S \cap Z = \mathcal{S}(q)$. If $\mathrm{conv}(S')$ is some closed interval $[0, c]$, then set $q_{r+1} = c - x_1$. Otherwise, i.e., if $\mathrm{conv}(S') = [0, \infty)$, set $q_{r+1} = 1$ (just to keep the notation uniform). Then $\mathcal{S}(q) = \mathcal{S}(q, q_{r+1})$ and $\mathcal{R}_{[d]}(q, q_{r+1}) \subseteq \mathcal{R}_{[d]}(q)$. By Proposition 3.18, and since S' has an interior point by assumption,

$$\mathcal{R}_{[d]}(q, q_{r+1}) = \mathrm{conv}(S')$$

holds if and only if every $\ell \in \mathbb{R}[x_1]$ of degree 1, that is nonnegative on S', belongs to $\mathrm{QM}_{[d]}(q, q_{r+1})$. So consider $\ell = x_1$, which is nonnegative on S', and suppose there exists a representation

$$x_1 = \sigma + \sum_{i \in I} \sigma_i q_i + \sum_{i \in J} \sigma_i q_i,$$

where $i \in I$ if $p_i(0) = 0$ and $i \in J$ if $p_i(0) > 0$ (and σ, σ_i are sums of squares). For $i \in J$, q_i has a positive constant term, so σ_i cannot have a constant term, and its homogeneous part of minimal degree must thus be at least quadratic. The same is true for σ. So none of the elements σ and $\sigma_i q_i$, where $i \in J$, contains the monomial x_1. But by hypothesis, $\nabla p_i(0)$ is orthogonal to the x_1-axis for all $i \in I$, which implies that the terms of q_i have all degree at least 2. This is a contradiction, and thus $\mathcal{R}_{[d]}(q, q_{r+1})$ strictly contains $\mathrm{conv}(S')$.

Since $q_{r+1} \in \mathrm{QM}_{[d]}(q, q_{r+1})$, a point larger than c does not lie in $\mathcal{R}_{[d]}(q, q_{r+1})$. Thus there exists some negative b with

$$b \in \mathcal{R}_{[d]}(q, q_{r+1}) \subseteq \mathcal{R}_{[d]}(q) \subseteq \mathcal{R}_{[d]}(p) \cap Z.$$

But since $(b, 0, \ldots, 0) \notin \mathrm{clos}(\mathrm{conv}(S))$ by hypothesis, this proves $\mathrm{clos}(\mathrm{conv}(S)) \subsetneq \mathcal{R}_{[d]}(p)$. □

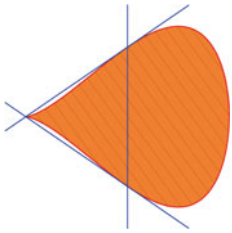
Corollary 3.61
Let $S = \mathcal{S}(p)$ have non-empty interior, and suppose there exists

$$u_0 \in \big(\partial\, \mathrm{conv}(S)\big) \cap S$$

such that $\nabla p_i(u_0) = 0$ whenever $p_i(u_0) = 0$. Then all Lasserre-Parrilo relaxations $\mathcal{R}_{[d]}(p)$ strictly contain $\mathrm{clos}(\mathrm{conv}(S))$.

Proof Apply Proposition 3.60 to any line Z through u_0 and $\operatorname{int}(S)$. □

Example 3.62
Let $p = x_1^3(1 - x_1) - x_2^2$ and put $S = \mathcal{S}(p)$. The origin is a singular point of $\mathcal{V}(p)$, i.e., $p(0) = 0$ and $\nabla p(0) = 0$. So no Lasserre-Parrilo relaxation of $\operatorname{conv}(S)$ with respect to p is exact.



But a quick computation shows that

$$\operatorname{conv}(S) = \mathcal{S}(p, 2x_1 - 1) \cup \mathcal{S}(2x_2 + x_1, x_1 - 2x_2, 1 - 2x_1).$$

The second set in the union is just a triangle, while the other possesses an exact Lasserre relaxation by Corollary 3.51 (see Exercise 3.63).

Exercise 3.63
Show that $\operatorname{conv}(S)$ in Example 3.62 is a spectrahedral shadow.

The necessary condition for exactness in Proposition 3.60 and Corollary 3.61 depends on the description of the basic closed set S. But there is also a more intrinsic geometric condition. We have seen in Theorem 2.26 that spectrahedra have only exposed faces. This does not carry over to spectrahedral shadows in general, since projections of exposed faces need not be exposed (see Example 3.65 below). But in fact it carries over to convex sets having an exact Lasserre-Parrilo relaxation. This was first shown in [68], but can now easily be deduced from Proposition 3.60.

Theorem 3.64
Let $S = \mathcal{S}(p)$ be convex with non-empty interior. If S has a non-exposed face, then no Lasserre-Parrilo relaxation of S with respect to p is exact.

Proof Let $F \subseteq S$ be a non-exposed face. By Proposition B.18 (6), there exists some face F' of S containing F, such that $\ell|_{F'} = 0$ holds for all $\ell \in \mathbb{R}[x]_1$ with $\ell|_F = 0$ and $\ell|_S \geqslant 0$. Choose $u_0 \in \operatorname{relint}(F), u_1 \in \operatorname{relint}(F')$, and let Z be the line through u_0 and u_1. Then $S \cap Z$ has non-empty interior in Z,

and u_0 is a point in the relative boundary of $S \cap Z$. So the claim follows from Proposition 3.60, once we have verified the condition on the gradients. Suppose that $p_i(u_0) = 0$ for some i. Since S is convex and $p_i|_S \geqslant 0$, we must have $\nabla p_i(u_0)^T(u - u_0) \geqslant 0$ for all $u \in S$. In other words, the polynomial

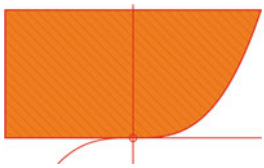
$$\ell := \nabla p_i(u_0)^T(x - u_0) \in \mathbb{R}[x]_1$$

is nonnegative on S. Since $u_0 \in \operatorname{relint}(F)$ and $\ell(u_0) = 0$, we must have $\ell|_F = 0$ and hence also $\ell|_{F'} = 0$. So ℓ vanishes in both u_0 and u_1, and thus on all of Z. But this implies that $\nabla p_i(u_0)$ is orthogonal to Z. □

Example 3.65
Returning to the convex set

$$S = \mathcal{S}(x_2 - x_1^3, 1 + x_1, 1 - x_1, x_2, 1 - x_2).$$

from Example 2.28, we conclude from Theorem 3.64 that S cannot have an exact Lasserre-Parrilo relaxation with respect to the given defining polynomials, or any other description as a basic closed semialgebraic set. This is because the origin is a non-exposed face of S.



However, S is the convex hull of the square $[-1, 0] \times [0, 1]$ and the basic closed semialgebraic set $S \cap \mathcal{S}(x_1)$, both of which are spectrahedral shadows. This is clear for the first, while for the second we gave an explicit proof in Example 3.21. So by Theorem 3.5 (6), S is a spectrahedral shadow.

Exercise 3.66
Find an example of a speactrahedron in $\mathbb{R}^3$, whose projection onto the first two coordinates has a non-exposed face.

▶ **Remarks 3.67**

1. Note that Theorem 3.64 only applies when $S = \mathcal{S}(p)$ is already convex. It may in fact happen that $\operatorname{conv}(S)$ has an exact Lasserre-Parrilo relaxation with respect to p, even when $\operatorname{conv}(S)$ has a non-exposed face. The football stadium from Exercise 2.24 can be written as $\operatorname{conv}(\mathcal{S}(p))$ with

$$p = -((x_1 + 1)^2 + x_2^2 - 1)((x_1 - 1)^2 + x_2^2 - 1).$$

Here, Proposition 3.18 and an explicit calculation can be used to show

$$\operatorname{conv}(\mathcal{S}(p)) = \mathcal{R}_{[4]}(p),$$

see [31], Proposition 4.12.

2. Furthermore, Theorem 3.64 only applies when the Lasserre-Parrilo relaxation is indeed exact. An inexact Lasserre-Parrilo relaxation of a convex basic closed set may well have non-exposed faces. For example, the set in Example 3.65 has a Lasserre-Parrilo relaxation with a non-exposed face, see [31], Corollary 4.11.

3.8 Generalized Relaxations and Scheiderer's Counterexamples

As we have seen in Sect. 3.7, the existence of an exact Lasserre-Parrilo relaxation is only sufficient, but not necessary for a set to be a spectrahedral shadow. Only when generalizing the above notion of relaxation, both properties indeed become equivalent. This approach can then be used to show that certain convex semialgebraic sets are indeed not spectrahedral shadows, answering a question from [65], and refuting the Helton-Nie conjecture [41, 42]. This was first achieved by Scheiderer in [99]. Our presentation follows the similar but slightly more elementary approach from [26].

The first result can be seen as a generalization of Proposition 3.18, using semialgebraic functions instead of just polynomials to construct a relaxation. In this setting, the existence of bounded degree sums-of-squares representations for linear polynomials is really equivalent to being a spectrahedral shadow. For simplicity, we restrict to convex cones.

Proposition 3.68
Let $S \subseteq \mathbb{R}^n$ be a semialgebraic set, and let $\mathcal{F}_{\mathrm{sa}}(S)$ denote the $\mathbb{R}$-algebra of real-valued semialgebraic functions on S. Then the following are equivalent:

1. $\operatorname{clos}(\operatorname{cone}(S))$ *is a spectrahedral shadow.*
2. *There exists a finite-dimensional subspace $W \subseteq \mathcal{F}_{\mathrm{sa}}(S)$ with $\ell_{|S} \in \Sigma W^2$ for every $\ell \in S^\vee$.*

Proof We have seen that (2) implies (1) in several variants throughout this book already: The set ΣW^2 is a spectrahedral shadow, by a similar construction as Example 2.2 (4). By assumption (2), the cone $S^\vee$ is the inverse image of ΣW^2

under the linear restriction map

$$\begin{aligned}\left(\mathbb{R}^n\right)^* &\to \mathcal{F}_{\mathrm{sa}}(S)\\ \ell &\mapsto \ell_{|S}\end{aligned}$$

and thus also a spectrahedral shadow, by Theorem 3.5 (3). Thus

$$\left(S^\vee\right)^\vee = \operatorname{clos}(\operatorname{cone}(S))$$

is also a spectrahedral shadow, using Theorem 3.5 (9) and the Biduality Theorem B.5.

For the converse, we can assume there exist a strictly feasible homogeneous linear matrix polynomial $M(x) + N(y)$, of some size s, such that

$$\operatorname{clos}(\operatorname{cone}(S)) = \left\{u \in \mathbb{R}^n \mid \exists v \in \mathbb{R}^m \colon M(u) + N(v) \succcurlyeq 0\right\}.$$

Now whenever $\ell \in S^\vee$ is pulled back to a linear functional on $\mathbb{R}^n \times \mathbb{R}^m$ via the projection map (i.e., using the first coordinates of points only), it is nonnegative on the spectrahedron defined by $M(x) + N(y) \succcurlyeq 0$. The proof of Theorem 3.5 (8) and (9) shows that there exists some $P \in \mathrm{Sym}_s^+$ with

$$\ell(x_1, \ldots, x_n) = x_1 \operatorname{Tr}(PM_1) + \cdots + x_n \operatorname{Tr}(PM_n) + y_1 \underbrace{\operatorname{Tr}(PN_1)}_{=0} + \cdots + y_m \underbrace{\operatorname{Tr}(PN_m)}_{=0}.$$

We write $P = \sum_k \xi_k \xi_k^T$ for certain $\xi_k \in \mathbb{R}^s$, and choose for each $u \in S$ some $v \in \mathbb{R}^m$ with $Q_{u,v} := M(u) + N(v) \succcurlyeq 0$. We now compute

$$\begin{aligned}\ell(u) &= \sum_i u_i \operatorname{Tr}(PM_i) + \sum_j v_j \operatorname{Tr}(PN_j)\\ &= \operatorname{Tr}(P(M(u) + N(v)))\\ &= \operatorname{Tr}\left(P\sqrt{Q}_{u,v}^2\right)\\ &= \sum_k \left(\sqrt{Q}_{u,v}\xi_k\right)^T \left(\sqrt{Q}_{u,v}\xi_k\right).\end{aligned}$$

We now observe that we can choose v as a semialgebraic function of u, and that the (unique) positive semidefinite square-root of a positive semidefinite matrix is also a semialgebraic function in the matrix entries (Exercise A.3). In total, all entries of $\sqrt{Q}_{u,v}$ are semialgebraic functions in u. So if $W \subseteq \mathcal{F}_{\mathrm{sa}}(S)$ is the space spanned by all these functions, the claim follows. □

▶ **Remark 3.69** Every finite-dimensional space $W \subseteq \mathcal{F}_{\mathrm{sa}}(S)$ gives rise to an outer relaxation of clos (cone(S)) by a spectrahedral shadow, by the construction in the first part of the proof. To be more explicit, we take all linear functionals on $\mathbb{R}^n$ that belong to ΣW^2, and consider their joint nonnegativity set. This is a spectrahedral shadow containing clos (cone(S)).
If $S = \mathcal{S}(p)$ is basic closed, we can take W to be the space spanned by all functions of the form

$$q\sqrt{p_i} \in \mathcal{F}_{\mathrm{sa}}(S)$$

where q is a polynomial with $\deg(q) \leqslant \frac{1}{2}(d - \deg(p_i))$. Then ΣW^2 contains all functions from $\mathrm{QM}_{[d]}(p)$, and the generalized relaxation defined by W thus contains the Lasserre-Parrilo relaxation defined by $\mathrm{QM}_{[d]}(p)$, in the sense of Remark 3.23. In this sense we consider generalized relaxations, and we have shown that being a spectrahedral shadow is really equivalent to having an exact generalized relaxation.

The following theorem is the first result that formulates an unavoidable obstruction to being a spectrahedral shadow. We will see explicit examples of such sets below.

Theorem 3.70
Let $h \in \mathbb{R}[x]$ be homogeneous, globally nonnegative, but not a sum of squares in $\mathbb{R}[x]$. Let $V = \mathrm{span}\{q_1, \ldots, q_m\} \subseteq \mathbb{R}[x]$ be a finite-dimensional subspace that contains all translates

$$h_u(x) := h(x - u)$$

for $u \in \mathbb{R}^n$, and let $q = (q_1, \ldots, q_m)\colon \mathbb{R}^n \to \mathbb{R}^m$ be the corresponding polynomial map. Then for every semialgebraic set $S \subseteq \mathbb{R}^n$ with nonempty interior, the set

$$\mathrm{clos}\,(\mathrm{cone}(q(S))) \subseteq \mathbb{R}^m$$

is a closed semialgebraic convex cone, but not a spectrahedral shadow.

Proof Assume for contradiction that clos (cone$(q(S))$) is a spectrahedral shadow. We apply Proposition 3.68 to $q(S)$, pull everything back to S via q, and obtain a finite-dimensional subspace $X \subseteq \mathcal{F}_{\mathrm{sa}}(S)$, such that every polynomial from V that is nonnegative on S belongs to ΣX^2.

Let X be spanned by the semialgebraic functions $f_1, \ldots, f_r$. Since semialgebraic functions are smooth almost everywhere (see Theorem A.6), there is a point $u \in S$ at

which all f_i are smooth. The polynomial h_u is nonnegative on S and therefore a sum of squares of elements from X. It is also homogeneous with respect to u, and after a shift we can thus assume without loss of generality that $u = 0$ (here we have used that V contains all translates of h). If a homogeneous polynomial is a sum of squares of functions that are smooth at the origin, it is a sum of squares of polynomials already. This follows by simply comparing the nontrivial homogeneous terms of lowest degree of the Taylor expansions. This yields the desired contradiction. □

One can get rid of the homogeneity assumption on h in the last theorem, at the cost of some more technical difficulties. This will reduce the size of n by one in the following corollaries. We refer the reader to [26, 99] for these improvements.

Corollary 3.71
Let either $n \geqslant 3$ and $d \geqslant 6$, or $n \geqslant 4$ and $d \geqslant 4$, and let $S \subseteq \mathbb{R}^n$ be a semialgebraic set with nonempty interior. Then

$$\mathcal{P}(S) \cap \mathbb{R}[x]_d = \left\{p \in \mathbb{R}[x] \mid \deg(p) \leqslant d,\, p_{|S} \geqslant 0\right\}$$

is a closed semialgebraic convex cone, but not a spectrahedral shadow. This applies in particular to $S = \mathbb{R}^n$.

Proof In the specified cases, there exist nonnegative homogeneous polynomials in n variables of degree d, which are not sums of squares of polynomial. Examples are the Motzkin form

$$x_1^4x_2^2 + x_1^2x_2^4 - 3x_1^2x_2^2x_3^2 + x_3^6$$

and the Choi-Lam form

$$x_1^2x_2^2 + x_2^2x_3^2 + x_3^2x_1^2 - 4x_1x_2x_3x_4 + x_4^4,$$

see for example [63]. So for such n and d, the space

$$V = \mathbb{R}[x]_d = \operatorname{span}\left\{x^\alpha \mid |\alpha| \leqslant d\right\}$$

of polynomial of degree at most d fulfills the assumption of Theorem 3.70. The corresponding polynomial map is the Veronese embedding

$$\begin{aligned} v_{n,d} \colon \mathbb{R}^n &\to \mathbb{R}^m \\ u &\mapsto \left(u^\alpha\right)_{|\alpha| \leqslant d} \end{aligned}$$

where $m = \binom{n+d}{n}$. So

$$\operatorname{clos}\left(\operatorname{cone}(v_{n,d}(S)\right) \subseteq \mathbb{R}^m$$

is not a spectrahedral shadow, by Theorem 3.70. In view of biduality Theorems B.5 and 3.5 (9), also

$$v_{n,d}(S)^\vee = \left\{ (p_\alpha)_{|\alpha|\leqslant d} \mid \forall u \in S\colon \sum_\alpha p_\alpha u^\alpha \geqslant 0 \right\}$$

is not a spectrahedral shadow. This set can clearly be identified with $\mathcal{P}(S) \cap \mathbb{R}[x]_d$. □

Corollary 3.72
For $n \geqslant 4$, the set of globally nonnegative biquadratic forms in

$$\mathbb{R}[x_1, \ldots, x_n, y_1, \ldots, y_n]$$

is a closed semialgebraic convex cone, but not a spectrahedral shadow.

Proof There exists a nonnegative biquadratic form in $\mathbb{R}[x_1, \ldots, x_{n-1}, y_1, \ldots, y_{n-1}]$, that is not a sums of squares of bilinear forms, and thus not a sum of squares of any polynomials (Exercise 3.73). One such example is the Choi form [17]

$$x_1^2y_1^2+x_2^2y_2^2+x_3^2y_3^2-2(x_1x_2y_1y_2+x_2x_3y_2y_3+x_3x_1y_3y_1)+2(x_1^2y_2^2+x_2^2y_3^2+x_3^2y_1^2).$$

We can thus apply Theorem 3.70 to the space $V \subseteq \mathbb{R}[x_1, \ldots, x_{n-1}, y_1, \ldots, x_{n-1}]$ spanned by all monomials of degree at most 2 in both x and y. The dual of the image $q(\mathbb{R}^{n-1} \times \mathbb{R}^{n-1})$ of the corresponding map is

$$\{p \in \mathbb{R}[x_1, \ldots, x_{n-1}, y_1, \ldots, y_{n-1}] \mid \deg_x(p) \leqslant 2, \deg_y(p) \leqslant 2, p \geqslant 0 \text{ on } \mathbb{R}^{n-1} \times \mathbb{R}^{n-1}\},$$

which is not a spectrahedral shadow, by the same arguments as in the proof of Corollary 3.71. After bi-homogenization with x_n and y_n, i.e., applying the isomorphism

$$p \mapsto x_n^2y_n^2 \cdot p\,(x/x_n, y/y_n)\,,$$

we conclude that the set of nonnegative biquadratic forms in $x_1, \ldots, x_n, y_1, \ldots, y_n$ is not a spectrahedral shadow. □

Exercise 3.73
Show that if a biquadratic form is a sum of squares of polynomials, then it is a sum of squares of bilinear forms.

Definition 3.74

For $n \geqslant 1$, inside the space $\mathrm{Her}_n \otimes \mathrm{Her}_n \cong \mathrm{Her}_{n^2}$ we consider the convex cone

$$\mathrm{Sep}_{n,n} := \left\{ \sum_i P_i \otimes Q_i \mid P_i, Q_i \in \mathrm{Her}_n^+ \right\}$$

and call its elements *separable psd matrices*.

Exercise 3.75
For $n \geqslant 1$, show the following:

1. $\mathrm{Sep}_{n,n} \subseteq \mathrm{Her}_{n^2}^+$, and for $n \geqslant 2$ the inclusion is strict.
   Hint: If you haven't encountered separable psd matrices before, search for the *Peres-Horodecki criterion*.
2. $\mathrm{Sep}_{n,n}$ is a closed and semialgebraic convex cone.
   Hint: Both statements might require to use Carathéodory's Theorem, see for example [3] Theorem 2.3.

Corollary 3.76
For $n \geqslant 4$, the cone $\mathrm{Sep}_{n,n} \subseteq \mathrm{Her}_{n^2}$ is not a spectrahedral shadow.

Proof If $\mathrm{Sep}_{n,n}$ was a spectrahedral shadow, so would be its dual in $\mathrm{Her}_n \otimes \mathrm{Her}_n$ (where we use the trace inner product on matrix spaces). The dual is the cone of so-called *block positive matrices*, i.e., matrices which are nonnegative as quadratic forms only on elementary tensors $v \otimes w$ of vectors. This uses the fact that $\mathrm{Sep}_{n,n}$ is spanned as a convex cone by matrices of the form

$$vv^* \otimes ww^*$$

with $v, w \in \mathbb{C}^n$.

The space $\mathrm{Her}_n \otimes \mathrm{Her}_n$ canonically projects onto the space of biquadratic forms in $\mathbb{R}[x, y]$, by contracting the first tensor factor with the vector of variables x, and the second with y. The cone $\mathrm{Sep}_{n,n}^\vee$ of block-positive matrices is thereby mapped onto the cone of nonnegative biquadratic forms. In Corollary 3.72 we have seen that this is not a spectrahedral shadow, and so neither is $\mathrm{Sep}_{n,n}^\vee$ and $\mathrm{Sep}_{n,n}$. □

Note that the last statement is clearly also true for symmetric instead of Hermitian matrices. Since separable matrices play an important role in quantum physics, and it is common to consider Hermitian matrices there, we stated the results for Hermitian matrices only.

▶ **Remark 3.77** The given counterexamples show an interesting pattern. A common theme in polynomial optimization has recently been to transform hard optimization problems to easier semidefinite problems, by using sums-of-squares certificates instead of positivity (see for example [58] for a comprehensive overview). However, this approach is in general not equivalent to the initial problem, since sums-of-squares certificates usually fail to exist for all nonnegative polynomials. The Peres-Horodecki criterion for separability [48, 80] is of the same spirit. Corollaries 3.71 and 3.76 (to be precise: the versions with n reduced by one, which we did not prove) state that exactly in the cases where the approaches become non-equivalent, the corresponding set of nonnegative polynomials and separable psd matrices are also not a spectrahedral shadows. So it seems like the hard problems resist being made simple.

▶ **Remark 3.78** A completely different approach to the problem in this section, using the Tarski principle and convex separation over real closed fields, has recently been developed by Bodirsky et al. in [11]. This has also produced new counterexamples, notably the cone of copositive matrices of size 5×5.

Real Algebraic Geometry

A

This section contains a short introduction to concepts and results from real algebra and geometry, with a special focus on (Matrix-)Positivstellensätze, stability of quadratic modules, and sums of squares on curves. The reader who wants to learn more about this exciting topic is referred to the books [10, 63, 83].

A.1 Semialgebraic Sets, Semialgebraic Mappings and Dimension

Recall that a *basic closed semialgebraic* set is defined as

$$\mathcal{S}(p) = \mathcal{S}(p_1, \ldots, p_r) = \left\{u \in \mathbb{R}^n \mid p_1(u) \geqslant 0, \ldots, p_r(u) \geqslant 0\right\},$$

for $p = (p_1, \ldots, p_r)$ with all $p_i \in \mathbb{R}[x]$. A general *semialgebraic set* is a (finite) Boolean combination of basic closed semialgebraic sets. The Projection Theorem 1.2 states that projections of semialgebraic sets are again semialgebraic. This innocent-looking result is in fact not completely trivial to prove, and has many strong implications, see [10, 63, 83] for more details. For our purpose, the most relevant implications are that closure, interior, convex hulls, and many more constructions with semialgebraic sets again produce semialgebraic sets (see Example 1.4).

Definition A.1

The *dimension* $\dim(S)$ of a semialgebraic set is the dimension of the $\mathbb{R}$-Zariski closure of S, as an affine $\mathbb{R}$-variety. This coincides for example with the maximal length of a chain of prime ideals in the ring $\mathbb{R}[x]/\mathcal{I}(S)$, where $\mathcal{I}(S)$ is the ideal of real polynomials vanishing on S.

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For example, if $S \subseteq \mathbb{R}^n$ has nonempty interior (with respect to the Euclidean topology), then $\dim(S) = n$.

Definition A.2

Let $S \subseteq \mathbb{R}^n$ be a semialgebraic set. Then $f\colon S \to \mathbb{R}^m$ is called a *semialgebraic mapping* if its graph

$$\Gamma(f) := \{(u, f(u)) \mid u \in S\}$$

is a semialgebraic subset of $\mathbb{R}^n \times \mathbb{R}^m = \mathbb{R}^{n+m}$.

Exercise A.3
Show the following:

1. Componentwise polynomial mappings are semialgebraic.
2. The square-root function $\sqrt{}\colon [0, \infty) \to [0, \infty)$ is semialgebraic.
3. The square-root mapping $\sqrt{}\colon \mathrm{Sym}_s^+ \to \mathrm{Sym}_s^+$ is semialgebraic (where both sets are understood as subsets of $\mathrm{Sym}_s \cong \mathbb{R}^{s(s+1)/2}$).
4. Let $T \subseteq \mathbb{R}^{n+m}$ be a closed semialgebraic set, and $S := \pi_x(T) \subseteq \mathbb{R}^n$. Then there exists a semialgebraic mapping $s\colon S \to \mathbb{R}^{n+m}$ with

$$s(u) \in T \text{ and } \pi_x(s(u)) = u$$

for all $u \in S$.

Theorem A.4
If $f\colon S \to \mathbb{R}^m$ is a semialgebraic mapping, then $\dim(f(S)) \leqslant \dim(S)$.

Proof See [10], Theorem 2.8.8. □

Proposition A.5
For a semialgebraic set S, the set

$$\mathcal{F}_{\mathrm{sa}}(S) := \{f\colon S \to \mathbb{R} \mid f \text{ semialgebraic}\}$$

of all semialgebraic functions on S is an $\mathbb{R}$-algebra, with pointwise operations.

Proof See for example [10], Proposition 2.2.6. □

Theorem A.6
Let $S \subseteq \mathbb{R}^n$ be a semialgebraic set with nonempty interior and let $f : S \to \mathbb{R}$ be a semialgebraic function. Then f is smooth in all points of S outside a semialgebraic subset of dimension $< n$.

Proof This follows for example from [39], Theorem 1.7, or from more general decomposition results such as [10], § 9.1, applied to the graph of f. □

A.2 Positive Polynomials and Quadratic Modules

Throughout this section let $p = (p_1, \ldots, p_r)$ be a tuple of polynomials from $\mathbb{R}[x]$, and let $S = \mathcal{S}(p) \subseteq \mathbb{R}^n$ be the basic closed set defined by p. Let

$$\mathcal{P}(S) := \left\{ p \in \mathbb{R}[x] \mid \ p(u) \geqslant 0 \text{ for all } u \in S \right\}$$

be the set of non-negative polynomials on S (which we also call the *saturated preordering of* S). Understanding the structure of $\mathcal{P}(S)$ is a central goal of real algebra and geometry, for which we introduce some standard notions.

Definition A.7

A *quadratic module* in $\mathbb{R}[x]$ (or indeed in any commutative ring with 1) is a subset $Q \subseteq \mathbb{R}[x]$ such that

1. $1 \in Q$,
2. $Q + Q \subseteq Q$,
3. $p^2 Q \subseteq Q$ for all $p \in \mathbb{R}[x]$.

A quadratic module is called a *preordering* if

4. $Q \cdot Q \subseteq Q$.

It is clear that $\mathcal{P}(S)$ above is a preordering. The cone Σ of sums of squares is the smallest preordering, contained in all quadratic modules in $\mathbb{R}[x]$. More generally, the set

$$\mathrm{QM}(p) := \left\{ \sigma_0 + \sigma_1 p_1 + \cdots + \sigma_r p_r \mid \sigma_0, \ldots, \sigma_r \in \Sigma \right\}$$

is clearly the smallest quadratic module containing all p_i, called the quadratic module *generated by* p. The preordering generated by p is the quadratic module generated by all finite products of the p_i:

$$\mathrm{PO}(p) := \left\{ \sum_{e\in\{0,1\}^r} \sigma_e p_1^{e_1} \cdots p_r^{e_r} \mid \sigma_i \in \Sigma \right\}.$$

In particular, it is also finitely generated as a quadratic module.

Notation A.8 *We will always implicitly define* $p_0 = 1$*, so that an element of* $\mathrm{QM}(p)$ *can be written in the form* $\sum_{i=0}^{r} \sigma_i p_i$ *with* $\sigma_i \in \Sigma$.

Now it is clear from the definition that

$$\mathrm{QM}(p) \subseteq \mathrm{PO}(p) \subseteq \mathcal{P}(\mathcal{S}(p))$$

holds. Much research in real algebra and geometry revolves around the question of how close these inclusions are to equality.

Example A.9

1. Every non-negative polynomial in one variable is a sum of (two) squares of polynomials, so for $S = \mathbb{R}$ we have $\mathcal{P}(S) = \Sigma$. This becomes false in higher dimensions, i.e., $\Sigma \subsetneq \mathcal{P}(\mathbb{R}^n)$ for $n \geqslant 2$, by a classical result of Hilbert.
2. Every $q \in \mathbb{R}[x_1]$ with $q|_{[0,1]} \geqslant 0$ is contained in $\mathrm{QM}(x_1(1-x_1))$, i.e., has a representation

$$q = \sigma_0 + \sigma_1 \cdot x(1-x)$$

for $\sigma_0, \sigma_1 \in \Sigma$. For general subsets of the line, see [63], §2.7.

Exercise A.10
Show that $\mathcal{P}(\mathbb{R}) = \Sigma$ and $\mathcal{P}([0,1]) = \mathrm{QM}(x_1(1-x_1))$ as claimed above.

An important general result, which triggered a lot of further research on positive polynomials and sums of squares, is the following, often called *Schmüdgen's Positivstellensatz*.

Theorem A.11 (Schmüdgen's Positivstellensatz)
If $\mathcal{S}(p) \subseteq \mathbb{R}^n$ *is bounded, the preordering* $\mathrm{PO}(p)$ *contains all polynomials* $q \in \mathbb{R}[x]$ *with* $q(u) > 0$ *for all* $u \in \mathcal{S}(p)$.

Proof See for example [63], Corollary 6.1.2, or [83], Theorem 5.2.9. □

Definition A.12

A quadratic module Q in $\mathbb{R}[x]$ is called *Archimedean* if it contains a polynomial g such that $\mathcal{S}(g)$ is bounded. If $\mathrm{QM}(p)$ is Archimedean, we also say that p provides an *Archimedean description* of the (compact) set $\mathcal{S}(p)$.

Note that this Archimedean property is often defined in a different way, but our definition is equivalent and most easy to state (see [63], Theorem 7.1.1, or [83], Theorem 5.3.8). A version of Schmüdgen's Positivstellensatz for quadratic modules is the following. Instead of boundedness of $\mathcal{S}(p)$ it requires the quadratic module $\mathrm{QM}(p)$ to be Archimedean.

Theorem A.13 (Putinar's Positivstellensatz)
If the quadratic module $\mathrm{QM}(p)$ *is Archimedean, then it contains all polynomials* $q \in \mathbb{R}[x]$ *with* $q(u) > 0$ *for all* $u \in \mathcal{S}(p)$.

Proof See for example [63], Theorem 5.6.1, or [83], Theorem 5.3.8. □

Corollary A.14
A finitely generated quadratic module Q *in* $\mathbb{R}[x]$ *is Archimedean if and only if there exists a number* λ *such that* $\lambda^2 - \sum_{i=1}^{n} x_i^2$ *is contained in* Q.

Proof One direction is obvious from the definition, the other one follows directly from Theorem A.13. □

▶ **Remark A.15** Given Putinar's Theorem, Schmüdgen's Theorem can be rephrased as saying that $\mathrm{PO}(p)$ is Archimedean whenever $\mathcal{S}(p)$ is bounded. One can find examples of such p for which $\mathrm{QM}(p)$ is not Archimedean, which shows that Schmüdgen's Theorem does not extend to quadratic modules without additional assumptions.
For practical purposes, if S is bounded, the assumption that Q is Archimedean is often considered quite mild, since one can just add the polynomial $\lambda^2 - \sum_{i=1}^{r} x_i^2$ to the description of S, if $S \subseteq B_\lambda(0)$. Since the representation of a positive polynomial in the quadratic module is simpler than in the preordering ($r + 1$ summands instead of 2^r), the use of quadratic modules is often preferred.

On the other hand, we have the following negative result by Scheiderer.

Theorem A.16
If S is of dimension at least 3, then $\mathcal{P}(S)$ is not a finitely generated quadratic module. In other words, for any description $S = \mathcal{S}(p)$ there exists $q \in \mathcal{P}(S)$ with $q \notin \mathrm{QM}(p)$.

Proof See [63], Proposition 2.6.2. □

Of course, if S is bounded, then any $q \in \mathcal{P}(S) \setminus \mathrm{PO}(p)$ must necessarily have a zero somewhere in S, by Schmügen's Theorem.

Note that the cone $\mathcal{P}(S)$ lives in the infinite-dimensional vector space $\mathbb{R}[x]$. What if instead we consider the finite-dimensional slices?

Proposition A.17
The set

$$\mathcal{P}(S)_d := \mathcal{P}(S) \cap \mathbb{R}[x]_d$$

is a closed and semialgebraic convex cone in the finite-dimensional real vector space $\mathbb{R}[x]_d$.

Proof For each point $u \in \mathbb{R}^n$, let $L_u \in \mathbb{R}[x]_d^*$ be the linear functional $q \mapsto q(u)$. Then we can write

$$\mathcal{P}(S)_d = \bigcap_{u \in S} L_u^{-1}([0, \infty)).$$

Since the functionals L_u are continuous, this expresses $\mathcal{P}(S)_d$ as an intersection of closed halfspaces, hence it is a closed convex cone. Furthermore, the projection onto the first factor of the set

$$\left\{(q, u, q(u)) \mid q \in \mathbb{R}[x]_d, u \in \mathbb{R}^n\right\} \cap (\mathbb{R}[x]_d \times S \times (-\infty, 0)) \subseteq \mathbb{R}[x]_d \times \mathbb{R}^n \times \mathbb{R}$$

is semialgebraic, and it is the complement of $\mathcal{P}(S)_d$ inside $\mathbb{R}[x]_d$. So $\mathcal{P}(S)_d$ is semialgebraic. □

While $\mathcal{P}(S)_d$ is a closed and semialgebraic convex cone, it is usually not even a spectrahedral shadow, as shown in Corollary 3.71. So how about finitely generated quadratic modules instead? Let us consider

$$\mathrm{QM}_d(p) := \mathrm{QM}(p) \cap \mathbb{R}[x]_d.$$

and compare it to the cone

$$\mathrm{QM}_{[d]}(p) := \left\{ \sum_{i=0}^{r} \sigma_i p_i \mid \deg(\sigma_i p_i) \leqslant d \text{ for all } i = 0, \ldots, r \right\}$$

in $\mathbb{R}[x]_d$, which we call the *truncation of degree d* of $\mathrm{QM}(p)$ (with respect to p). For general quadratic modules, it is not true that $\mathrm{QM}_{[d]}(p) = \mathrm{QM}_d(p)$ or even that $\mathrm{QM}_d(p)$ is contained in $\mathrm{QM}_{[e]}(p)$ for some $e \geqslant d$.

Example A.18
Let $p = (x(1-x))^3$ and $Q = \mathrm{QM}(p) \subseteq \mathbb{R}[x]$ in the polynomial ring in one variable, describing the closed interval $\mathcal{S}(p) = [0, 1]$. It is not hard to check that $x \notin Q$. For suppose we had

$$x = \sigma_0 + \sigma_1 p$$

with $\sigma_0, \sigma_1 \in \Sigma$, then since x divides p, we could conclude that x also divides σ_0. Since σ_0 is a sum of squares, this really implies that x^2 divides σ_0 (give an argument why!). So the right hand side would be divisible by x^2, a contradiction.
On the other hand, Q is a preordering (since there is only one generator) and $\mathcal{S}(p) = [0, 1]$ is bounded, so Q contains all strictly positive polynomials by Schmüdgen's Theorem A.11. In particular, $x + \varepsilon \in Q$ holds for all $\varepsilon > 0$. Now, if we had $x + \varepsilon \in \mathrm{QM}_{[e]}(p)$ for all $\varepsilon > 0$ and fixed $e \geqslant 0$, we could write

$$x + \varepsilon = \sigma_{0,\varepsilon} + \sigma_{1,\varepsilon} p$$

with all $\deg(\sigma_{0,\varepsilon}), \deg(\sigma_{1,\varepsilon}) + 3 \leqslant e$. We could then (carefully!) take limits as $\varepsilon \to 0$ and conclude $x \in Q$, a contradiction. That last argument is made precise in the following proposition.

Proposition A.19
If $\mathrm{QM}(p) \cap -\mathrm{QM}(p) = \{0\}$, *then the convex cone* $\mathrm{QM}_{[d]}(p)$ *is closed in* $\mathbb{R}[x]_d$, *for all* $d \geqslant 0$.

Proof See for example [63], Lemma 4.1.4. □

Definition A.20

Let $Q = \mathrm{QM}(p)$ be a finitely generated quadratic module. For each integer d, we say that Q is *d-stable* with respect to p, if there exists an integer $e \geqslant 0$ such that

$$\mathrm{QM}_d(p) \subseteq \mathrm{QM}_{[e]}(p).$$

So this means that the elements of degree at most d from Q all admit representations with degrees of all terms $\sigma_i p_i$ bounded by e. In particular, the degrees of the sums of squares σ_i are bounded by $e - \deg(p_i)$.
We say that Q is *stable* if it is d-stable for all $d \geqslant 0$. To rephrase, this means that every polynomial p in Q must admit a representation $p = p_0\sigma_0 + \cdots + p_r\sigma_r$ in which the degree of each summand $\sigma_i p_i$ is bounded by a function depending only on the degree of p.

Note that if $\mathrm{QM}_d(p) \subseteq \mathrm{QM}_{[e]}(p)$, then

$$\mathrm{QM}_d(p) = \mathrm{QM}_{[e]}(p) \cap \mathbb{R}[x]_d.$$

It is also not hard to show that whether Q is stable does not depend on the choice of generators p. However, the notion of d-stability for fixed d *does* depend on the choice of generators.

Exercise A.21
Show that stability is independent of the choice of generators.

We will discuss a model-theoretic characterization of stability in Sect. A.4 below. For now let us mention that for bounded sets $\mathcal{S}(p)$, stability of the corresponding quadratic module $\mathrm{QM}(p)$ is only possible in exceptional situations of small dimension:

Exercise A.22
Use Theorem A.11, Theorem A.16 and Proposition A.19 to show: If $\mathcal{S}(p)$ is bounded of dimension at least 3, then $\mathrm{PO}(p)$ is not stable (and neither is $\mathrm{QM}(p)$, if the description is Archimedean.) This is in fact also true in dimension 2, by a result of Scheiderer [95].

In Chap. 3.3, we showed that proving exactness of the Lasserre-Parrilo relaxation for the convex hull of a bounded basic closed semialgebraic set is equivalent to establishing 1-stability for the corresponding (Archimedean) quadratic module. This is basically the statement of Proposition 3.18, using also Putinar's Theorem A.13. In other words, we need uniform degree bounds for representations of supporting hyperplanes obtained from Putinar's (or Schmüdgen's) Theorem. On the other hand, we have already seen, in Example A.18 and much more generally in Exercise A.22,

that such degree bounds cannot exist in great generality, i.e., for polynomials of arbitrary degree in arbitrary dimensions.

However, there do in fact exist non-uniform degree bounds, i.e., depending on other data than just the degree of the represented polynomial. For some of the results in Chap. 3, this was indeed all we needed to know. The following result is from [74]. We indicate below the much more elementary proof from [52].

Theorem A.23 (Putinar's Positivstellensatz with Degree Bounds)
Let $p = (p_1, \ldots, p_r)$ be an Archimedean description of the set $S = \mathcal{S}(p)$. Given $\delta > 0$, every polynomial $q \in \mathbb{R}[x]$ satisfying $q(u) \geqslant \delta$ for all $u \in S$ admits a representation

$$q = \sigma_0 + \sigma_1 p_1 + \cdots + \sigma_r p_r$$

where $\sigma_0, \ldots, \sigma_r \in \Sigma$ fulfill

$$\deg(\sigma_i) \leqslant D(p, \deg(q), \|q\|, \delta)$$

for $i = 0, \ldots, r$. Any norm $\|\cdot\|$ can be chosen here, but the function D will depend on that choice.

The following exercise contains the crucial idea from [52].

Exercise A.24
Let $C_1 \subseteq C_2 \subseteq \cdots$ be an increasing chain of convex sets in the real vectorspace V, and set $C = \bigcup_{d \geqslant 1} C_d$. Let W be a finite-dimensional subspace of V, and let $K \subseteq \operatorname{int}(C \cap W)$ be a compact set (all with respect to the Euclidean topology on W). Show that there exists some $d \geqslant 1$ with $K \subseteq C_d$.

Exercise A.25
Give a proof of Theorem A.23, using Theorem A.13 and Exercise A.24.

The above arguments just provide the existence of degree bounds. However, note that Nie and Schweighofer have also determined the complexity of the function D in [74].

This result by itself does not help us much: To prove exactness of the Lasserre-Parrilo relaxation, we would need to show that the degree bound can be chosen independently of δ and $\|q\|$, at least if q has degree 1. The insight of Helton and Nie (as explained in Sect. 3.5) was that this could be proved under suitable regularity assumptions with the use of Lagrange functions. Their approach requires a version of Putinar's Theorem for (non-linear) matrix polynomials with degree bounds. We will prove this result in Sect. A.3 below.

But first we will need a version of Putinar's Theorem for affine varieties. We briefly recall the correspondence between affine $\mathbb{R}$-varieties, (real) radical ideals, and affine $\mathbb{R}$-algebras. For any ideal I in $\mathbb{R}[x]$, the set

$$V = \mathcal{V}_{\mathbb{C}}(I) \subseteq \mathbb{C}^n$$

of common complex zeros of elements in I is the *affine* $\mathbb{R}$*-variety* defined by I. We denote by $V(\mathbb{R}) = \mathcal{V}_{\mathbb{R}}(I)$ its real points. Conversely, for any subset $S \subseteq \mathbb{C}^n$, we write $\mathcal{I}(S)$ for the vanishing ideal of S in $\mathbb{R}[x]$. *Hilbert's Nullstellensatz* (see for example [25]) says that

$$\mathcal{I}(\mathcal{V}_{\mathbb{C}}(I)) = \sqrt{I} = \{p \in \mathbb{R}[x] \mid \exists k \geqslant 0 \colon p^k \in I\}$$

is the radical of the ideal I. The *real coordinate ring/algebra* of the affine variety $V = \mathcal{V}_{\mathbb{C}}(I)$ is the residue class algebra

$$\mathbb{R}[V] := \mathbb{R}[x]/\sqrt{I}.$$

It consists of all real polynomial mappings on V, also called *(real) regular functions*.

The real analogue of Hilbert's Nullstellensatz is the *real Nullstellensatz*, which says that

$$\mathcal{I}(\mathcal{V}_{\mathbb{R}}(I)) = \sqrt[\mathrm{re}]{I}$$

is the *real radical*, defined by

$$\sqrt[re]{I} := \{p \in \mathbb{R}[x] \mid \exists k \geqslant 0, \sigma \in \Sigma \colon p^{2k} + \sigma \in I\}.$$

It is easy to see that a vanishing ideal $\mathcal{I}(S)$ for $S \subseteq \mathbb{R}^n$ is a real radical ideal , i.e.

$$\sqrt[\mathrm{re}]{\mathcal{I}(S)} = \mathcal{I}(S).$$

A standard result in real algebraic geometry says that the radical ideal $\sqrt{I}$ is real radical if and only if $V(\mathbb{R})$ is $\mathbb{R}$-Zariski dense in V.

The algebra $\mathbb{R}[V]$ is a reduced, finitely generated $\mathbb{R}$-algebra. Conversely, given any such algebra $\mathcal{A}$, fix finitely many generators $y_1, \ldots, y_n$ of $\mathcal{A}$ and consider the surjective ring homomorphism

$$\begin{aligned} \varphi \colon \mathbb{R}[x] &\to \mathcal{A} \\ x_i &\mapsto y_i. \end{aligned}$$

Since $\mathcal{A}$ is reduced, $I = \ker(\varphi)$ is a radical ideal in $\mathbb{R}[x]$, and $\mathcal{A}$ is thus isomorphic to the coordinate ring of the affine $\mathbb{R}$-variety $\mathcal{V}_{\mathbb{C}}(I)$. Furthermore, this can be made independent of the choice of generators $y_1, \ldots, y_n$, by indentifying points in $\mathcal{V}_{\mathbb{C}}(I)$

with $\mathbb{R}$-algebra homomorphisms $\mathcal{A} \to \mathbb{C}$. In this way, the set $\mathrm{Hom}_{\mathbb{R}}(\mathcal{A}, \mathbb{C})$ can be regarded as the abstract variety V corresponding to $\mathcal{A}$, with real points $V(\mathbb{R}) = \mathrm{Hom}_{\mathbb{R}}(\mathcal{A}, \mathbb{R})$.

Given an affine $\mathbb{R}$-variety V and elements $p_1, \ldots, p_r \in \mathbb{R}[V]$, we have a corresponding semialgebraic subset

$$\mathcal{S}_V(p) = \mathcal{S}_V(p_1, \ldots, p_r) := \{u \in V(\mathbb{R}) \mid p_1(u) \geqslant 0, \ldots, p_r(u) \geqslant 0\}.$$

Just as in the polynomial ring, we call a quadratic module $Q \subseteq \mathbb{R}[V]$ *Archimedean,* if it contains an element g such that $\mathcal{S}_V(g)$ is bounded. We need the following generalization of Putinar's Theorem to this setup.

Corollary A.26 (Putinar's Positivstellensatz for Affine Varieties)
Let V be an affine $\mathbb{R}$-variety with coordinate ring $\mathbb{R}[V]$, and assume $Q = \mathrm{QM}(p) \subseteq \mathbb{R}[V]$ is a finitely generated Archimedean quadratic module. Then Q contains all elements $q \in \mathbb{R}[V]$ such that $q(u) > 0$ for all $u \in \mathcal{S}_V(p)$.

Proof We can fix an embedding of V into affine space and just interpret Putinar's Theorem A.13 modulo the vanishing ideal: Let $y_1, \ldots, y_n$ be generators of $\mathbb{R}[V]$, and let $I \subseteq \mathbb{R}[x]$ be the kernel of the surjective homomorphism

$$\begin{aligned} \varphi\colon \mathbb{R}[x] &\to \mathcal{A} \\ x_i &\mapsto y_i, \end{aligned}$$

so that $V(\mathbb{R})$ is identified with the algebraic subset $\mathcal{V}_{\mathbb{R}}(I)$ of $\mathbb{R}^n$. Let $q_1, \ldots, q_r \in \mathbb{R}[x]$ be such that $\varphi(q_i) = p_i$ in $\mathbb{R}[V]$, and let $h_1, \ldots, h_s \in \mathbb{R}[x]$ be generators of the ideal I. Consider

$$Q_0 := \varphi^{-1}(Q) = Q + I.$$

Then

$$Q_0 = \mathrm{QM}(q_1, \ldots, q_r, h_1, \ldots, h_s, -h_1, \ldots, -h_s)$$

is also a finitely generated quadratic module. This follows from the observation that any element from $\mathbb{R}[x]$ is a difference of two sums of squares, see Exercise 3.27 (or even a difference of two squares if the degree does not matter).

We next show that Q_0 is Archimedean. Since Q is Archimedean, there is some $g \in \mathbb{R}[x]$ such that $\varphi(g) \in Q$ and $\mathcal{S}_V(\varphi(g)) \subseteq V(\mathbb{R})$ is bounded. By Schmüdgen's Theorem A.11, the preordering

$$P := \mathrm{PO}(g, h_1, \ldots, h_s, -h_1, \ldots, -h_s) \subseteq \mathbb{R}[x]$$

contains all polynomials that are strictly positive on the bounded set

$$\mathcal{S}(g, h_1, \ldots, h_s, -h_1, \ldots, -h_s) = \mathcal{S}(g) \cap \mathcal{V}_{\mathbb{R}}(I) \subseteq \mathbb{R}^n.$$

Hence there is some $g' \in P$ such that $\mathcal{S}(g') \subseteq \mathbb{R}^n$ is bounded. On the other hand, we have $\mathrm{PO}(\varphi(g)) = \mathrm{QM}(\varphi(g)) \subseteq Q$, and thus

$$P = \varphi^{-1}(\mathrm{PO}(\varphi(g))) \subseteq \varphi^{-1}(Q) = Q_0,$$

which shows that Q_0 is indeed Archimedean.

Finally, if $q \in \mathbb{R}[x]$ satisfies $\varphi(q)(u) > 0$ for all $u \in \mathcal{S}_V(Q)$, this implies $q(u) > 0$ for all $u \in \mathcal{S}(Q_0)$, and therefore $q \in Q_0$, by Putinar's Theorem A.13. Hence $\varphi(q) \in \varphi(Q_0) = Q$. □

▶ **Remark A.27** In the last theorem, it is not necessary for the algebra to be reduced, i.e., to be the coordinate algebra of V. The statement remains true if it is finitely generated and V is the associated affine variety.

A.3 Positive Matrix Polynomials

In this section, we explain Putinar's Positivstellensatz for matrix polynomials, with and without degree bounds. It is a crucial ingredient in the proofs of the results of Helton and Nie in Sect. 3.5.

By a *sum of squares* in the non-commutative ring $\mathrm{Mat}_s\, \mathbb{R}[x]$, we mean a matrix polynomial of the form $Q^T Q$, where $Q \in \mathrm{Mat}_{r \times s}\, \mathbb{R}[x]$ is a matrix polynomial, for some $r \geqslant 1$. Clearly, if a matrix polynomial is a sum of squares, it is symmetric and positive semidefinite in every point of $\mathbb{R}^n$. We start with some exercises to familiarize ourselves with the notion.

Exercise A.28
Show that $F \in \mathrm{Sym}_s\, \mathbb{R}[x]$ is a sum of squares if and only if there exist matrices $Q_1, \ldots, Q_r \in \mathrm{Mat}_s\, \mathbb{R}[x]$, for some $r \geqslant 1$, such that $F = Q_1^T Q_1 + \cdots + Q_r^T Q_r$.

Exercise A.29
Let $F \in \mathrm{Sym}_s\, \mathbb{R}[x]$ be a matrix polynomial. Show that F is a sum of squares in $\mathrm{Mat}_s\, \mathbb{R}[x]$ if and only if the polynomial yFy^T, in the variables

$$(x, y) = (x_1, \ldots, x_n, y_1, \ldots, y_s),$$

is a sum of squares in the polynomial ring $\mathbb{R}[x, y]$.

Exercise A.30
Show that if $F \in \mathrm{Sym}_s \mathbb{R}[x]$ is a sum of squares, then $\det(F)$ (and hence all principal minors of F) are sums of squares in $\mathbb{R}[x]$. What about the converse?

Exercise A.31
Show that every globally positive semidefinite matrix polynomial in one variable is a sum of squares. *Hint:* If you get stuck, take a look at [18], Theorem 7.1.

To prove Putinar's Theorem for matrix polynomials, we use an idea of Klep and Schweighofer, as presented in [49] and [45]. Let $F \in \mathrm{Sym}_s \mathbb{R}[x]$ be a matrix polynomial and let $\mathcal{A}_F$ be the commutative $\mathbb{R}[x]$-subalgebra of $\mathrm{Sym}_s \mathbb{R}[x]$ generated by F. Explicitly, $\mathcal{A}_F$ consists of all expressions of the form $p(x, F)$, where $p \in \mathbb{R}[x, t]$ is a polynomial in x and one additional variable t.

Lemma A.32
Let $F \in \mathrm{Sym}_s \mathbb{R}[x]$ be a matrix polynomial, and consider the map

$$\begin{aligned} \varphi \colon \mathbb{R}[x,t] &\to \mathcal{A}_F \\ p(x,t) &\mapsto p(x,F). \end{aligned}$$

Then the following holds:

1. *The minimal polynomial μ_F of F in the polynomial ring $\mathbb{R}(x)[t]$ in one variable is contained in $\mathbb{R}[x,t]$, and it generates the ideal $\ker(\varphi)$.*
2. *The real points $\mathcal{V}_{\mathbb{R}}(\ker\varphi)$ of the variety corresponding to $\mathcal{A}_F$ form the hypersurface consisting of all points $(u,\lambda) \in \mathbb{R}^n \times \mathbb{R}$ such that λ is an eigenvalue of $F(u)$.*

Proof

(1) Recall that μ_F is the unique monic polynomial in $\mathbb{R}(x)[t]$ of minimal degree in t such that $\mu_F(F) = 0$. By the Cayley-Hamilton Theorem, it is a factor of the characteristic polynomial

$$\chi_F(t) = \det(tI_s - F) \in \mathbb{R}[x][t],$$

say

$$\chi_F = \mu_F \cdot r$$

with $r \in \mathbb{R}(x)[t]$ monic. Let $c \in \mathbb{R}[x]$ be the least common multiple of the denominators of the (maximally reduced) coefficients of μ_F with respect to t, so that $c\mu_F \in \mathbb{R}[x][t]$ is primitive. By Gauss's Lemma (in the form of Lemma A.33 below), we have

$$\frac{1}{c} \cdot r \in \mathbb{R}[x][t].$$

Since r is monic, this implies $\frac{1}{c} \in \mathbb{R}[x]$, hence $c \in \mathbb{R}^\times$ and $\mu_F \in \mathbb{R}[x][t]$. It now follows from Lemma A.33 that μ_F divides any element of $\ker(\varphi)$ in $\mathbb{R}[x,t]$ and is therefore a generator. (2) follows from (1), since $\mathcal{V}_\mathbb{R}(\mu_F) = \mathcal{V}_\mathbb{R}(\chi_F)$. □

Lemma A.33
Let R be a unique factorization domain with field of fractions K, and let $p, q \in R[t]$ be polynomials with q primitive. If $p = qr$ for some $r \in K[t]$, then $r \in R[t]$.

Proof This is a direct consequence of Gauss's Lemma (see [55], Corollary IV.2.2). □

We will now state and prove Putinar's Positivstellensatz for matrix polynomials. Although the statement looks like a non-commutative result, we can in fact reduce everything to a commutative algebra, mainly because *only one* matrix polynomial is involved.

Theorem A.34 (Putinar's Positivistellensatz for Matrix Polynomials)
Let $p = (p_1, \ldots, p_r)$ be an Archimedean description of the set $S = \mathcal{S}(p)$. Then every matrix polynomial $F \in \mathrm{Sym}_s \mathbb{R}[x]$, such that $F(u)$ is positive definite for all $u \in S$, admits a representation

$$F = S_0 + p_1 S_1 + \cdots + p_r S_r$$

where $S_0, \ldots, S_r$ are sums of squares in the algebra $\mathcal{A}_F \subseteq \mathrm{Sym}_s \mathbb{R}[x]$.

Proof Let $\varphi\colon \mathbb{R}[x,t] \to \mathcal{A}_F$ be as above, with corresponding $\mathbb{R}$-variety $V = \mathcal{V}(\ker\varphi)$. To apply Putinar's Positivstellensatz for affine algebras, we first need to check that the quadratic module generated by $p_1, \ldots, p_r$ in $\mathcal{A}_F$ is Archimedean. By assumption, $\mathrm{QM}(p)$ is Archimedean, so it contains some $g \in \mathbb{R}[x]$ such that

$\mathcal{S}(g) \subseteq \mathbb{R}^n$ is bounded. Now

$$\mathcal{S}_V(\varphi(g)) = \left\{(u, \lambda) \in \mathbb{R}^n \times \mathbb{R} \mid g(u) \geqslant 0, \lambda \text{ an eigenvalue of } F(u)\right\} \subseteq V(\mathbb{R})$$

is also bounded, because the spectral radius of $F(u)$ (largest absolute value of an eigenvalue) is bounded on the bounded set $\mathcal{S}(g)$. Thus $\mathrm{QM}_{\mathcal{A}_F}(p)$ is also Archimedean (see [45], Satz 5.2.1, for a more careful version of this argument).

Now $F \in \mathcal{A}_F$, regarded as a function on $\mathcal{V}_{\mathbb{R}}(\ker(\varphi)) \subseteq \mathbb{R}^{n+1}$, is just the polynomial t, i.e., it is the function

$$V(\mathbb{R}) \ni (u, \lambda) \mapsto \lambda.$$

Since λ is an eigenvalue of $F(u)$ by Lemma A.32 (2), and since F is positive definite on S, this is a strictly positive function on $\mathcal{S}_V(p)$, and we can apply Putinar's Positivstellensatz in the form of Corollary A.26 to conclude that F is contained in the quadratic module generated by p in $\mathcal{A}_F$. □

We finally obtain the existence of degree bounds for the matrix polynomial version of Putinar's Theorem just as for Theorem A.23 above, using Kriel's idea from Exercise A.24. This result was first proven in [42], in a less elementary way. There are some more approaches to the result, which even contain some information about the degree-bounding function D. The theses [49] and [45] give very good accounts of these results.

Here, the degree of a matrix polynomial is defined as the maximum degree of its entries (which is the same as the usual degree, when considered as a polynomial with matrix coefficients).

Theorem A.35 (Putinar's Theorem for Matrix Polynomials with Degree Bounds)
Let $p = (p_1, \ldots, p_r)$ be an Archimedean description of the set $S = \mathcal{S}(p)$. Given $\delta > 0$, every matrix polynomial $F \in \mathrm{Sym}_s \mathbb{R}[x]$ satisfying $F(u) \geqslant \delta I_s$ for all $u \in S$ admits a representation

$$F = S_0 + p_1 S_1 + \cdots + p_r S_r$$

where $S_0, \ldots, S_r \in \mathrm{Sym}_s \mathbb{R}[x]$ are sums of squares, with degrees bounded by

$$\deg(S_i) \leqslant D(p, s, \deg(F), \|F\|, \delta)$$

for $i = 0, \ldots, r$. Again, any norm $\|\cdot\|$ can be chosen, but the function D depends on that choice.

A.4 Model-Theoretic Characterization of Stability

In this section, we will describe some abstract tools that can be used to prove the existence of degree bounds for sums of squares and quadratic modules. We will assume some familiarity with the theory of real closed fields, in particular the Tarski Transfer Principle, which says that a first-order sentence in the language of ordered rings holds in one real closed field, say $\mathbb{R}$, if and only if it holds in *every* real closed field (see for example [10, 63, 83] and also [84]). In fact, we are only interested in extension fields of $\mathbb{R}$ and the corresponding extensions of semialgebraic sets: If S is a semialgebraic subset of $\mathbb{R}^n$, and R is any real closed extension field of $\mathbb{R}$, we write $S(\mathrm{R})$ for the *base extension* of S to R, which is just the subset of R^n described by the same formula[1] as S. Now an important consequence of the Transfer Principle is that $S(\mathrm{R})$ is non-empty if and only if S is non-empty. So unlike the complex numbers, or the algebraic field extensions of $\mathbb{Q}$ studied in number theory, the purpose of real closed extension fields of $\mathbb{R}$ is not to add solutions to polynomial systems of equations and inequalities. The point is rather that solvability remains the same, even though the underlying field may be radically different from $\mathbb{R}$ in other respects. To understand this, recall that in a first-order formula we cannot quantify over the natural numbers or over subsets. This has two important consequences:

(1) The Archimedean axiom

$$\text{“}\forall a \in \mathbb{R}\ \exists n \in \mathbb{N}\colon |a| < n\text{”}$$

is *not* a first order formula, and indeed the interesting real closed extension fields of $\mathbb{R}$ are *non-Archimedean*, in other words, they contain infinitely large and infinitesimal elements.

(2) A statement of the form

“There exists a polynomial such that…”

cannot be encoded in a first-order formula. But instead, for fixed $d \geqslant 0$, a statement of the form

“There exists a polynomial of degree d such that…”

can be encoded, by quantifying its finitely many coefficients. This provides the connection to the degree bounds in quadratic modules that we want to study.

Now, in precise technical terms, here is the statement we will need.

[1] A general semialgebraic set S in $\mathbb{R}^n$ is of the form $S = \bigcup_{i=1}^{k}\{u \in \mathbb{R}^n \mid g_{ij}(u) > 0, h_i(u) = 0 \text{ for all } j = 1,\dots,l\}$ for some finite family of polynomials $g_{ij}, h_i \in \mathbb{R}[x]$. Then the base extension is simply the semialgebraic set $S(\mathrm{R}) := \bigcup_{i=1}^{k}\{u \in \mathrm{R}^n \mid g_{ij}(u) > 0, h_{ij}(u) = 0 \text{ for all } j = 1,\dots,l\}$. Since R is real closed, this is independent of the description, by Tarski's Transfer Principle.

Theorem A.36 ($\aleph_1$-Saturation)
There exists a real closed extension field $\mathbb{R}^*$ of $\mathbb{R}$ with the following property: Every countable cover of a semialgebraic subset of $(\mathbb{R}^*)^n$ by semialgebraic subsets defined over $\mathbb{R}$ possesses a finite subcover. More precisely, any ultrapower $\mathbb{R}^* = \mathbb{R}^{\mathbb{N}}/\mathcal{F}$, where $\mathcal{F}$ is a non-principal ultrafilter on $\mathbb{N}$, has this property.

Proof See [83], Theorem 2.2.11. □

The ultrapower $\mathbb{R}^*$ can be written down more or less explicitly, assuming that a non-principal ultrafilter $\mathcal{F}$ on $\mathbb{N}$ is given (but it lies in the nature of non-principal ultrafilters that they exist only by virtue of the axiom of choice, so one cannot actually write one down). To relieve the somewhat ethereal nature of the argument, and to help understand what is going on here, we look at a more concrete example.

Example A.37
Let $S_i = [-i, i] \subseteq \mathbb{R}$ and consider the ascending chain of closed intervals $S_1 \subseteq S_2 \subseteq \cdots$ in $\mathbb{R}$. Of course, this chain is not stationary, yet

$$\bigcup_{i \in \mathbb{N}} S_i = \mathbb{R}$$

is semialgebraic. So $\mathbb{R}$ itself is clearly not $\aleph_1$-saturated. However, it is not hard to show that the rational function field $\mathbb{R}(t) = \mathrm{Quot}(\mathbb{R}[t])$ has a unique ordering in which t is infinitely large, i.e., larger than any constant from $\mathbb{R}$. Using this ordering, we consider

$$\bigcup_{i \in \mathbb{N}} S_i(\mathbb{R}(t)) = \{f \in \mathbb{R}(t) \mid \exists i \in \mathbb{N}\colon \ \pm f < i\},$$

which is also called the *order-convex hull* of $\mathbb{Z}$ in $\mathbb{R}(t)$. This is *not* a semialgebraic subset of $\mathbb{R}(t)$, by Exercise A.38. The non-Archimedean field $\mathbb{R}(t)$ is tiny compared to the ultrapower $\mathbb{R}^*$, but this example captures the nature of the compactness of $\mathbb{R}^*$ in the above theorem.

Exercise A.38
Prove that the order-convex hull of $\mathbb{Z}$ in $\mathbb{R}(t)$ is not semialgebraic.

We now present an application to our problem of stability of quadratic modules. Given a finitely generated quadratic module $Q = \mathrm{QM}(p)$ in $\mathbb{R}[x]$, and a real closed extension field R of $\mathbb{R}$, consider the quadratic module $\mathrm{QM}_{\mathrm{R}}(p)$ generated by p in $\mathrm{R}[x]$. It turns out that Q is stable if and only if $\mathrm{QM}_{\mathrm{R}}(p)$ is the base extension of Q to R (suitably defined), for all $\mathrm{R}/\mathbb{R}$. The result is taken from [95].

Proposition A.39
For a finitely generated quadratic module $Q = \mathrm{QM}(p)$ in $\mathbb{R}[x]$, the following are equivalent:

1. *Q is stable.*
2. *For all real closed extension fields R of $\mathbb{R}$, and for all $d \geqslant 0$, the set*

$$\mathrm{QM}_{\mathrm{R}}(p) \cap \mathrm{R}[x]_d$$

   is semialgebraic over $\mathbb{R}$.
3. *The cone $Q_d := Q \cap \mathbb{R}[x]_d$ is semialgebraic for all $d \geqslant 0$, and for all real closed extension fields R of $\mathbb{R}$, $\mathrm{QM}_{\mathrm{R}}(p) \cap \mathrm{R}[x]_d$ coincides with the base extension $Q_d(\mathrm{R})$ of Q_d to R.*

Proof (1)⇒(3): If Q is stable, we have $Q_d = \mathrm{QM}_{[e]}(p) \cap \mathbb{R}[x]_d$ for some $e \in \mathbb{N}$. So we can define Q_d by an explicit first order formula, saying

"There exists a quadratic module representation with degrees bounded by e."

In particular, Q_d is semialgebraic. In addition, for each $k \in \mathbb{N}$, there is a valid statement over $\mathbb{R}$, saying

"Whenever a polynomial of degree d has a representation in $\mathrm{QM}_{[k]}(p)$, then it has one in $\mathrm{QM}_{[e]}(p)$."

By Tarski's Transfer Principle, these statements thus also hold over R, and we immediately obtain the desired statement

$$\mathrm{QM}_{\mathrm{R}}(p) \cap \mathrm{R}[x]_d = Q_d(\mathrm{R}).$$

(3)⇒(2) is clear. For (2)⇒(1), let $\mathrm{R} = \mathbb{R}^*$ be a saturated real closed extension field as in Theorem A.36. Now $\mathrm{QM}_{\mathrm{R}}(p) \cap \mathrm{R}[x]_d$ is semialgebraic over $\mathbb{R}$ by assumption, and it is clearly covered by the ascending chain of semialgebraic sets defined by the formulas saying

"There exists a quadratic module representation with degrees bounded by k".

So by Theorem A.36 one fixed such k suffices, and the same is then true over $\mathbb{R}$, by Tarski's Transfer Principle. This is stability of Q. □

A.5 Sums of Squares on Compact Curves and Base Extension

The goal of this section is to sketch the proof of Theorem 3.30, which says that the saturated preordering of a compact subset of a smooth curve over $\mathbb{R}$ remains saturated when going up to a real closed extension field. Using the results from the previous section, this even implies stability of these finitely generated preorderings.

Example A.40
Consider the situation on the real line, where everything is completely elementary. For the unit interval $S = [-1, 1]$, the saturated preordering $P = \mathcal{P}(S)$ in $\mathbb{R}[x]$ is generated by the polynomial $1 - x^2$. This is not hard to show directly. If $\mathrm{R}/\mathbb{R}$ is a real closed field extension, the preordering $\mathrm{PO}_{\mathrm{R}}(1 - x^2)$ in $\mathrm{R}[x]$ is still saturated. But that is because we can just do the same direct proof over R, not because of what we know over $\mathbb{R}$.

If Z is an affine curve over $\mathbb{R}$, the situation is far more complicated. One of the main results in [94] says, that if Z is smooth and $P = \mathrm{PO}(p_1, \ldots, p_r)$ is a finitely generated preordering in $\mathbb{R}[Z]$ defining a bounded set $S = \mathcal{S}(p_1, \ldots, p_r) \subseteq Z(\mathbb{R})$, then P is saturated if and only if in each boundary point u of S in $Z(\mathbb{R})$ one of the generators p_i vanishes to order 1 (or two generators with opposing sign changes if u is an isolated point, see [94], Theorem 5.17). We can see this in the above example too: The polynomial $1 - x^2$ has simple roots 1 and -1, whereas $(1 - x^2)^3$ has triple roots and therefore does not generate $\mathcal{P}([-1, 1])$, as we showed in Example A.18. However, the same statement does not hold for curves over a non-Archimedean real closed field R. The proof of Theorem 3.30 requires completely new techniques. But it turns out that finding certain elements in P_{R} with simple zeros on the boundary of S still plays a role in the proof.

Let A be a commutative ring with $\frac{1}{2} \in A$, and let P be a preordering of A. In this generality, P is called *Archimedean* if for every $a \in A$ there exists $n \in \mathbb{Z}$ such that $n \pm a \in P$. It is a consequence of Schmüdgen's Theorem that this definition agrees with the one given earlier for preorderings in the polynomial ring. If $\mathfrak{p}$ is a prime ideal in A, we let $P_{\mathfrak{p}}$ denote the preordering generated by P in the localisation $A_{\mathfrak{p}}$. Explicitly, we have

$$P_{\mathfrak{p}} = \left\{ \frac{a}{b^2} \;\middle|\; a \in P, b \in A \setminus \mathfrak{p} \right\}.$$

We will use the following local-global principle, developed in [96].

Theorem A.41 (Archimedean Local-Global Principle)
Let A be a ring containing $\frac{1}{2}$, and let P be an Archimedean preordering of A. Then an element $a \in A$ is contained in P if and only if it is contained in $P_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$ of A.

Proof See [96], Theorem 2.8, or [63], Theorem 9.6.2. □

To study the local preorderings $P_{\mathfrak{m}}$, we will need to work in the real spectrum. We quickly recall the basics and fix notations: The points of the *real spectrum* $\operatorname{Sper} A$ of A are the pairs

$$(\mathfrak{p}, \leqslant)$$

where $\mathfrak{p}$ is a prime ideal of A, and $\leqslant$ is an ordering of the residue field $\operatorname{Quot}(A/\mathfrak{p})$. If $\alpha = (\mathfrak{p}, \leqslant)$ is a point in $\operatorname{Sper} A$, the prime ideal $\mathfrak{p}$ is called the *support* of α, denoted $\operatorname{supp}(\alpha)$. An element $a \in A$ is regarded as a function on $\operatorname{Sper} A$, and

$$a(\alpha) \geqslant 0 \text{ or } a(\alpha) > 0$$

means that the class of a in

$$\operatorname{Quot}(A/\operatorname{supp}(\alpha))$$

is non-negative (strictly positive) with respect to the ordering given by α.

If $A_{\mathfrak{p}}$ is the localization of A in a prime ideal $\mathfrak{p}$, the prime ideals of $A_{\mathfrak{p}}$ are in canonical bijection with the prime ideals of A contained in $\mathfrak{p}$. Accordingly, the real spectrum $\operatorname{Sper} A_{\mathfrak{p}}$ can be considered as an open subset of $\operatorname{Sper} A$, consisting of those points $\alpha \in \operatorname{Sper} A$ with $\operatorname{supp}(\alpha) \subseteq \mathfrak{p}$. Given a preordering P of A, we write

$$X_P := \left\{\alpha \in \operatorname{Sper} A \mid a(\alpha) \geqslant 0 \text{ for all } a \in P\right\}$$

for the subset of $\operatorname{Sper} A$ defined by P.

If V is an affine $\mathbb{R}$-variety, the real points $V(\mathbb{R})$ can be identified with a subset of $\operatorname{Sper} \mathbb{R}[V]$. Namely, each point $v \in V(\mathbb{R})$ corresponds to a maximal ideal $\mathfrak{m}_v$ in $\mathbb{R}[V]$, and the residue field

$$\mathbb{R}[V]/\mathfrak{m}_v \cong \mathbb{R}$$

has a unique order, so that there is a unique point $\alpha_v \in \operatorname{Sper} \mathbb{R}[V]$ with $\operatorname{supp}(\alpha_v) = \mathfrak{m}_v$. If P is a finitely generated preordering in $\mathbb{R}[V]$, then clearly $\mathcal{S}(P) \subseteq X_P$ under this identification.

The following proposition will be very important to reduce the number of maximal ideals we need to consider, when applying the Archimedean local global principle.

Proposition A.42
Let A be a local ring with $\frac{1}{2} \in A$, and let P be a preordering in A. Then every $a \in A$ with $a > 0$ on $X_P \subseteq \operatorname{Sper} A$ is contained in P.

Proof See [97], Proposition 2.1. □

Let $\mathrm{R}/\mathbb{R}$ be a real closed field extension. Again we denote by $\mathcal{O}$ the convex hull of $\mathbb{Z}$ in R. It is a subring of R, which clearly has the property that $a \in \mathcal{O}$ or $a^{-1} \in \mathcal{O}$ holds for every $a \in \mathrm{R} \setminus \{0\}$. It is therefore a *valuation ring*, i.e., there is a valuation

$$\nu \colon \mathrm{R} \to \Gamma \cup \{\infty\}$$

into some ordered group Γ, such that

$$\mathcal{O} = \{a \in \mathrm{R} \mid \nu(a) \geqslant 0\}$$

is the valuation ring of ν. The residue field of $\mathcal{O}$ modulo its maximal ideal

$$\mathfrak{m} = \{a \in \mathcal{O} \mid \nu(a) > 0\}$$

is just $\mathbb{R}$, and we denote the residue class map $\mathcal{O} \to \mathcal{O}/\mathfrak{m} \cong \mathbb{R}$ by $a \mapsto \overline{a}$.

Now let V be an irreducible affine variety over $\mathbb{R}$ with coordinate ring $\mathbb{R}[V]$, and consider the base extension of V to R with coordinate ring $\mathrm{R}[V] = \mathbb{R}[V] \otimes_{\mathbb{R}} \mathrm{R}$. The valuation ν extends to a map $\omega \colon \mathrm{R}[V] \to \Gamma \cup \{\infty\}$ as follows: Given an element

$$p = \sum_{i=1}^{r} p_i \otimes a_i \in \mathrm{R}[V] = \mathbb{R}[V] \otimes_{\mathbb{R}} \mathrm{R}$$

with $p_i \in \mathbb{R}[V]$, $a_i \in \mathrm{R}$, we define

$$\omega(p) = \min\{\nu(a_i) \mid i = 1, \ldots, r\}.$$

One can check that ω has the same formal properties as ν, i.e.,

$$\omega(p+q) \geqslant \min\{\omega(p), \omega(q)\} \text{ and}$$
$$\omega(pq) = \omega(p) + \omega(q)$$

for all $p, q \in \mathrm{R}[V]$. The first is clear from the definition. The second is not difficult to show, but uses the fact that $\mathbb{R}[V]$ is integral.

We will mostly work in the coordinate ring of V with coefficients in the valuation ring, which is the ring

$$\mathcal{O}[V] = \mathbb{R}[V] \otimes_{\mathbb{R}} \mathcal{O} = \{p \in \mathrm{R}[V] \mid \omega(f) \geqslant 0\}.$$

The residue map of $\mathcal{O}$ extends to a homomorphism $\mathcal{O}[V] \to \mathbb{R}[V]$, denoted $p \mapsto \overline{p}$, defined coefficient wise, i.e.,

$$\overline{\sum_i p_i \otimes a_i} = \sum_i p_i \otimes \overline{a}_i = \sum_i \overline{a}_i p_i \in \mathbb{R}[V].$$

Clearly, $\overline{p} = 0$ if and only if $\omega(p) > 0$. Likewise, the residue map also induces a map on points

$$V(\mathcal{O}) \to V(\mathbb{R})$$
$$v \mapsto \overline{v}.$$

In fact, if $v \in V(\mathcal{O})$ is regarded as an $\mathbb{R}$-algebra homomorphism $v\colon \mathbb{R}[V] \to \mathcal{O}$, then $\overline{v}$ is just the composition of v with the residue map $\mathcal{O} \to \mathbb{R}$. If coordinates are fixed and $v = (v_1, \ldots, v_n)$ is regarded as a tuple in $\mathcal{O}^n$, then $\overline{v} = (\overline{v}_1, \ldots, \overline{v}_n) \in \mathbb{R}^n$.

Lemma A.43
Let A be an $\mathbb{R}$-algebra, and P an Archimedean preordering in A. Let $\mathrm{R}/\mathbb{R}$ be a real closed extension, and $\mathcal{O}$ the convex hull of $\mathbb{Z}$ in R. *Then the preordering $P_{\mathcal{O}}$ generated by P in $A \otimes_{\mathbb{R}} \mathcal{O}$ is again Archimedean.*

Proof Note first that we have $P - P = A$. For given $a \in A$, choose $c \in \mathbb{R}$ with $c \pm a \in P$, and observe

$$a = \frac{1}{2}(c + a) - \frac{1}{2}(c - a) \in P - P.$$

It follows that any $p \in A \otimes_{\mathbb{R}} \mathcal{O}$ can be written in the form $p = \sum_{i=1}^{r} p_i \otimes a_i$ with $p_i \in P$ and $a_i \in \mathcal{O}$ for $i = 1, \ldots, r$. Now choose $0 < c_1 \in \mathbb{R}$ with $c_1 - p_i \in P$ for all i. Also, by definition of $\mathcal{O}$, there exists $c_2 \in \mathbb{R}$ with $a_i < c_2$ for all i, hence $c_2 - a_i$ is a square in $\mathcal{O}$. Now we can write

$$rc_1c_2 - p = \sum_{i=1}^{r} c_1c_2 \otimes 1 - \sum_{i=1}^{r} p_i \otimes a_i \underbrace{- \sum_{i=1}^{r} c_2 p_i \otimes 1 + \sum_{i=1}^{r} p_i \otimes c_2}_{=0}$$
$$= c_2 \sum_{i=1}^{r} (c_1 - p_i) \otimes 1 + \sum_{i=1}^{r} p_i \otimes (c_2 - a_i)$$

which is an element of $P_{\mathcal{O}}$, showing that $P_{\mathcal{O}}$ is Archimedean. □

We are now ready to present at least the main ideas in the proof of Theorem 3.30, which we restate.

Theorem
Let Z be a smooth affine curve over $\mathbb{R}$, let $S \subseteq Z(\mathbb{R})$ be a compact semialgebraic subset, and let $P = \mathcal{P}(S) \subseteq \mathbb{R}[Z]$ be the saturated preordering of S. For any real closed field extension $\mathrm{R}/\mathbb{R}$, the preordering P_{R} generated by P in $\mathrm{R}[Z]$ is again saturated.

Proof Let $p \in \mathrm{R}[Z]$ with $p \geqslant 0$ on $S(\mathrm{R}) = \mathcal{S}(P_{\mathrm{R}})$. First, we can find $c \in \mathrm{R}$ with $\omega(p) = \nu(c^2)$, which implies $\omega(c^{-2}p) = 0$ and $c^{-2}p \in \mathcal{O}[Z]$. Since $c^{-2}p \in P_{\mathcal{O}}$ clearly implies $p \in P_{\mathrm{R}}$, we may assume $p \in \mathcal{O}[Z]$ with $\omega(p) = 0$. We show $p \in P_{\mathcal{O}}$ in several steps.

1. We wish to apply the Archimedean local global principle Theorem A.41 for $P_{\mathcal{O}} \subseteq \mathcal{O}[Z]$. Since S is bounded, $P = \mathcal{P}(S)$ is Archimedean, and so $P_{\mathcal{O}}$ is Archimedean by Lemma A.43. So we only need to show that p is contained in $(P_{\mathcal{O}})_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$ of $\mathcal{O}[Z]$.
2. The next step is to show that since p is non-negative on $S(\mathrm{R}) \subseteq Z(\mathrm{R})$, it is indeed non-negative on the corresponding subset $X_{P_{\mathcal{O}}}$ of $\operatorname{Sper}\mathcal{O}[Z]$. We will omit the proof of this fact (see [98], Lemma 3.7).
3. Now localizing in a maximal ideal $\mathfrak{m}$, we know that p is non-negative on the subset

$$X_{P,\mathfrak{m}} := X_{P_{\mathcal{O}}} \cap \operatorname{Sper}\mathcal{O}[Z]_{\mathfrak{m}}$$

of $\operatorname{Sper}\mathcal{O}[Z]$. If $p > 0$ on $X_{P,\mathfrak{m}}$, then $p \in P_{\mathfrak{m}}$ by Proposition A.42. So we need only consider those maximal ideals $\mathfrak{m}$ of $\mathcal{O}[Z]$ that contain p and with $X_{P,\mathfrak{m}} \neq \emptyset$. We claim that such a maximal ideal $\mathfrak{m}$ is necessarily of the form

$$\mathfrak{m}_z = \left\{ \sum_i p_i \otimes a_i \mid \sum_i \overline{a}_i p_i(z) = 0 \text{ in } \mathbb{R} \right\},$$

the maximal ideal of $\mathcal{O}[Z]$ corresponding to a point $z \in S$. To see this, let $\alpha \in X_{P,\mathfrak{m}}$ with $\operatorname{supp}(\alpha) \subseteq \mathfrak{m}$ and let

$$\mathfrak{p} = \operatorname{supp}(\alpha) \cap \mathbb{R}[Z],$$

a prime ideal in $\mathbb{R}[Z]$. Since $w(p) = 0$, we cannot have $\operatorname{supp}(\alpha) \subseteq \mathbb{R}[Z] \otimes \mathfrak{m}$, and so $\mathfrak{p} \neq (0)$. Since Z is a curve, this implies that $\mathfrak{p}$ is a maximal ideal of $\mathbb{R}[Z]$

and therefore corresponds to a point $z \in Z$. Since $\alpha \in X_{P,\mathfrak{m}}$, the point z must be real and contained in S. Now $I := \mathfrak{p} \otimes \mathcal{O}$ is an ideal of $\mathcal{O}[Z]$ with

$$I \subseteq \operatorname{supp}(\alpha) \subseteq \mathfrak{m}$$

and $\mathcal{O}[Z]/I = \mathcal{O}$. So $\mathcal{O}[Z]/\mathfrak{m}$ is a field containing $\mathbb{R}$ and contained in $\mathcal{O}$, hence it is Archimedean and thus coincides with $\mathbb{R}$. Thus $\mathfrak{m}$ corresponds to a real point, which must be the point z.

4. Our goal is to find an element $q \in P_{\mathcal{O}}$ with the property that $g = p/q$ is a unit in $\mathcal{O}[Z]_{\mathfrak{m}}$, and such that g is positive on $X_{P,\mathfrak{m}}$. When that is done, we can apply Lemma A.42 and conclude $g \in P_{\mathfrak{m}}$, hence $p = gq \in P_{\mathfrak{m}}$, completing the proof.
5. Showing the existence of q as in (4) is the most technical part of the proof and we will only give an outline. First, let $\mathcal{O}^c := \mathcal{O}[\sqrt{-1}]$ and consider the set of $\mathcal{O}^c$-points

$$U(z) = \{\zeta \in Z(\mathcal{O}^c) \mid \overline{\zeta} = z\}$$

that map to z under the residue map. We split the real zeros of p in $U(z)$ into two groups

$$\{\zeta \in U(z) \cap Z(\mathrm{R}) \mid p(\zeta) = 0\} = \{\eta_1, \ldots, \eta_r\} \cup \{\zeta_1, \ldots, \zeta_s\}$$

in such a way that $\eta_1, \ldots, \eta_r \in \operatorname{int}(S(\mathrm{R}))$ and $\zeta_1, \ldots, \zeta_s \notin \operatorname{int}(S(\mathrm{R}))$. Furthermore, let $\{\omega_1, \ldots, \omega_t\} \subseteq U(z)$ be a subset containing exactly one representative from each pair of non-real complex conjugate zeros of p in $U(z)$.

Next, one computes the order of vanishing of $\overline{p}$ at z and finds (see [98], Proposition 3.15)

$$\operatorname{ord}_z(\overline{p}) = \sum_{j=1}^{r} \operatorname{ord}_{\eta_j}(p) + \sum_{k=1}^{s} \operatorname{ord}_{\zeta_k}(p) + 2\sum_{l=1}^{t} \operatorname{ord}_{\omega_l}(p).$$

The crucial point is now to show that for every point $\zeta \in U(z)$ with $p(\zeta) = 0$, there exists an element $q_\zeta \in P_{\mathcal{O}}$ with $\omega(q_\zeta) = 0$, $q_\zeta(\zeta) = 0$ and

$$\operatorname{ord}_z(\overline{q_\zeta}) = \begin{cases} 1 & \text{if } \zeta \in Z(\mathrm{R}), \zeta \notin \operatorname{int}(S(\mathrm{R})) \\ 2 & \text{if } \zeta \in \operatorname{int}(S(\mathrm{R})) \text{ or } \zeta \notin Z(\mathrm{R}). \end{cases}$$

This is proven in [98], Lemma 4.9. Given this, we can define

$$q := \prod_{j=1}^{r} (q_{\eta_j})^{\frac{1}{2}\operatorname{ord}_{\eta_j}(p)} \cdot \prod_{k=1}^{s} (q_{\zeta_k})^{\operatorname{ord}_{\zeta_k}(p)} \cdot \prod_{l=1}^{t} (q_{\omega_l})^{\operatorname{ord}_{\omega_l}(p)} \in P_{\mathcal{O}}.$$

which, by the above computation of $\mathrm{ord}_z(\overline{p})$, satisfies $\mathrm{ord}_z(\overline{q}) = \mathrm{ord}_z(\overline{p})$. This implies that $g = p/q$ is a unit in $\mathcal{O}[Z]_{\mathfrak{m}}$. Since $q \in \mathcal{P}_{\mathcal{O}}$, it is clear that g is non-negative in all points of $X_{P,\mathfrak{m}}$ where q does not vanish. In the zeros of q, one can argue with continuity in $U(z) \cap Z(\mathrm{R})$, except when there are isolated points, which require an additional adjustment of q. □

Convexity

B

In this section we collect some important notions and results from convexity theory and optimization. As general references on convexity, we recommend the books [3, 90, 103], for optimization we refer to [5, 13, 65].

B.1 Convex Cones and Duality

This section covers some basics from convexity theory, mostly about cones and duality. Throughout, let V be a finite-dimensional real vector space. By a *cone* in V, we will always mean a convex cone, i.e., a non-empty subset $C \subseteq V$ such that $u + v \in C$ and $\alpha v \in C$ hold for all $u, v \in C$ and $\alpha \in \mathbb{R}$ with $\alpha \geqslant 0$. In particular, a cone always contains 0. A cone C is called *pointed* if $C \cap (-C) = \{0\}$. For example, Sym_s^+ is a pointed cone in Sym_s, while any non-zero linear subspace of V is an example of a non-pointed cone. Given a subset S of V, we write

$$\mathrm{cone}(S) = \left\{ \sum_{i=1}^{k} \alpha_i u_i \mid u_i \in S, \alpha_i \geqslant 0, k \in \mathbb{N} \right\}$$

for the *conic hull* of S in V, the smallest cone in V containing S.

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T. Netzer, D. Plaumann, *Geometry of Linear Matrix Inequalities*, Compact Textbooks in Mathematics, https://doi.org/10.1007/978-3-031-26455-9

Proposition B.1
If the cone $C \subseteq V$ is closed and pointed, then there exists an affine hyperplane H of V with $0 \notin H$, $C \cap H$ compact, and

$$C = \operatorname{cone}(C \cap H).$$

Conversely, for any compact convex set $S \subseteq \mathbb{R}^n$ with $0 \notin S$, the cone $\operatorname{cone}(S)$ is closed and pointed.

Exercise B.2
Prove Proposition B.1.

In general, if $C \subseteq \mathbb{R}^n$ is non-empty, closed and convex, there is a unique linear subspace $V \subseteq \mathbb{R}^n$, called the *lineality space* of C, with the property

$$C = (C \cap V^\perp) + V$$

and such that $C \cap V^\perp$ does not contain any affine-linear subspace. The proof is contained in the following exercise.

Exercise B.3
Let $C \subseteq \mathbb{R}^n$ be non-empty, closed and convex. Prove the following:

1. If $V \subseteq \mathbb{R}^n$ is a linear subspace with $u + V \subseteq C$ for some $u \in C$, then $u + V \subseteq C$ for *all* $u \in C$.
2. If V is a subspace with the property in (1), then $C = (C \cap V^\perp) + V$.
3. There is a unique maximal subspace with the property in (1), called the *lineality space* of C.
4. If V is the lineality space, then $C \cap V^\perp$ does not contain any affine-linear subspace.

We will write

$$V^* := \operatorname{Hom}(V, \mathbb{R}) = \{\ell \colon V \to \mathbb{R} \text{ linear}\}$$

for the *dual space* of V, whose elements are the *linear functionals* on V. If $\varphi \colon V \to W$ is a linear map between vector spaces, the map

$$\begin{aligned} \varphi^* \colon W^* &\to V^* \\ \ell &\mapsto \ell \circ \varphi \end{aligned}$$

is again linear. By definition, it has the property

$$\ell(\varphi(v)) = (\varphi^*(\ell))(v)$$

for all $v \in V$, $\ell \in W^*$. Exactness of duality or direct computation show

$$\operatorname{im}(\varphi^*) = \{\ell \in V^* \mid \ell|_{\ker(\varphi)} = 0\}. \tag{B.1}$$

A crucial tool in convexity theory is the Separation Theorem for convex sets, see for example [3] for a proof.

Theorem B.4 (Separation Theorem)
Let $S \subseteq V$ be a closed convex set.

1. *For any $u \in V \setminus S$ there exists $\ell \in V^*$ and $\alpha \in \mathbb{R}$ with $\ell(s) > \alpha$ for all $s \in S$ and $\ell(u) < \alpha$.*
2. *For any $u \in V \setminus \operatorname{relint}(S)$ there exists $\ell \in V^*$ and $\alpha \in \mathbb{R}$ with $\ell(s) > \alpha$ for all $s \in \operatorname{relint}(S)$ and $\ell(u) = \alpha$.*

For a subset $S \subseteq V$ we denote by

$$S^\circ := \left\{\ell \in V^* \mid \forall s \in S\colon\ \ell(s) \geqslant -1\right\}$$

the *polar* of S, and by

$$S^\vee := \left\{\ell \in V^* \mid \forall s \in S\colon \ell(s) \geqslant 0\right\}$$

the dual cone of S. Note that $S^\vee \subseteq S^\circ$, S° is a closed convex set containing 0, and $S^\vee$ is a closed convex cone. If $C \subseteq V$ is a cone, then

$$C^\circ = C^\vee,$$

since given $\ell \in C^\circ$ and $c \in C$, we must have

$$\alpha\ell(c) = \ell(\alpha c) \geqslant -1$$

for all $\alpha \geqslant 0$, so $\ell(c) \geqslant 0$. Note also that if $U \subseteq V$ is a linear subspace, then

$$U^\circ = U^\vee = \{\ell \in V^* \mid \ell|_U = 0\} =: U^\perp.$$

But the real connection between polar and dual only becomes clear when looking at homogenizations. If $W \subseteq V$ is a hyperplane and $v \in V \setminus W$, then for $S \subseteq W$ we

have

$$\left\{\ell_{|W} \mid \ell \in V^*, \ell \in (v+S)^\vee, \ell(v) = 1\right\} = \left\{\ell' \in W^* \mid \ell' \in S^\circ\right\}.$$

From this point of view, polar and dual are basically the same.

Note that V is canonically isomorphic to $(V^*)^*$, since V is finite-dimensional. Under this identification, a fundamental fact is *biduality* and *bipolarity*, which can easily deduced from the Separation Theorem B.4 (1).

Theorem B.5 (Bipolar/Biduality Theorem)
For any subset S of V, we have

$$(S^\circ)^\circ = \operatorname{clos}\left(\operatorname{conv}(S \cup \{0\})\right) \text{ and } \left(S^\vee\right)^\vee = \operatorname{clos}\left(\operatorname{cone}(S)\right).$$

In particular, if $S \subseteq V$ is a closed convex subset containing 0, then $(S^\circ)^\circ = S$, and if $C \subseteq V$ is a closed cone, then $\left(C^\vee\right)^\vee = C$.

If V is a finite-dimensional Euclidean space with inner product $\langle \cdot, \cdot \rangle$, we can identify the dual space V^* with V, using the map

$$\begin{aligned} V &\to V^* \\ u &\mapsto \langle u, \cdot \rangle. \end{aligned}$$

Via this identification, the polar/dual of a set $S \subseteq V$ are

$$S^\circ = \{u \in V \mid \forall s \in S\colon\ \langle u, s\rangle \geqslant -1\} \text{ and } S^\vee = \{u \in V \mid \forall s \in S\colon\ \langle u, s\rangle \geqslant 0\}.$$

A cone C is called *selfdual* if $C^\vee = C$.

On a space of matrices, the standard scalar product is given by the trace via

$$\langle A, B\rangle = \operatorname{tr}\left(AB^T\right)$$

for $A, B \in \operatorname{Mat}_{m\times n}$.

Proposition B.6
The cone of real positive semidefinite matrices is selfdual.

Proof Let $A, B \in \mathrm{Sym}_s^+$ and write $A = P^T P$, $B = Q^T Q$ (see Exercise 1.1). Then

$$\langle A, B\rangle = \mathrm{tr}\left(P^T P Q^T Q\right) = \mathrm{tr}\left(Q P^T P Q^T\right) = \mathrm{tr}\underbrace{\left(Q P^T \left(Q P^T\right)^T\right)}_{\geqslant 0} \geqslant 0.$$

Here, we used that the trace is invariant under cyclic permutations, and that positive semidefinite matrices have non-negative trace. Conversely, let $A \in \mathrm{Sym}_s$ with $\langle A, B\rangle \geqslant 0$ for all $B \in \mathrm{Sym}_s^+$. Then

$$v^T A v = \mathrm{tr}(v^T A v) = \mathrm{tr}(A v v^T) = \langle A, v v^T\rangle \geqslant 0$$

for all $v \in \mathbb{R}^s$, so A is positive semidefinite. □

Exercise B.7
Let V and W be finite-dimensional euclidean spaces with inner products $\langle\cdot,\cdot\rangle_V$ and $\langle\cdot,\cdot\rangle_W$. Given a linear map $\varphi\colon V \to W$, show that there is a unique linear map $\varphi^*\colon W \to V$ with

$$\langle\varphi(v), w\rangle_W = \langle v, \varphi^*(w)\rangle_V$$

for all $v \in V$, $w \in W$. Verify that this corresponds to the dual map of φ under the identification $V = V^*$, $W = W^*$ via the scalar product.

Lemma B.8
Let C_1 and C_2 be cones in V. Then

1. $(C_1 + C_2)^\vee = C_1^\vee \cap C_2^\vee$.
2. If C_1 and C_2 are closed, then $(C_1 \cap C_2)^\vee = \mathrm{clos}(C_1^\vee + C_2^\vee)$.

Proof (1) is immediate. For (2), we use biduality and (1) to conclude

$$\mathrm{clos}\left(C_1^\vee + C_2^\vee\right) = \left(C_1^\vee + C_2^\vee\right)^{\vee\vee} = \left(C_1^{\vee\vee} \cap C_2^{\vee\vee}\right)^\vee = (C_1 \cap C_2)^\vee. \qquad \square$$

Exercise B.9
Find an example of two closed cones C_1 and C_2 in $\mathbb{R}^3$, such that $C_1 + C_2$ is not closed.

Lemma B.10
Let $C \subseteq V$ and $D \subseteq W$ be cones and $\varphi\colon V \to W$ a linear map. Then

1. $\varphi(C)^\vee = (\varphi^*)^{-1}\left(C^\vee\right)$.
2. $\varphi^{-1}(D)^\vee = \varphi^*\left((D \cap \mathrm{im}(\varphi))^\vee\right)$.

Proof (1) follows directly from the equality $\ell(\varphi(v)) = \varphi^*(\ell)(v)$ for all $v \in V$, $\ell \in W^*$.

For (2) let $\ell \in (D \cap \mathrm{im}(\varphi))^\vee \subseteq W^*$. Then

$$\varphi^*(\ell)(v) = \ell(\varphi(v)) \geqslant 0$$

holds for all $v \in \varphi^{-1}(D)$, hence $\varphi^*(\ell) \in \varphi^{-1}(D)^\vee$. Conversely, since $\ker(\varphi) \subseteq \varphi^{-1}(D)$, any $\ell \in \varphi^{-1}(D)^\vee$ must vanish on $\ker(\varphi)$, and is therefore in the image of φ^* (see Eq. (B.1)). Then if $\ell = \varphi^*(\ell')$, we have

$$\ell'(\varphi(v)) = \varphi^*(\ell')(v) = \ell(v) \geqslant 0$$

whenever $\varphi(v) \in D$, hence $\ell' \in (D \cap \mathrm{im}(\varphi))^\vee$. □

Lemma B.11
Let C be a closed cone in V and U a linear subspace of V.

1. *If $\mathrm{int}(C) \cap U \neq \emptyset$, then $C^\vee \cap U^\perp = \{0\}$.*
2. *If C is pointed and $C \cap U = \{0\}$, then $C + U$ is closed.*

Proof Let $\pi : V \to V/U$ be the canonical projection.

(1) If $u \in \mathrm{int}(C) \cap U$, then $0 = \pi(u)$ is an interior point of $\pi(C)$ in V/U. Thus $\pi(C) = V/U$ and $C + U = \pi^{-1}(\pi(C)) = V$. Thus

$$C^\vee \cap U^\perp = (C + U)^\vee = V^\vee = \{0\},$$

where we have used Lemma B.8.

(2) Since the cone C is closed and pointed, there is a compact convex subset S of V, not containing 0, with $C = \mathrm{cone}(S)$, by Proposition B.1. It follows from the hypothesis that

$$S \cap \ker(\pi) = \emptyset,$$

so that $\pi(S)$ is again compact and convex with $0 \notin \pi(S)$. Hence $\pi(C) = \mathrm{cone}(\pi(S))$ is closed by Proposition B.1, and so is $C + U = \pi^{-1}(\pi(C))$.

□

B.2 Faces and Dimension

We now explain the notion of dimension for convex sets, as well the concept of faces. Both notions are often very helpful when examining convex sets. Again, let V be a finite-dimensional real vector space.

Definition B.12

Let $S \subseteq V$ be convex. We define the *dimension* $\dim(S)$ *of* S as the dimension of the affine hull $\mathrm{aff}(S)$ of S in V.

Exercise B.13
Prove that the convex set S has nonempty interior in $\mathrm{aff}(S)$.

Definition B.14

Let $S \subseteq V$ be closed and convex.

1. A *face* of S is a convex subset $F \subseteq S$, with the property that $\alpha u + (1-\alpha)v \in F$ for $u, v \in S$ and some $\alpha \in (0, 1)$ implies $u, v \in F$.
   In words, if a line segment within S has interior points in F, then it is fully contained in F.
2. An *extreme point of* S is a point $u \in S$ such that $\{u\}$ is a face of S.
3. A face F is called *proper* if $F \neq \emptyset, S$. It is called *exposed* if it is cut out by a supporting hyperplane, i.e., if there exists $\ell \in V^*$ and $\alpha \in \mathbb{R}$ with $\ell|_S \geqslant \alpha$ and

$$F = \{u \in S \mid \ell(u) = \alpha\}.$$

   Otherwise, F is called *non-exposed*.[1]

Example B.15
The faces of a polyhedron $\mathcal{S}(\ell_1, \ldots, \ell_r)$ are exactly the vertices, edges, etc., defined by the vanishing of a subset of $\ell_1, \ldots, \ell_r$, and they are all exposed.

[1] Unfortunately, the terminology here is not uniform in the literature. It is equally common to call *face* what we call *exposed face* and use another term (e.g. *facelet* or *extremal convex subset*) for what we call *face*.

Exercise B.16
Let $S \subseteq V$ be a closed convex set.

1. Show that any intersection of faces of S is a face of S.
2. Show that no proper face of S contains a point of $\operatorname{relint}(S)$.
3. Show that a convex subset F of S is a face if and only if $S \setminus F$ is convex and any convex subset of S containing F has strictly greater dimension than F.

Exercise B.17
Let $S \subseteq V$ be a closed convex set.

1. Show that for a proper face $F \subsetneq S$ we have $\dim(F) < \dim(S)$.
2. Show that any intersection of a family of faces of S is already obtained as a finite intersection of faces from the family.
3. Show that any intersection of exposed faces of S is an exposed face of S.

Proposition B.18
Let $S \subseteq V$ be closed and convex, and let F be a face of S. Then the following holds:

1. *F is closed.*
2. *Every face of F is also a face of S.*
3. *If $u \in \operatorname{relint}(F)$ and $\ell \in V^*$ with $\ell|_S \geqslant \alpha$ and $\ell(u) = \alpha$, then $\ell|_F \equiv \alpha$.*
4. *Any point in the relative boundary of S is contained in a proper exposed face of S.*
5. *For any point $u \in S$ there exists a unique face of S containing u in its relative interior. This is precisely the smallest face of S containing u.*
6. *F is exposed if and only if for every face F' strictly containing F, there is $\ell \in V^*$ with $\ell|_S \geqslant \alpha$, such that $\ell|_F \equiv \alpha$ but $\ell|_{F'} \not\equiv \alpha$.*

Proof (1)–(3) Exercise.

(4) Let $u \in S \setminus \operatorname{relint}(S)$. By the Separation Theorem B.4 (2), there exists $\ell \in V^*$ and $\alpha \in \mathbb{R}$ with $\ell(u) = \alpha$ and $\ell > \alpha$ on $\operatorname{relint}(S)$. Thus

$$\{s \in S \mid \ell(s) = \alpha\}$$

is a suitable exposed face of S.

(5) Let F be the intersection of all faces containing u. Then F is a face of S by Exercise B.16 (1), and therefore obviously the smallest face containing u. If u were not contained in the relative interior of F, it would be contained in a proper face of F (by (4)), which would also be a face of S (by (2)), a contradiction. If F' is another

face containing u, then $F \subseteq F'$ by definition of F. So if $F \subsetneq F'$, then u cannot be contained in the relative interior of F'.

(6) If F is exposed, there exists such ℓ by definition. Conversely, let F be a face of S with this property. We claim that F is the intersection of all exposed faces of S that contain F. The statement then follows from Exercise B.17 (3). Denote this intersection by F'. Then $F \subseteq F'$ is clear, using (4). In case $F \subsetneq F'$ we could choose $\ell \in V$ as in the assumption. Then the exposed face of S defined by ℓ contains F but not F', a contradiction. □

Exercise B.19
Let $S \subseteq V$ be a closed convex set. Show that all maximal proper faces of S are exposed.

B.3 Semidefinite Programming

We now give a very brief introduction into the theory of semidefinite programming. Since spectrahedra and their shadows are precisely the feasible sets for these types of optimization problems (as we will see), a short such section must not be missing from this book. A *semidefinite program* is a convex optimization problem that can be written in the following form, usually called the *primal form*:

$$\text{(P)} \qquad \left.\begin{array}{ll} \text{Find} & \inf\langle B, X\rangle \\ \text{subject to} & \langle A_i, X\rangle = c_i \text{ for } i = 1, \ldots, n \\ & X \succcurlyeq 0. \end{array}\right\} \text{ in the variable } X \in \mathrm{Sym}_s,$$

where $A_1, \ldots, A_n, B \in \mathrm{Sym}_s$ and $c \in \mathbb{R}^n$ are given. Thus the problem is to compute the minimum (or infimum) of the linear function $X \mapsto \langle B, X\rangle$ on the space of symmetric matrices of size s, under the constraint that X should be contained in the spectrahedron defined by the linear equations $\langle A_i, X\rangle = c_i$. Here, the spectrahedron is seen as an affine section of the psd cone, i.e., as a set of matrices as in Remark 2.7 (2).

Starting in the 1990s, efficient algorithms for solving semidefinite programs have been developed, based on so-called *interior-point methods*. This is the main reason for the current interest in spectrahedra, but is outside the scope of this book (see for example [27, 78, 101, 102, 105] for much more information). But let us mention that duality plays an extremely important role in solving semidefinite programs. To the primal semidefinite program (P) above, there is a corresponding *dual program*

$$\text{(D)} \qquad \left.\begin{array}{ll} \text{Find} & \sup\langle c, y\rangle \\ \text{subject to} & B - \sum_{i=1}^n y_i A_i \succcurlyeq 0 \end{array}\right\} \text{ in the variable } y \in \mathbb{R}^n.$$

Here we optimize the linear function $y \mapsto \langle c, y\rangle$ over the spectrahedron in $\mathbb{R}^n$ defined by the linear matrix inequality $B - \sum_i y_i A_i \succcurlyeq 0$, as in Remark 2.7 (1). It is easily checked that the optimal value of the dual program is always less or equal to the optimal value of the primal problem. This already provides an error bound when

solving both problems at the same time numerically. The usefulness of duality in solving semidefinite programs also stems from optimality results like the following.

Theorem B.20 (Duality Theorem)
Consider the semidefinite programs (P) and (D) above. Assume that the matrices $A_1, \ldots, A_n$ are linearly independent and that both (P) and (D) possess strictly feasible points. Then there is no duality gap, meaning the optimal values of both problems coincide. Furthermore, X' is an optimal solution of (P) and y' an optimal solution of (D) if and only if

$$\langle X', B - \sum_{i=1}^{n} y_i' A_i \rangle = 0.$$

Proof See for example [3], Section IV.10. □

Here, a strictly feasible point of (P) is a positive definite matrix X satisfying the constraints in (P), and for (D) it is some $y \in \mathbb{R}^n$ with $B - \sum_i y_i A_i > 0$. This duality theorem provides only one example of various assumptions one can make on the dual pair (P), (D) implying that there is no duality gap. The condition here is usually called the *interior point* or *Karush-Kuhn-Tucker* condition.

Now finally suppose we are given a general convex programming problem of the form

$$\left.\begin{array}{ll} \text{Find} & \inf \ell(u) \\ \text{subject to} & u \in S \end{array}\right\} \text{ in the variable } u \in \mathbb{R}^n$$

where S is some convex subset of $\mathbb{R}^n$ and ℓ a linear functional. If we wish to apply semidefinite programming methods, it makes little difference whether we can represent S as a spectrahedron $S = \mathcal{S}(M)$, or only as a spectrahedral shadow $S = \pi_x \mathcal{S}(M(x, y))$. In either case, we just solve the program for the spectrahedron $\mathcal{S}(M)$. What matters much more is how and whether we can actually find the representing linear matrix polynomial M, and whether the size of the matrices and the number of extra variables y are not too large. So we should always keep the following in mind:

For optimization, lifted representations are no worse than non-lifted representations, even though the geometry is very different.

Let us conclude this section with an exercise relating semidefinite programming to general conic programming.

Exercise B.21
Let $C_1 \subseteq V_1$ and $C_2 \subseteq V_2$ be cones in finite-dimensional Euclidean spaces V_1 and V_2 with inner products $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$. A linear map $\varphi \colon V_1 \to V_2$ and elements $b \in V_1$ and $c \in V_2$ define the following dual pair of optimization problems:

$$\left.\begin{array}{ll} \text{Find} & \inf \langle b, x \rangle_1 \\ \text{subject to} & \varphi(x) - c \in C_2 \\ & x \in C_1 \end{array}\right\} \text{ in } x \in V_1 \qquad \left.\begin{array}{ll} \text{Find} & \sup \langle c, y \rangle_2 \\ \text{subject to} & b - \varphi^*(y) \in C_1^\vee \\ & y \in C_2^\vee \end{array}\right\} \text{ in } y \in V_2.$$

Verify that the optimal value of the second problem is always less or equal to the optimal value of the first problem. Then check that duality of semidefinite programming is a special case of this general setup.

B.4 Lagrange Multipliers and Convex Optimization

The basic idea in Sect. 3.5 is to study representations of (non-linear) Lagrange functions of a linear polynomial, in order to represent it as an element of a quadratic module. To explain this, we need a bit of background and terminology from general (polynomial) optimization. The following is a special case of the *Karush-Kuhn-Tucker* theorem.

Theorem B.22
Let $S = \mathcal{S}(p_1, \ldots, p_r) \subseteq \mathbb{R}^n$ be a basic closed set. Let $q \in \mathbb{R}[x]$ and assume that $u \in S$ is a point in which q attains its minimum on S. Assume further that there is $v \in \mathbb{R}^n$ with

$$\begin{cases} \langle \nabla p_i(u), v \rangle > 0 \text{ if } \deg(p_i) \geqslant 2 \\ \langle \nabla p_i(u), v \rangle \geqslant 0 \text{ if } \deg(p_i) \leqslant 1 \end{cases}$$

whenever $p_i(u) = 0$. Then there exist $\lambda_1, \ldots, \lambda_r \geqslant 0$ such that

$$\nabla q(u) = \sum_{i=1}^{r} \lambda_i \nabla p_i(u)$$

$$\lambda_i p_i(u) = 0 \quad \text{for all } i = 1, \ldots, r.$$

(The same holds if $q, p_1, \ldots, p_r$ are just continuously differentiable functions.)

The constants $\lambda_1, \ldots, \lambda_r$ are called *Lagrange multipliers* for q at the *minimizer* u. The second statement, which just says that the Lagrange multipliers for inactive inequalities are zero, is called *complementary slackness*. There are a number of

conditions, called *constraint qualifications*, implying the existence of Lagrange multipliers in a minimizer. The one stated here is the *Mangasarian-Fromowitz* constraint qualification. We prove the theorem only in the special case where the objective function q is linear. This is all we need in Sect. 3.5. See for example [27], Theorem 2.2.5 et seq. for a full proof.

Proof of Theorem B.22 Let $p = \ell \in \mathbb{R}[x]_1$ and let $u \in S$ be a minimiser of ℓ on S. It is not restrictive to assume that $u = 0$ and $\ell(u) = 0$. Furthermore, we may assume that $p_1(0) = \cdots = p_m(0) = 0$ and $p_{m+1}(0), \ldots, p_r(0) > 0$ for some $m \geqslant 1$, i.e., exactly the first m inequalities are active at 0. We show that there are $\lambda_1, \ldots, \lambda_m \geqslant 0$ with

$$\nabla \ell(0) = \sum_{i=1}^{m} \lambda_i \nabla p_i(0)$$

and set $\lambda_{m+1} = \cdots = \lambda_r = 0$. Suppose for contradiction that such λ_i do not exist. We may then apply Farkas's Lemma B.23, and conclude that there exists $w \in \mathbb{R}^n$ such that

$$\langle \nabla p_i(0), w \rangle \geqslant 0 \text{ for } i = 1, \ldots, m$$

but $\langle \nabla \ell(0), w \rangle < 0$. On the other hand, we may pick $v \in \mathbb{R}^n$ as in the hypothesis. Choose $\varepsilon > 0$ with

$$\langle \nabla \ell(0), w + \varepsilon v \rangle < 0.$$

Now if $1 \leqslant i \leqslant m$ and $\deg(p_i) \geqslant 2$, then $\langle \nabla p_i(0), w + \varepsilon v \rangle > 0$ for all $\varepsilon > 0$, which implies

$$p_i(\delta(w + \varepsilon v)) \geqslant 0$$

for all sufficiently small $\delta > 0$, since $p_i(0) = 0$. The same holds if $\deg(p_i) = 1$, since in this case we have

$$p_i(\delta(w + \varepsilon v)) = \delta \langle \nabla p_i(0), w + \varepsilon v \rangle \geqslant 0.$$

Finally, we may also assume that the same holds for the inactive inequalities (for $i = m+1, \ldots, r$), by making δ smaller if necessary. But this implies that $\delta(w + \varepsilon v)$ is a point in S for which

$$\ell(\delta(w + \varepsilon v)) = \delta \langle \nabla \ell(0), w + \varepsilon v \rangle < 0,$$

contradicting the fact that $0 = \ell(0)$ is the minimum of ℓ on S. □

The following basic lemma was used in the proof (and in fact also earlier in Example 3.21).

Lemma B.23 (Farkas's Lemma)
Let $c_1, \ldots, c_m \in \mathbb{R}^n$ and let

$$P = \left\{u \in \mathbb{R}^n \mid \langle c_i, u\rangle \geqslant 0 \text{ for all } i = 1, \ldots, m\right\}.$$

Then

$$P^\vee = \operatorname{cone}(c_1, \ldots, c_m).$$

In other words, for $c \in \mathbb{R}^n$ there either exists $u \in P$ such that $\langle c, u\rangle < 0$, or there exist $\lambda_1, \ldots, \lambda_m \geqslant 0$ with $c = \sum_{i=1}^m \lambda_i c_i$.

Proof By definition we have $\operatorname{cone}(c_1, \ldots, c_m)^\vee = P$, so that

$$P^\vee = \operatorname{cone}(c_1, \ldots, c_m)^{\vee\vee} = \operatorname{clos}(\operatorname{cone}(c_1, \ldots, c_m))$$

by Theorem B.5. However, a finitely generated convex cone is always closed, which can for example be deduced by reducing to the positive orthant via Carathéodory's Theorem (see for example [3] Theorem 2.3). This finishes the proof. □

Recall that a function $f : \mathbb{R}^n \to \mathbb{R}$ is *concave* on a convex subset $S \subseteq \mathbb{R}^n$ if

$$f(\lambda u + (1-\lambda)v) \geqslant \lambda f(u) + (1-\lambda) f(v)$$

holds for all $u, v \in S$ and $\lambda \in [0, 1]$. If the inequality is strict for all $u \neq v, \lambda \in (0, 1)$, f is called *strictly concave*. It is called (strictly) *convex* if the opposite inequality holds, i.e., if $-f$ is (strictly) concave. Some essential facts are contained in the following exercises.

Exercise B.24
Let $S \subseteq \mathbb{R}^n$ be a convex set and $f : \mathbb{R}^n \to \mathbb{R}$ a function.

1. If f is continuously differentiable, then f is concave on S if and only if

$$f(v) \leqslant f(u) + \langle \nabla f(u), v - u\rangle$$

holds for all $u, v \in S$.

2. If f is twice continuously differentiable, then f is concave on S if and only if its Hessian $\nabla^2 f$ is negative semidefinite in all points $u \in S$. Furthermore, if $(\nabla^2 f)(u)$ is negative definite for all $u \in S$, then f is strictly concave on S. Give an example showing that the converse is false.

Exercise B.25
Let $S = \mathcal{S}(p_1, \ldots, p_r)$ be a basic closed set, and suppose that $p_1, \ldots, p_r$ are concave on S. Show that S is convex.

Exercise B.26
If $S \subseteq \mathbb{R}^n$ is compact and convex, and $f : \mathbb{R}^n \to \mathbb{R}$ is concave on S, then there is an extreme point of S in which f attains its minimum on S.

Exercise B.27

1. The set of extreme points of a compact convex subset of $\mathbb{R}^2$ is compact.
2. Give an example of a compact convex subset of $\mathbb{R}^3$ whose set of extreme points is not closed.

Corollary B.28
Let $S = \mathcal{S}(p_1, \ldots, p_r)$ be compact and convex with non-empty interior, and suppose that $p_1, \ldots, p_r$ are concave on S. Then Lagrange multipliers exist for any linear polynomial at any minimizer on S.

Proof We can assume $p_i \neq 0$ for all $i = 1, \ldots, r$. Since S has non-empty interior, there exists a point $u_0 \in S$ with $p_i(u_0) > 0$ for $i = 1, \ldots, r$, since otherwise the product $p_1 \cdots p_r$ would vanish on S, contradicting the nonempty interior assumption.

Now let $\ell \in \mathbb{R}[x]_1$ and let $u \in S$ be a minimizer of ℓ on S. If $p_i(u) = 0$, then

$$0 < p_i(u_0) \leqslant p_i(u) + \langle \nabla p_i(u), u_0 - u \rangle = \langle \nabla p_i(u), u_0 - u \rangle,$$

since p_i is concave on S (using Exercise B.24 (1)). Hence the hypotheses of Theorem B.22 are satisfied for $v = u_0 - u$. □

Note that in the proof of the corollary we did not need the assumption that ℓ has degree 1. But in this way, it relies only on the special case of Theorem B.22 which we have actually proven.

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